# Key Features

## Processor

- Quad-core ARM Cortex™-A55
- Single-core E907 RISC-V MCU, up to 600 MHz
- Embedded E902 RISC-V
- Up to 2 TOPS NPU

## Memory

- 32-bit DDR3/DDR3L/LPDDR3/DDR4/LPDDR4/LPDDR4X interface with inline ECC, up to 8 GB
- SD3.0/eMMC5.1 interface
- 8-bit parallel NAND flash interface with maximum 80-bit/1KB BCH
- One octal SPI interface for flash devices

## Video Engine

- Video decoding
  - MJPEG decoder, up to 3840x2160@15fps
  - JPEG Baseline decoder, up to 1080P@60fps
- Video encoding
  - H.264 BP/MP/HP, up to 3840x2160@25fps
  - MJPEG, up to 3840x2160@15fps
  - JPEG, up to 8192 x 8192 resolution
- Video decoder and video encoder are time-multiplexed

## Video Input

- Parallel CSI
  - 8/10/12/16-bit
  - supporting BT.656 up to 4*720P@30fps and BT.1120 up to 4*1080P@30fps
- 4+4-lane/4+2+2-lane/2+2+2+2-lane MIPI CSI
  - Compliant with MIPI-CSI2 V1.1 and MIPI DPHY V1.2
  - Up to 2.0 Gbit/s per lane
  - Maximum video capture resolution of 8M@30fps (4-lane performance)

## Video Output

- One dual-link LVDS interfaces, up to 1920 x 1080@60fps
- One RGB interfaces with DE/SYNC mode, up to 1920 x 1200@60fps (reduced blanking)
- 4-lane MIPI-DSI output interface up to 1920 x 1200@60fps (reduced blanking)

## Audio

- 1 DAC
- 1 x audio output: LINEOUTP/N
- 4x I2S/PCM external interfaces
- Maximum 8 digital PDM microphones (DMIC)
- One OWA RX and one OWA TX, compliance with S/PDIF interface

## Peripherals Interfaces

- 1 x USB2.0 Host, 1 x USB2.0 DRD, 1 x USB3.1 DRD&PCIe2.1 Combo
- 2 x GMAC (10/100/1000 Mbps port with RGMII and RMII interfaces)
- 5 x IR RX, 1 x IR TX, 9 x TWI, 6 x SPI, 17 x UART, 4 x CAN-FD, 1 x Local Bus
- 4-ch TPADC, 34-ch PWM, 31-ch GPADC, 1-ch LRADC
- SDIO 3.0, LEDC

## Security Subsystem

- AES, DES, 3DES, and SM4 encryption and decryption algorithms
- MD5, SHA, and SM3 tamper proofing
- RSA, ECC, SM2 signature and verification algorithms

## Device Summary

| Orderable Device(s) | Package |
|---|---|
| T536MX-CEN2<br>T536MX-CEX<br>T536MX-CXX<br>T536LX-CXX | ED-FCCSP 533 balls<br>Body size: 15 mm x 15 mm<br>Maximum height: 1.056 mm<br>Ball pitch: 0.5 mm<br>Ball size: 0.3 mm |

See Table 2-1 Ordering Information for details.

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# Revision History

| Revision | Date | Author | Description |
|---|---|---|---|
| 1.00 | May 12, 2025 | AWA1896 | Second Major Release Version |

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# About This Document

## Purpose and Scope

The documentation describes features of each module, pin/signal characteristics, current consumption, interface timing, thermal and package of the T536 family. For details about register descriptions of each module, see the *T536_User_Manual*.

## Intended Audience

The document is intended for:

- Hardware designers and maintenance personnel for electronics
- Sales personnel for electronic parts and components

## Related Documentation

For a complete listing of related documentation and development-support tools for the device, contact the Allwinner FAE or access the Allwinner Customer Service Platform by visiting https://open.allwinnertech.com/.

## Revision Number Definition

**Revision 1.00-1.0x**

This document is released based on the verifications in small volume production. The information in this document may be modified, calibrated and supplemented hereafter.

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# Conventions

## Symbol Conventions

The symbols that may be found in this document are defined as follows.

| Symbol | Description |
|---|---|
|  CAUTION | A caution means that damage to equipment is possible. |
|  NOTE | Provides additional information to emphasize or supplement important points of the main text. |

## Table Content Conventions

The table content conventions that may be found in this document are defined as follows.

| Symbol | Description |
|---|---|
| - | The cell is blank. |

## Numerical System

The expressions of the data capacity, the frequency, and the data rate are described as follows.

| Type | Symbol | Value |
|---|---|---|
| Data capacity | K | 1024 |
| | M | 1,048,576 |
| | G | 1,073,741,824 |
| Frequency, data rate | k | 1000 |
| | M | 1,000,000 |
| | G | 1,000,000,000 |

# FT/QA/QC Test

All Allwinner chips provided for clients have passed the following tests.

| Test Item | Description |
|---|---|
| FT Test | FT test is the finished product testing after the chip is packaged, and it is a functional test of all modules for each produced chip. |
| QA Test | QA test is a system-level sampling test for good-quality chips. According to the application level of the chip, a certain percentage of good-quality chips are selected for system-level testing to make the chip work in a typical application scenario, and judge whether the chip works normally in this scenario. |

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| Test Item | Description |
| --- | --- |
| QC Test | QC test is a module-level sampling test for good-quality chips. According to the chip application level, a certain percentage of good-quality chips are selected for module-level functional testing to monitor whether the chip production process is normal. |

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# 1 Overview

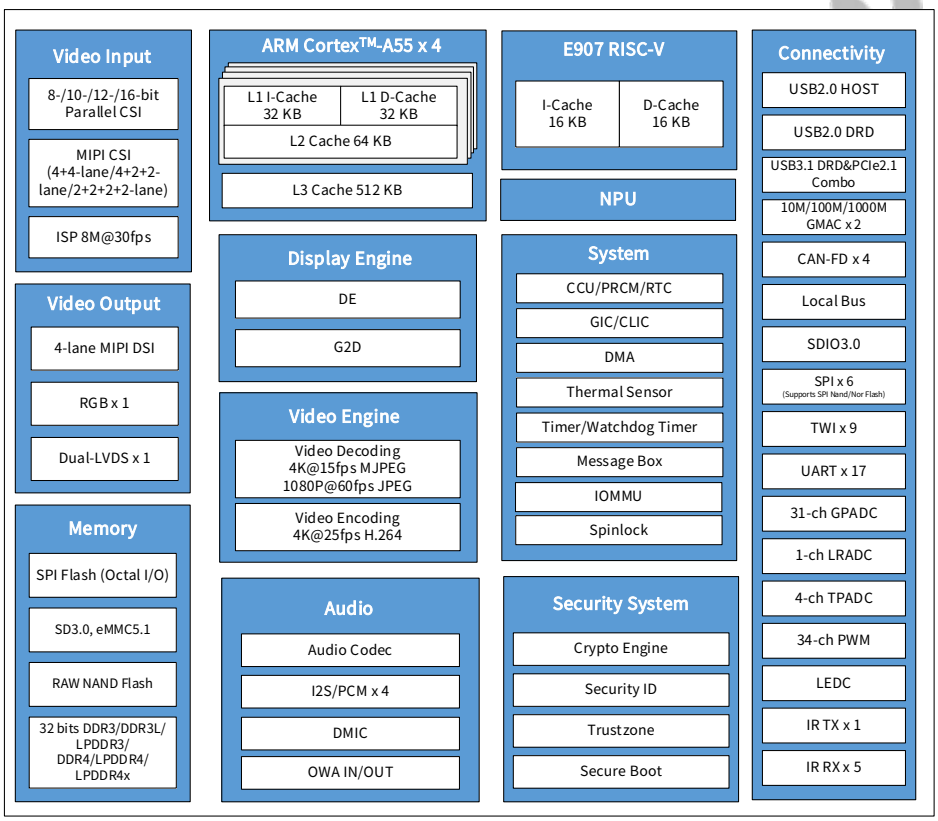
T536 family features high-performance quad-core Cortex™-A55 platform SoCs for the Industrial and intelligent hardware fields, which can be applied in interactive terminals, smart manufacturing, and other intelligent hardware and Industrial devices.

The T536 integrates a quad-core Cortex™-A55 CPU with independent L2 cache per core and a single-core E907 RISC-V with computing power scalability. T536 also has a neural processing unit (NPU) delivering up to 2 TOPS. With multiple heterogeneous expansion modes and multiple OS architectures. This processor family can meet the requirements of different application scenarios.

Besides, T536 family supports the combination of RGB/MIPI DSI/LVDS interfaces. The T536 also supplies high-speed interfaces for connectivity with 2 x GMAC and 1 x USB3.1 Gen1&PCIe2.1 Combo. In addition, the T536 processors contain 4 x CAN-FD and 1 x Local Bus for industrial application and expansion.

The following figure shows the system block diagram for the device.

Figure 1-1 System Block Diagram




(1) Video decoder and video encoder are time-multiplexed.

NOTE

Some modules shown in this block diagram are not offered on all devices, please refer to Table 2-1 Ordering Information for details.

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# 2 Ordering Information

The following table provides the differences of orderable device(s) covered by this document.

Table 2-1 Ordering Information

| Features | T536MX-CEN2 | T536MX-CEX | T536MX-CXX | T536LX-CXX |
|---|---|---|---|---|
| ARM CPU | 4 x A55, up to 1.6 GHz | 4 x A55, up to 1.6 GHz | 4 x A55, up to 1.6 GHz | 4 x A55, up to 1.4 GHz |
| RISC-V CPU | 1 x E907, up to 600 MHz<br>1 x E902, up to 200 MHz | 1 x E907, up to 600 MHz<br>1 x E902, up to 200 MHz | 1 x E907, up to 600 MHz<br>1 x E902, up to 200 MHz | 1 x E902, up to 200 MHz |
| NPU | Up to 2 TOPS | / | / | / |
| ECC Protection | Support | Support | / | / |
| CAN-FD | 4 | 4 | 4 | 2 |
| Local Bus | 1 | 1 | 1 | / |
| Qualification Level | Industrial | Industrial | Industrial | Industrial |
| Package | ED-FCCSP 533 balls, 15 mm x 15 mm | ED-FCCSP 533 balls, 15 mm x 15 mm | ED-FCCSP 533 balls, 15 mm x 15 mm | ED-FCCSP 533 balls, 15 mm x 15 mm |

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# 3 Features

## 3.1 Processors

- DynamIQ Architecture quad-core ARM Cortex™-A55
  - 32 KB L1 I/D-cache + 64 KB L2 cache per core
  - 512 KB L3 cache
  - ECC protection on L1 cache, L2 cache, and L3 cache
- E907 RISC-V MCU, up to 600 MHz
- E902 RISC-V for low-power management
- NPU
  - Up to 2 TOPS
  - Supports INT8, UINT8, and INT16

## 3.2 Memory

### 3.2.1 Internal Memory

- 160 KB SRAM A2
- 512 KB SHARED_SRAM (shared by NPU and system)

### 3.2.2 Boot ROM (BROM)

- On-chip memory
- Supports system boot from the following devices:
  - SD Card
  - eMMC
  - RAW NAND Flash
  - SPI NOR Flash (Quad Mode and Single Mode)
  - SPI NAND Flash
- Supports mandatory upgrade process through USB or SD card
- Supports GPADC pin and eFuse module to select the boot media type
- Supports normal booting and secure booting

### 3.2.3 Raw NAND Flash (NDFC)

- Up to 80-bit BCH per 1024 bytes
- 1K/2K/4K/8K/16K/32K bytes page size
- Up to 8-bit data bus width
- Supports SLC/MLC flash and EF-NAND

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- Supports SDR, ONFI DDR1.0, Toggle DDR1.0, ONFI DDR2.0, and Toggle DDR2.0 Raw NAND flash

### 3.2.4 SDRAM

- 32-bit DDR3/DDR3L/LPDDR3/DDR4/LPDDR4/LPDDR4X interface with inline ECC
  - Not supports 6GB (4 GB +2 GB) LPDDR3 address space when inline ECC is enabled
  - Not supports inline ECC when the DQ width of BL8 DDR(DDR3, DDR4, LPDDR3 and BL8 mode LPDDR4) is 16bits
- 2 ranks
- Memory capacity up to 8 GB
- Clock frequency up to 1066 MHz for DDR3, DDR3L, and LPDDR3
- Clock frequency up to 1600 MHz for DDR4, LPDDR4, and LPDDR4X

### 3.2.5 SD/MMC Host Controller (SMHC)

- Three SD/MMC Host Controller (SMHC) interfaces
  - SMHC0, compliant with the protocol Secure Digital Memory (up to SD3.0)
  - SMHC1, compliant with the protocol Secure Digital I/O (up to SDIO3.0)
  - SMHC2, compliant with the protocol Multimedia Card (up to eMMC5.1)
- The SMHC0 and the SMHC1 support the following:
  - 1-bit or 4-bit data width
  - ECC protection for SMHC0
  - Maximum performance:
    - SDR mode 200 MHz@1.8 V IO pad
    - DDR mode 50 MHz@1.8 V IO pad
    - SDR mode 50 MHz@3.3 V IO pad
- The SMHC2 supports the following:
  - 1-bit, 4-bit, or 8-bit data width
  - Supports ECC protection
  - Supports HS400 mode and HS200 mode
  - Maximum performance:
    - SDR mode 200 MHz@1.8V IO pad
    - DDR mode 200 MHz@1.8V IO pad
    - SDR mode 50 MHz@3.3V IO pad
    - DDR mode 50 MHz@3.3V IO pad
- Supports block size of 1 to 65535 bytes
- Supports hardware CRC generation and error detection

### 3.2.6 SPI Flash Controller (SPIF)

- Supports multiple SPI modes
  - Standard SPI

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  - Dual-Input/Dual-Output SPI, and Dual-I/O SPI
  - Quad-Input/Quad-Output SPI, Quad-I/O SPI, and QPI
  - Octal-Input/Octal-Output SPI, Octal-I/O SPI, and OPI
  - 3-wire SPI with programmable serial data frame length of 1 bit to 32 bits
- Supports STR mode and DTR mode, and DTR mode supports DQS signal
- Supports prefetch mode to execute code from the Flash memory and to significantly reduce read time
- High Speed Clock Frequency
  - 150MHz for STR Mode
  - 100MHz for DTR Mode
- Supports mode0, mode1, mode2 and mode3
- Supports control signal configuration
  - One chip select to support multiple peripherals
  - Polarity and phase of the Chip Select (SPIF_CS) and SPI Clock (SPIF_CLK) are configurable

## 3.3 Image Processing

### 3.3.1 Image Signal Processor (ISP)

- Supports multiple sensor (up to 4) inputs
- Supports raw8/10/12/14/16
- Maximum resolution:
  - Online: 2688 x 4224
  - Offline: 4800 x 4224
- Performance of one sensor:
  - Online: 5M@30fps (linear), 5M@30fps(2f-wdr)
  - Offline: 8M@30fps (linear), 8M@30fps (2f-wdr)
- Supports multiple data stream outputs
  - Crop
  - 1 to 1/16 scaling for height and width
  - Up to 16 rectangle boxes
  - YUV422/YUV420/RGB888/RGB565 output
  - graphics mirror and flip
- Supports algorithm features
  - Crop
  - 2F-WDR
  - Dynamic Range Compression (DRC)
  - Local Tone Mapping
  - Black Level Correction
  - Digital Gain

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- White Balance
- Len Shading Correction (LSC)
- Gamma Correction
- Defect Pixel Correction (DPC)
- Cross Talk Correction (CTC)
- Chromatic Aberration Correction (CAC)
- Noise Reduction (2D/3D)
- Bayer Interpolation
- Sharpness
- Color Enhancement (3D-LUT)
- Adjustable 3A functions: Automatic White Balance (AWB), Automatic Exposure (AE), and Automatic Focus (AF)
- Supports Anti-Flick Detection Statistics
- Histogram Statistics

### 3.3.2 Encoder Pre-Processor (ENCPP)

- Geometric Distortion Correction (GDC), including Lens Distortion Correction (LDC) and Fish Eye Correction (FEC)
- Input data format: 8-bit image, YUV420/YUV422/ARGB8888
- Input resolution
  - Minimum: 64 x 64
  - Maximum: 8192 x8192
- Output resolution
  - Minimum: 64 x 64
  - Maximum: 8192 x 8192
- Rotation of 90/180/270 degrees clockwise and horizontal flip
- Watermark images overlaying
- Pixel-level cropping and virtual zoom
- Maximum unlimited 8x horizontal and vertical zoom-out
- Scaled image write back

## 3.4 Graphics Processing

### 3.4.1 Display Engine (DE)

- One display output, size up to 2048 x 2048
- Three alpha blending channels optional
- Four overlay layers in each channel, and has an independent scaler.
- Supports potter-duff compatible blending operation

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- Input format
  - Semi-planar of YUV422/YUV420/YUV411
  - Planar of YUV422/YUV420/ YUV411
  - ARGB8888/XRGB8888/RGB888/ARGB4444/ ARGB1555/RGB565
- Supports input format palette only for UI channel to save image buffer memory
- Support Frame Packing/Top-and-Bottom/Side-by-Side Full/Side-by-Side Half 3D format data.
- SmartColor2.0 for excellent display experience
  - Flesh tone protection
  - Hue gain, saturation gain, and value gain controller
  - Fully programmable color matrix
  - Dynamic gamma
- Supports write back for verification

### 3.4.2 Graphic 2D (G2D)

- Mixer supports layer size up to 2K pixels, and rotation supports layer size up to 4K pixels
- Mixer contains the following functions:
  - Multiple video formats and format conversion
    - YUV 4:2:0, YUV4:2:2, and YUV4:1:1
    - Supports conversion between any two of the above formats
  - Multiple pixel formats and 8/16/24/32-bit graphic layer
  - 1/16x to 32x resize ratio
  - Pre-multiplied alpha
  - Color key
  - Window clip
  - Porter-duff alpha blending
  - 32-phase 8-tap horizontal anti-alias filter and 2-phase 4-tap vertical anti-alias filter
  - FillRectangle, BitBlit, StretchBlit, and MaskBlit.
- Multiple rotation types
  - Horizontal flip and vertical flip
  - 0, 90, 180, or 270 degrees' rotation in clockwise direction

## 3.5 Video Engine

NOTE

Video decoder and video encoder are time-multiplexed.

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### 3.5.1 Video Decoder

- Motion JPEG, up to 3840 x 2160@15fps
- JPEG baseline
  - Supports 1920x1080@60fps
  - Maximum decoding resolution: 16384x16384
  - Supports two channels for output, the prior channel and the second channel.
    - The prior channel supports online scale-down: 1/2, 1/4, 1/8, 1/16, 1/32
    - The second channel supports online scale-down: 1/2, 1/4, 1/8
    - The second can be enabled when the prior channel output format is YUV420 and scale ratio is 1/1, 1/2 or 1/4. The prior channel scale ratio shall be larger than the second channel scale ratio.
  - Supports translation: YUVV444 to YUV420, YUV422 to YUV420, YUV420T to YUV420, YUV to RGB
  - Supports tile decoding (ROI Decoding, random rectangle region in the JPEG Picture. If tile decoding mode is enabled, the second shall be disabled)

### 3.5.2 Video Encoder

- H.264 BP/MP/HP encoding
  - Supports 3840x2160@25fps
  - Supports I/P frame type
  - Supports CBR, VBR and FIXEDQP modes
  - Supports region of interest(ROI) encoding, a maximum of eight ROIs
  - Maximum resolution: 4096x4096
- JPEG baseline encoding
  - Supports 3840x2160@15fps
  - Supports YUV420, YUV422 and YUV444 format
  - Maximum resolution: 8192x8192
- MJPEG baseline encoding up to 3840x2160@15fps

## 3.6 Video Input Interfaces

### 3.6.1 MIPI CSI

- Complaint with MIPI CSI-2 specification v1.1 and MIPI D-PHY specification v1.2
- Up to 2.0 Gbit/s per lane, 8M@30fps RAW12 2F-WDR (4-lane performance), size up to 3264(H) x 2448(V)
- Configurable number of data lanes and sequence of data lanes for MIPI interface:
  - 4+4-lane
  - 4+2+2-lane

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- 2+2+2+2-lane
- Pixel formats
  - RAW8, RAW10, RAW12, RAW14, and RAW16
  - 8-bit YUV420 and YUV422
  - RGB888 and RGB565

### 3.6.2 Parallel CSI

- Supports 8/10/12/16-bit width
- Supports BT.656, BT.601, BT.1120, and Digital Camera (DC) protocol
- Supports ITU-R BT.656 up to 4*720P@30fps
- Supports ITU-R BT.1120 up to 4*1080P@30fps
- Supports BT.656 2/4-channel time-multiplexed format
- Dual Data Rate (DDR) sample mode with maximum pixel clock of 148.5MHz

## 3.7 Video Output Interfaces

### 3.7.1 RGB and LVDS

- One RGB/BT.656/i8080 interface
  - DE/SYNC RGB mode, up to1920 x 1200@60fps (reduced blanking)
  - Serial/dummy RGB mode, up to 800 x 480@60fps
  - Dither function for RGB888, RGB666 and RGB565
  - Support BT.656 interface for NTSC and PAL
  - i8080 interface, up to 800x480@60fps
- Dual-link LVDS interface
  - Up to 1920x1080@60fps for dual-link
  - Up to 1366x768@60fps for single-link

### 3.7.2 MIPI DSI

- Compliant with MIPI DSI specification v1.02 and MIPI D-PHY specification v1.0
- Up to 1.0 Gbit/s per lane
- Supports one 4-lane MIPI DSI display, up to 1920 x 1200@60fps (reduced blanking)

## 3.8 System Peripherals

### 3.8.1 Clock Controller Unit (CCU)

- Supports clock configuration and clock generation for corresponding modules
- Supports software-controlled clock gating and software-controlled reset for corresponding modules

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### 3.8.2 Message Box (MSGBOX)

- Up to seven message boxes
  - Three message boxes for communication between A55 subsystem, E907 subsystem, and E902 subsystem through one way channels.
  - Four separate message boxes for priority adjustment in AMP mode among A55 cores through one way channels
  - Each MSGBOX has one receiver and two or three transmitters
- 4 channels between every 2 users
- The FIFO depth of a channel is 8
- The FIFO width of a channel is 32 bits
- Supports interrupts

### 3.8.3 Power Reset Clock Management (PRCM)

- CPUS domain clock configuration
- Bus gating, bus reset, and clock configuration
- VDD-SYS power interface control
- RAM configuration control

### 3.8.4 RTC

- One on-chip RC oscillator
- Supports one external 24 MHz DCXO and one external 32.768 kHz oscillator
- Provides a 16-bit counter for counting day, 5-bit counter for counting hour, 6-bit counter for counting minute, 6-bit counter for counting second
- Timer frequency is 1 kHz
- Configurable initial value by software anytime
- Supports fanout function of internal 32.768 kHz clock
- Supports timing alarm, and generates interrupt and wakeup the external devices
- 8 general purpose registers for storing power-off information in RTC domain

### 3.8.5 Spinlock

- Supports 32 lock units
- Two kinds of lock status: locked and unlocked
- Lock time of the processor is predictable

### 3.8.6 Thermal Sensor Controller (THS)

- Three thermal sensors
- Averaging filter for thermal sensor reading
- Supports over-temperature protection interrupt and over-temperature alarm interrupt

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- Support ±3°C error limit from 60°C to 125°C, and ±5°C from -40°C to 60°C

### 3.8.7 Timer

- Sixteen programmable 56-bit down timers
- Configurable counting clock: 32KHz, 24MHz, 16MHz, or 200MHz
- Two working modes: periodic mode and single count mode
- Generates an interrupt when the count is decreased to 0

### 3.8.8 Watchdog Timer (WDT)

- Three watchdog timers
- Supports 12 initial values
- Supports the generation of timeout interrupts
- Supports the generation of reset signals
- Supports watchdog restart

## 3.9 Audio

### 3.9.1 Audio codec

- One audio differential lineout output: LINEOUTP/N
- One audio digital-to-analog converter (DAC) channel
  - 16-bit and 20-bit sample resolution
  - 8 kHz to 192 kHz DAC sample rate
  - 95 ± 3 dB SNR@A-weight, -85 ± 3 dB THD+N
- Supports Dynamic Range Controller adjusting the DAC playback
- One 128x20-bits FIFO for DAC data transmit
- Programmable FIFO thresholds
- Supports interrupts and DMA

### 3.9.2 I2S/PCM

- Four I2S/PCM interfaces (I2S0, I2S1, I2S2, and I2S3)
- Compliant with standard Philips Inter-IC sound (I2S) bus specification
  - Left-justified, Right-justified, PCM mode, and TDM format
  - Programmable PCM frame width: 1 BCLK width (short frame) and 2 BCLKs width (long frame)
- Transmit and receive data FIFOs
  - Programmable FIFO thresholds
  - 128 x 32-bit TXFIFO, 64 x 32-bit RXFIFO
- Supports multiple function clocks

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  - Clock up to 24.576 MHz Data Output of I2S/PCM in Master mode (Only if the IO PAD and peripheral I2S/PCM satisfy timing parameters)
  - Clock up to 12.288 MHz Data Input of I2S/PCM in Master mode (Only if the IO PAD and peripheral I2S/PCM satisfy timing parameters)
- Supports TX/RX DMA slave interface
- Supports multiple application scenarios
  - Up to 16 channels (fs = 48 kHz) which has adjustable width from 8 bits to 32 bits
  - Sample rate from 8 kHz to 384 kHz (sample rate * channel * slot width ⩽ 24.576 MHz)
  - 8-bit u-law and 8-bit A-law companded sample
- Supports master/slave mode

### 3.9.3 DMIC

- Supports maximum 8 digital PDM microphones
- Supports sample rate from 8 kHz to 48 kHz

### 3.9.4 One Wire Audio (OWA)

- One OWA TX and One OWA RX, compliant with S/PDIF interface
- Compatible with IEC-60958 and IEC-61937
  - IEC-60958 supports data formats: 16 bits, 20 bits, and 24 bits
- Supports sample rate: 8 kHz – 384 kHz
- Transmit and receive data FIFOs
  - Programmable FIFO thresholds
  - One 128x24bits TXFIFO and one 64x24bits RXFIFO for audio data transfer

## 3.10 Peripheral Interfaces

### 3.10.1 SerDes System

SerDes contains one PCIe2.1 & USB3.1 DRD combo PHY, one PCIe2.1 controller, one USB3.1 DRD controller, one USB2.0 DRD controller, and one USB2.0 Host controller.

#### 3.10.1.1 PCIe2.1

- One 1-lane PCI Express 2.1 (PCIe) port
- Supports Gen1(2.5 Gbit/s), Gen2 (5.0 Gbit/s) speed
- Supports Dual Mode (DM) mode: Root Complex (RC) mode and EndPoint (EP) mode
- RAS DP (Data Protection) for RAMs using ECC
- 2 Virtual Channels (VCs)
- Multiple Functions: support 4 Functions for PCIe in EP Mode
- Configurable MAX_PAYLOAD_SIZE size and up to 256 bytes

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- Embedded DMA with Hardware Flow Control. Support 4 Write/Read Channels
- Internal Address Translation Unit. Support 16 Inbound Address Translation regions and 16 Outbound Address Translation regions
- MSI with Per-Vector Masking (PVM), Extended message data for MSI
- PCIe and USB2_U3(USB3.1 GEN2) share one COMB0 PHY in SerDes
- PCI Express Active State Power Management (ASPM)

#### 3.10.1.2 USB3.1 DRD

- One USB3.1 PIPE PHY (USB2_U3)
- One USB 2.0 UTMI+ PHY (USB2_U2). USB2_U2 has no separate physical PHY, and it share the PHY with USB2.0 DRD or USB2.0 Host.
- Compliant with USB 3.1 Gen1 Specification
- Compatible to the xHCI specification v1.1
- Static host and device operation
- USB device mode supports the following:
  - Super-Speed (SS, 5 Gbit/s) for USB3.1 PHY
  - High-Speed (HS, 480 Mbit/s) and Full-Speed (FS, 12-Mbit/s) for USB2.0 PHY
- USB host mode supports the following:
  - Super-Speed (SS, 5 Gbit/s) for USB3.1 PHY
  - High-Speed (HS, 480 Mbit/s), Full-Speed (FS, 12 Mbit/s) and Low-Speed (LS, 1.5 Mbit/s) for USB2.0 PHY
- 4 pairs of user endpoints share TX FIFO (22 KB) and RX FIFO (asynchronous transmission 8KB and periodic transmission 8 KB). The size of Endpoint 0 RX/TX FIFO is 64B.
- Supports device or host operation at a time
- Hardware handles all data transfer
- AXI interface for DMA operation
- Implements both static and dynamic power reduction techniques at multiple levels

#### 3.10.1.3 USB2.0 DRD

- One USB2.0 DRD (USB0), with integrated USB 2.0 analog PHY
- Complies with USB2.0 Specification
- Static host and device mode
- USB host that supports the following:
  - Compatible with Enhanced Host Controller Interface (EHCI) specification, version 1.0
  - Compatible with Open Host Controller Interface (OHCI) specification, version 1.0a
  - High-Speed (HS, 480 Mbit/s), Full-Speed (FS, 12 Mbit/s), and Low-Speed (LS, 1.5 Mbit/s)
  - Supports only 1 USB Root Port shared between EHCI and OHCI
- USB device that supports the following:

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  - High-Speed (HS, 480 Mbit/s), Full-Speed (FS, 12 Mbit/s)
  - Up to 8KB+64Bytes FIFO for EPs (including EP0)
- Supports an internal DMA controller for data transfer with memory
- Device and Host controller share an 8KB SRAM and a physical PHY
- Link low power mode: supports L0(On)/L2(Suspend)/L3(Off)

#### 3.10.1.4 USB2.0 Host

- One USB 2.0 Host (USB1), with integrated USB 2.0 analog PHY
- Compatible with Enhanced Host Controller Interface (EHCI) specification, version 1.0
- Compatible with Open Host Controller Interface (OHCI) specification, version 1.0a
- Supports High-Speed (HS, 480 Mbit/s), Full-Speed (FS, 12 Mbit/s) and Low-Speed (LS, 1.5 Mbit/s) Device
- An internal DMA Controller for data transfer with memory
- Supports only 1 USB Root Port shared between EHCI and OHCI
- Link low power mode: supports L0(On)/L2(Suspend)/L3(Off)

### 3.10.2 GMAC

- Two 10/100/1000 Mbit/s Ethernet ports with RGMII and RMII interfaces
- Compliant with IEEE 802.3-2015 standard
- IEEE 1588-2008 for precision networked clock synchronization
- Supports Ethernet packet timestamping described in IEEE 1588-2002 and IEEE 1588-2008
- Supports both full-duplex and half-duplex operations
- Supports Full-duplex flow control
- Programmable frame length to support Standard or Jumbo Ethernet frames with sizes up to 16 KB
- Double VLAN for VLAN insertion or VLAN replacement
- Transmitting TCP/IP Checksum Offload
- 4 KB TXFIFO for transmission packets and 8 KB RXFIFO for reception packets.
- Supports the management data input/output (MDIO) interface
- Supports memory ECC

### 3.10.3 CAN-FD

- Four CAN-FD ports (up to 64 data bytes)
- Conforms with ISO 11898-1:2015
- Supports CAN 2.0A and 2.0B protocol specification
- Supports AUTomotive Open System ARchitecture (AUTOSAR)
- Supports SAE J1939 standards
- CAN error logging

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- Improved acceptance filtering
- Two configurable receive FIFOs
- Separate signaling on reception of high priority messages
- Up to 64 dedicated receive buffers
- Up to 32 dedicated transmit buffers
- Configurable transmit FIFO
- Configurable transmit queue
- Configurable transmit event FIFO
- Direct message RAM access for host CPU
- Supports DMA Master

### 3.10.4 Local Bus Controller (LBC)

- 8/16/32-bit wide transmission
- Address and data multiplexed access
- Supports parity check
- Synchronous read/write 128 x 8 FIFO (AHB clock domain)
- Read/write 16 x 8 buffer (local bus clock domain)
- Up to 400 MHz AHB clock and up to 100 MHz local bus clock
- Up to four chip selects, and the polarity is configurable
- Supports burst mode

### 3.10.5 UART

- Up to 17 UART controllers
  - UART0, UART1, UART2, UART3, UART4, UART5, UART6, UART7, UART8, UART9, UART10, UART11, UART12, UART13, and UART14
  - S_UART0 and S_UART1
- Compatible with industry-standard 16450/16550 UARTs
- 5-8 data bits, 1/1.5/2 stop bits, programmable parity (even, odd, or no parity)
  - Hardware parity is not supported when RS485 half-duplex hardware auto flow control mode is enabled. See section 17.13.5.4 in T536_User_Manual for the solution to this problem.
- Programmable baud rates up to 10 Mbit/s (excluding S_UART0 and S_UART1)
- Two separate FIFOs: one is RX FIFO, and the other is TX FIFO
  - Each of them is 64 bytes for UART0, UART9, UART10, UART11, UART12, UART13, UART14, and S_UART0
  - Each of them is 128 bytes for UART1, UART2, UART3, UART4, UART5, UART6, UART7, UART8, and S_UART1
- Supports 9-bit/RS-485 format and RS-485 full duplex mode
- Supports Software/hardware flow control

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- Supports interrupts and DMA mode
- Supports auto-flow by using CTS & RTS (excluding UART0)
- Supports LIN mode (UART1/S_UART1)

### 3.10.6 PWM

- Up to 30 PWM channels and 4 S-PWM channels
  - Supports PWM continuous mode output
  - Supports PWM pulse mode output, and the pulse number is configurable
  - Output frequency range:
    - 0 to 24 MHz (when the clock source is DCXO)
    - 0 to 100 MHz (when the clock source is APB0 clock)
  - Various duty-cycle: 0% -100%
  - Minimum resolution: 1/65536
  - Up to 10-ch periodic counting of input pulses
- Maximum 5 complementary pairs output and the pairing methods are as follows:
  - PWM Channels

    PWM0-0+PWM0-1, PWM0-2+PWM0-3, PWM0-4+PWM0-5, PWM0-6+PWM0-7, PWM0-8+PWM0-9, PWM1-0+PWM1-1, PWM1-2+PWM1-3, PWM1-4+PWM1-5, PWM1-6+PWM1-7, PWM1-8+PWM1-9, PWM2-0+PWM2-1, PWM2-2+PWM2-3, PWM2-4+PWM2-5, PWM2-6+PWM2-7, PWM2-8+PWM2-9
  - S-PWM Channels

    S-PWM0-1 + S-PWM0-1, S-PWM0-2 + S-PWM0-3
  - Supports dead-zone generator, and the dead-zone time is configurable
- Maximum 4 group of PWM channel output for controlling stepping motors
  - Supports any plural channels to form a group, and output the same duty-cycle pulse
  - In group mode, the relative phase of the output waveform for each channel is configurable
- Maximum 10 channels capture input
  - Supports rising edge detection and falling edge detection for input waveform pulse
  - Supports pulse-width measurement for input waveform pulse

### 3.10.7 SPI and SPI_DBI

- Up to 6 SPI controllers
  - SPI0, SPI2, SPI3, SPI4, and S_SPI0, supporting SPI mode
  - SPI1, supporting SPI mode and display bus interface (DBI) mode

**SPI mode**

- Multiple SPI modes:
  - Master mode and slave mode for standard SPI

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  - Master mode for Dual-Output/Dual-Input SPI and Dual I/O SPI
  - Master mode for Quad-Output/Quad-Input SPI
  - Master mode for 3-wire SPI, with programmable serial data frame length of 1 bit to 32 bits
  - Camera slave mode for Single-Output/Single-Input SPI (only SPI2 and SPI3)
  - Camera slave mode for Dual-Input SPI (only SPI2 and SPI3)
  - Camera slave mode for Quad-Input SPI (only SPI2 and SPI3)
  - Ordinary slave mode for Single-Output/Single-Input SPI
- Supports vertical synchronization and frame header detection for camera slave mode
- Supports mode0, mode1, mode2, and mode3 for master mode and both two slave modes
- TX/RX DMA Slave interface
- Transmit data FIFO with 8-bit wide and 64-entry
- Receive data FIFO with 8-bit wide and 64-entry
- Transmit data buffer with 8-bit wide and 128-entry
- Receive data buffer with 8-bit wide and 128-entry
- Supports control signal configuration for master mode and both two slave modes
  - Two chip selects for SPI0 and one chip select for SPI1, SPI2, SPI3, SPI4, S_SPI0 to support multiple peripherals
  - Polarity and phase of the Chip Select (SPI_CS) and SPI Clock (SPI_CLK) are configurable

DBI mode

- DBI Type C 3 Line/4 Line Interface Mode
- 2 Data Lane Interface Mode
- RGB111/444/565/666/888 video format
- Maximum resolution of RGB666 240 x 320@30Hz with single data lane
- Maximum resolution of RGB888 240 x 320@60Hz or 320 x 480@30Hz with dual data lane
- Tearing effect
- Software flexible control video frame rate

### 3.10.8 Two Wire Interface (TWI)

- Up to 9 TWI controllers
  - TWI0, TWI1, TWI2, TWI3, TWI4, TWI5, and TWI6
  - S_TWI0 and S_TWI1
- Compliant with I2C bus standard
- master mode and slave mode
- 7-bit and 10-bit device addressing modes
- Standard mode (up to 100 Kbit/s) and fast mode (up to 400 Kbit/s)
- Supports multi-master system

### 3.10.9 General Purpose ADC (GPADC)

- Three 10-ch ADCs (GPADC0-2) and one 1-ch ADC (GPADC3), total 31 channels

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- 12-bit sampling resolution
- Maximum sampling frequency
  - GPADC0-2: up to 1 MHz
  - GPADC3: up to 2 MHz
- Supports three operation modes:
  - Single conversion mode
  - Continuous conversion mode
  - Burst conversion mode (GPADC2 does not support this mode)
- Analog input range: 0 to 1.8 V

### 3.10.10 Low Rate ADC (LRADC)

- 1-ch LRADC with 6-bit resolution and maximum 2kHz sampling rate
- Supports hold key and general key
- Supports normal, continue and single work mode
- Power supply voltage:1.8 V; power reference voltage:1.35 V
- Voltage input range between 0 to 1.35 V

### 3.10.11 Touch Panel ADC (TPADC)

- 12-bit SAR type A/D converter
- 32x12 FIFO for storing A/D conversion result
- Supports DMA slave interface
- Supports AUX ADC, up to 4 channels
- 4-wire resistive touch panel input detection
  - Single touch detection
  - Dual touch detection
  - Touch pressure measurement with programmable threshold
  - Pen down detection with programmable sensitivity
  - Median and averaging filter for noise reduction
  - X/Y coordinate exchange
- Maximum sampling rate of 750 kHz

### 3.10.12 LED Controller (LEDC)

- LEDC is used to control the external intelligent control LED lamp
- Configurable LED output high/low level width and reset time
- LEDC data supports DMA configuration mode and CPU configuration mode
- Maximum 1024 LEDs serial connect
- Configurable interval time between data packets and frame data
- Configurable RGB display mode
- Configurable default level of non-data output

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### 3.10.13Consumer Infrared Transmitter (IR_TX)

- One IR_TX interface
- NEC format infra data
- RLC encoding
- Arbitrary wave generator
- Configurable carrier frequency
- 64x8 bits FIFO for data buffer
- Supports Interrupts and DMA mode
- Supports DMA shake and wait mode

### 3.10.14Consumer Infrared Receiver (IR_RX)

- Five IR_RX interfaces
- NEC format infra data
- RLC encoding
- 64x8 bits FIFO for data buffer
- Sample clock up to 1 MHz

## 3.11 Security

### 3.11.1 Crypto Engine (CE)

- Symmetrical algorithm:
  - AES symmetrical algorithm
    - Key size: 128/192/256 bits
    - CFB mode includes: CFB1, CFB8, CFB64, and CFB128
    - CTR mode includes: CTR16, CTR32, CTR64, and CTR128
    - Supports ECB, CBC, CTS, OFB, CBC-MAC, GCM, and XTS modes
  - Support XTS-AES symmetrical algorithm: key size: 256/512 bits
  - DES symmetrical algorithm
    - CTR mode, includes: CTR16, CTR32, and CTR64
    - Supports ECB, CBC, and CBC-MAC mode
  - 3DES, SM4 symmetrical algorithm
- Hash algorithms
  - Support MD5, SHA1, SHA224, SHA256, SHA384, SHA512, and SM3
  - Support HMAC-SHA1, HMAC-SHA256
- Random bit generator algorithms
  - Support PRNG, 175 bits seed width, and output with multiple of 5 words
  - Support TRNG, post-process by hardware with SHA256, output with multiple of 8 words

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- Public key algorithms
  - Support RSA public key algorithms: 512/1024/2048/3072/4096-bit width
  - Support ECC public key algorithms: 160/224/256/384/521-bit width
  - Support SM2 algorithms

### 3.11.2 Security ID (SID)

- 2048-bit eFuse
- Supports secure and non-secure world in eFuse
- Backup eFuse information by using SID_SRAM
- One-time programming
- Selecting double-bit check by parameter definition
- Data scrambling
- Reading and writing protection
- Password for secure debug

### 3.11.3 Secure Memory Control (SMC)

- The SMC is always secure, only secure CPU can access the SMC
- Sets secure area of DRAM
- Supports Master and address protection
- Sets secure property that Master accesses to DRAM
- Maximum 16 regions and Master has access to each region

### 3.11.4 Secure Peripherals Control (SPC)

- The SPC is always secure, only secure CPU can access the SPC
- Sets secure property of peripherals

### 3.11.5 TrustZone Memory Access (TZMA)

- The TZMA is always secure and can only be accessed by secure CPU
- Sets the secure area of SRAM

## 3.12 Package

ED-FCCSP 533 balls, 15 mm x 15 mm body size, 1.056 mm height (maximum), 0.5 mm ball pitch, 0.3 mm ball size.

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# 4 Pin Description

## 4.1 Pin Characteristics

This section lists the characteristics of the device pins from the following seven aspects.

[1] **Ball#**: Package ball numbers associated with each signal.

[2] **Pin Name**: The name of the package pin.

NC means these pins are not connected.

[3] **Type**: Denotes the signal direction

I (Input),

O (Output),

I/O (Input/Output),

OD (Open-Drain),

A (Analog),

AI (Analog Input),

AO (Analog Output),

P (Power),

G (Ground)

N/A (Not Applicable)

[4] **Ball Reset State**: The state of the terminal at reset.

PU: Pull Up

PD: Pull Down

Z: High Impedance

N/A: Not Applicable

[5] **Pull Up/Down**: Denotes the presence of an internal pull-up or pull-down resistor. Pull-up and pull-down resistors can be enabled or disabled via software.

PU: Internal pullup

PD: Internal pulldown

PU/PD: Internal pullup and pulldown

N/A: Not Applicable

[6] **Default Buffer Strength**: Defines the default drive strength of the associated output buffer. The maximum drive strength of each GPIO is 6 mA.

N/A means Not Applicable.

[7] **I/O Power Supply**: The voltage supplies for the IO buffers of the terminal.

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N/A means Not Applicable.

### 4.1.1 SDRAM

Table 4-1 SDRAM Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AC13 | SRST | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| V18 | SZQ | AI | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH12 | SA0 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG12 | SA1 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AC15 | SA2 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH10 | SA3 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH6 | SA4 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH11 | SA5 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AB11 | SA6 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AB15 | SA7 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AB9 | SA8 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF13 | SA9 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG13 | SA10 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH13 | SA11 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG11 | SA12 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD11 | SA13 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AC9 | SA14 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD9 | SA15 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG10 | SA16 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF11 | SA17 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AC11 | SACT | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD7 | SBA0 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG6 | SBA1 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF9 | SBG0 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD13 | SBG1 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF7 | SCKE0 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AH9 | SCKN | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG9 | SCKP | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG8 | SCS0 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH8 | SCS1 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF15 | SDQM0 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH19 | SDQM1 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH2 | SDQM2 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AE2 | SDQM3 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH14 | SDQ0 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG14 | SDQ1 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD15 | SDQ2 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH16 | SDQ3 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG16 | SDQ4 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH17 | SDQ5 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AB17 | SDQ6 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AC17 | SDQ7 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG17 | SDQ8 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF17 | SDQ9 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG19 | SDQ10 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD17 | SDQ11 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF19 | SDQ12 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD19 | SDQ13 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG20 | SDQ14 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH20 | SDQ15 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF3 | SDQ16 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG1 | SDQ17 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG2 | SDQ18 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG3 | SDQ19 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG4 | SDQ20 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH4 | SDQ21 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AG5 | SDQ22 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH5 | SDQ23 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AB2 | SDQ24 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AC3 | SDQ25 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AB3 | SDQ26 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AC2 | SDQ27 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AF2 | SDQ28 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AE3 | SDQ29 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AE5 | SDQ30 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD5 | SDQ31 | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH15 | SDQS0N | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG15 | SDQS0P | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH18 | SDQS1N | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG18 | SDQS1P | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AJ3 | SDQS2N | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH3 | SDQS2P | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD3 | SDQS3N | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AD2 | SDQS3P | I/O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AG7 | SODT0 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| AH7 | SODT1 | O | N/A | N/A | N/A | VCC-DRAM/VCC-DRAML |
| W9 | VCC-DRAM | P | N/A | N/A | N/A | N/A |
| W12 | VCC-DRAM | P | N/A | N/A | N/A | N/A |
| W13 | VCC-DRAM | P | N/A | N/A | N/A | N/A |
| W14 | VCC-DRAM | P | N/A | N/A | N/A | N/A |
| W15 | VCC-DRAM | P | N/A | N/A | N/A | N/A |
| Y12 | VCC-DRAM | P | N/A | N/A | N/A | N/A |
| V8 | VCC-DRAML | P | N/A | N/A | N/A | N/A |
| V10 | VCC-DRAML | P | N/A | N/A | N/A | N/A |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| V11 | VCC-DRAML | P | N/A | N/A | N/A | N/A |
| V14 | VCC-DRAML | P | N/A | N/A | N/A | N/A |
| Y15 | VDD18-DRAM | P | N/A | N/A | N/A | N/A |

### 4.1.2 System Control

Table 4-2 System Control Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AD21 | FEL | I | PU | PU/PD | N/A | VCC-IO |
| AE21 | JTAG-SEL | I | PU | PU/PD | N/A | VCC-IO |
| AE28 | NMI | I/O, OD | N/A | N/A | N/A | VCC-RTC |
| AF27 | RESET | I/O, OD | N/A | N/A | N/A | VCC-RTC |

### 4.1.3 RTC&PLL

Table 4-3 RTC&PLL Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AF28 | X32KFOUT | AO, OD | N/A | N/A | N/A | VCC-PM |
| AG29 | X32KIN | AI | N/A | N/A | N/A | VCC-RTC |
| AG28 | X32KOUT | AO | N/A | N/A | N/A | VCC-RTC |
| AA21 | PLLTEST | AO, OD | N/A | N/A | N/A | VCC-PLL |
| R14 | CPU-PLLTEST | AO, OD | N/A | N/A | N/A | VCC-CPU-PLL |
| R13 | VCC-CPU-PLL | P | N/A | N/A | N/A | N/A |
| AB24 | VCC-RTC | P | N/A | N/A | N/A | N/A |
| AA23 | VCC-PLL | P | N/A | N/A | N/A | N/A |

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### 4.1.4 DCXO

Table 4-4 DCXO Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AJ27 | DXIN | AI | N/A | N/A | N/A | VCC-PLL |
| AH27 | DXOUT | AO | N/A | N/A | N/A | VCC-PLL |
| AH28 | REFCLK-OUT | AO | N/A | N/A | N/A | VCC-PLL |

### 4.1.5 GPIO Groups

#### 4.1.5.1 Port A

Table 4-5 Port A Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| H24 | PA0 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| H25 | PA1 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| F27 | PA2 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| E28 | PA3 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| E29 | PA4 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| F28 | PA5 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| G29 | PA6 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| G28 | PA7 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| G27 | PA8 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| H26 | PA9 | I/O | Z | PU/PD | 4 mA | VCC18-ADC/VCC-PA |
| K22 | VCC-PA | P | N/A | N/A | N/A | N/A |

#### 4.1.5.2 Port B

Table 4-6 Port B Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AC21 | PB0 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AB21 | PB1 | I/O | Z | PU/PD | 4 mA | VCC-IO |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AG24 | PB2 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AF23 | PB3 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AE23 | PB4 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AD23 | PB5 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AD24 | PB6 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AH24 | PB7 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AG25 | PB8 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AH26 | PB9 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AG26 | PB10 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AF25 | PB11 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AC23 | PB12 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AJ25 | PB13 | I/O | Z | PU/PD | 4 mA | VCC-IO |
| AH25 | PB14 | I/O | Z | PU/PD | 4 mA | VCC-IO |

#### 4.1.5.3 Port C

Table 4-7 Port C Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| V4 | PC0 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| V3 | PC1 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| V5 | PC2 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| Y3 | PC3 | I/O | PU | PU/PD | 4 mA | VCC-PC |
| Y4 | PC4 | I/O | PU | PU/PD | 4 mA | VCC-PC |
| V7 | PC5 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| V6 | PC6 | I/O | PU | PU/PD | 4 mA | VCC-PC |
| Y7 | PC7 | I/O | PU | PU/PD | 4 mA | VCC-PC |
| V1 | PC8 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| V2 | PC9 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| W1 | PC10 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| W2 | PC11 | I/O | Z | PU/PD | 4 mA | VCC-PC |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| Y5 | PC12 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| Y1 | PC13 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| Y2 | PC14 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| AA1 | PC15 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| AA2 | PC16 | I/O | Z | PU/PD | 4 mA | VCC-PC |
| U8 | VCC-PC | P | N/A | N/A | N/A | N/A |

#### 4.1.5.4 Port D

Table 4-8 Port D Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| B5 | PD0 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| A5 | PD1 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| B4 | PD2 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| A4 | PD3 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| B3 | PD4 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| A3 | PD5 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| C1 | PD6 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| C2 | PD7 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| D1 | PD8 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| D2 | PD9 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| E1 | PD10 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| E2 | PD11 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| F1 | PD12 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| F2 | PD13 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| G1 | PD14 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| G2 | PD15 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| H1 | PD16 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| H2 | PD17 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| J1 | PD18 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| J2 | PD19 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| K6 | PD20 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| H6 | PD21 | I/O | Z | PU/PD | 4 mA | VCC-PD/VCC18-LVDS |
| H5 | PD22 | I/O | Z | PU/PD | 4 mA | VCC-PD |
| H4 | PD23 | I/O | Z | PU/PD | 4 mA | VCC-PD |
| K8 | VCC-PD | P | N/A | N/A | N/A | N/A |
| L9 | VCC18-LVDS | P | N/A | N/A | N/A | N/A |

#### 4.1.5.5 Port E

Table 4-9 Port E Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| B9 | PE0 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| G9 | PE1 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| F9 | PE2 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| G11 | PE3 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| F11 | PE4 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| E11 | PE5 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| D13 | PE6 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| E13 | PE7 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| E9 | PE8 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| D11 | PE9 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| D9 | PE10 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| C10 | PE11 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| A10 | PE12 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| C9 | PE13 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| B10 | PE14 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| H13 | PE15 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| G13 | PE16 | I/O | Z | PU/PD | 4 mA | VCC-PE |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| F13 | PE17 | I/O | Z | PU/PD | 4 mA | VCC-PE |
| K11 | VCC-PE | P | N/A | N/A | N/A | N/A |

#### 4.1.5.6 Port F

Table 4-10 Port F Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AH21 | PF0 | I/O | Z | PU/PD | 4 mA | VCC-IO/VCC18-PF |
| AG21 | PF1 | I/O | Z | PU/PD | 4 mA | VCC-IO/VCC18-PF |
| AH22 | PF2 | I/O | Z | PU/PD | 4 mA | VCC-IO/VCC18-PF |
| AG22 | PF3 | I/O | Z | PU/PD | 4 mA | VCC-IO/VCC18-PF |
| AJ23 | PF4 | I/O | Z | PU/PD | 4 mA | VCC-IO/VCC18-PF |
| AH23 | PF5 | I/O | Z | PU/PD | 4 mA | VCC-IO/VCC18-PF |
| AF21 | PF6 | I/O | Z | PU/PD | 4 mA | VCC-IO/VCC18-PF |
| V19 | VCC18-PF | P | N/A | N/A | N/A | N/A |

#### 4.1.5.7 Port G

Table 4-11 Port G Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| R29 | PG0 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| T27 | PG1 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| R28 | PG2 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| R27 | PG3 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| U29 | PG4 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| U28 | PG5 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| T24 | PG6 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| T25 | PG7 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| V27 | PG8 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| V28 | PG9 | I/O | Z | PU/PD | 4 mA | VCC-PG |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| V26 | PG10 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| V25 | PG11 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| T26 | PG12 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| U27 | PG13 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| P24 | PG14 | I/O | Z | PU/PD | 4 mA | VCC-PG |
| P23 | VCC-PG | P | N/A | N/A | N/A | N/A |

#### 4.1.5.8 Port H

Table 4-12 Port H Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| G5 | PH0 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| F5 | PH1 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| E5 | PH2 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| D5 | PH3 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| B6 | PH4 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| A6 | PH5 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| C6 | PH6 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| B7 | PH7 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| G7 | PH8 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| F7 | PH9 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| E7 | PH10 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| D7 | PH11 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| C7 | PH12 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| B8 | PH13 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| C8 | PH14 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| A8 | PH15 | I/O | Z | PU/PD | 4 mA | VCC-PH |
| K10 | VCC-PH | P | N/A | N/A | N/A | N/A |

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#### 4.1.5.9 Port I

Table 4-13 Port I Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| B27 | PI0 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| B28 | PI1 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| C28 | PI2 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| C27 | PI3 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| C29 | PI4 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| D27 | PI5 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| A27 | PI6 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| B26 | PI7 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| D25 | PI8 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| E25 | PI9 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| D21 | PI10 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| E21 | PI11 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| F21 | PI12 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| D23 | PI13 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| E23 | PI14 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| A26 | PI15 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| B25 | PI16 | I/O | Z | PU/PD | 4 mA | VCC-PI |
| K20 | VCC-PI | P | N/A | N/A | N/A | N/A |

#### 4.1.5.10 Port J

Table 4-14 Port J Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| P1 | PJ0 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| N3 | PJ1 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| P3 | PJ2 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| U3 | PJ3 | I/O | Z | PU/PD | 4 mA | VCC-PJ |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| P4 | PJ4 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| T3 | PJ5 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| T7 | PJ6 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| T6 | PJ7 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| P5 | PJ8 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| K5 | PJ9 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| U2 | PJ10 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| N2 | PJ11 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| P2 | PJ12 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| T2 | PJ13 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| T4 | PJ14 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| T5 | PJ15 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| M1 | PJ16 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| M3 | PJ17 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| M2 | PJ18 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| L1 | PJ19 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| R3 | PJ20 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| R2 | PJ21 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| P7 | PJ22 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| P6 | PJ23 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| K4 | PJ24 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| K3 | PJ25 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| M5 | PJ26 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| M4 | PJ27 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| K1 | PJ28 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| K2 | PJ29 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| M7 | PJ30 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| M6 | PJ31 | I/O | Z | PU/PD | 4 mA | VCC-PJ |
| P8 | VCC-PJ | P | N/A | N/A | N/A | N/A |

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#### 4.1.5.11 Port K

Table 4-15 Port K Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| B13 | PK0 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A13 | PK1 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B11 | PK2 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A11 | PK3 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B12 | PK4 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A12 | PK5 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B16 | PK6 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A16 | PK7 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B14 | PK8 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A14 | PK9 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B15 | PK10 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A15 | PK11 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B19 | PK12 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A19 | PK13 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B17 | PK14 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A17 | PK15 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| B18 | PK16 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| A18 | PK17 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| D19 | PK18 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| E19 | PK19 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| E15 | PK20 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| D15 | PK21 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| E17 | PK22 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| D17 | PK23 | I/O | Z | PU/PD | 4 mA | VCC-PK/VCC18-MCSI |
| J13 | VCC-PK | P | N/A | N/A | N/A | N/A |
| H15 | VCC18-MCSI | P | N/A | N/A | N/A | N/A |

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#### 4.1.5.12 Port L

Table 4-16 Port L Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| W27 | PL0 | I/O | PU | PU/PD | 4 mA | VCC-PL |
| Y28 | PL1 | I/O | PU | PU/PD | 4 mA | VCC-PL |
| W28 | PL2 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| W29 | PL3 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| AA29 | PL4 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| Y27 | PL5 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| Y26 | PL6 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| Y25 | PL7 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| AB28 | PL8 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| AB27 | PL9 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| AA27 | PL10 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| AA28 | PL11 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| AB26 | PL12 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| AB25 | PL13 | I/O | Z | PU/PD | 4 mA | VCC-PL |
| T21 | VCC-PL | P | N/A | N/A | N/A | N/A |

#### 4.1.5.13 Port M

Table 4-17 Port M Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| AD27 | PM0 | I/O | Z | PU/PD | 4 mA | VCC-PM |
| AE29 | PM1 | I/O | Z | PU/PD | 4 mA | VCC-PM |
| AD28 | PM2 | I/O | Z | PU/PD | 4 mA | VCC-PM |
| AC27 | PM3 | I/O | Z | PU/PD | 4 mA | VCC-PM |
| AC29 | PM4 | I/O | Z | PU/PD | 4 mA | VCC-PM |
| AC28 | PM5 | I/O | Z | PU/PD | 4 mA | VCC-PM |
| V21 | VCC-PM | P | N/A | N/A | N/A | N/A |

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### 4.1.6 SerDes

Table 4-18 SerDes Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| A24 | PCIE-REF-CLKN | A I/O | N/A | N/A | N/A | VCC18-COMB |
| B24 | PCIE-REF-CLKP | A I/O | N/A | N/A | N/A | VCC18-COMB |
| A25 | PCIE-REXT | A I/O | N/A | N/A | N/A | VCC18-COMB |
| A23 | PCIE-RX0-DN/USB2-U3-RXN | AI | N/A | N/A | N/A | VCC18-COMB |
| B23 | PCIE-RX0-DP/USB2-U3-RXP | AI | N/A | N/A | N/A | VCC18-COMB |
| A22 | PCIE-TX0-DN/USB2-U3-TXN | AO | N/A | N/A | N/A | VCC18-COMB |
| B22 | PCIE-TX0-DP/USB2-U3-TXP | AO | N/A | N/A | N/A | VCC18-COMB |
| B20 | USB0-DM | A I/O | N/A | N/A | N/A | VCC33-USB0-USB1 |
| A20 | USB0-DP | A I/O | N/A | N/A | N/A | VCC33-USB0-USB1 |
| B21 | USB1-DM | A I/O | N/A | N/A | N/A | VCC33-USB0-USB1 |
| A21 | USB1-DP | A I/O | N/A | N/A | N/A | VCC33-USB0-USB1 |
| H19 | VCC18-COMB | P | N/A | N/A | N/A | N/A |
| K16 | VCC33-USB0-USB1 | P | N/A | N/A | N/A | N/A |
| J18 | VDD09-COMB | P | N/A | N/A | N/A | N/A |
| J16 | VDD09-USB0-USB1 | P | N/A | N/A | N/A | N/A |

### 4.1.7 Audio Codec

Table 4-19 Audio Codec Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| D29 | LINEOUTN | AO | N/A | N/A | N/A | AVCC |
| D28 | LINEOUTP | AO | N/A | N/A | N/A | AVCC |
| H22 | VRA1 | AO | N/A | N/A | N/A | AVCC |
| K21 | AVCC | P | N/A | N/A | N/A | N/A |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| H23 | AGND | G | N/A | N/A | N/A | N/A |

### 4.1.8 GPADC

Table 4-20 GPADC Pin Characteristics(1)

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| K28 | BOOT-SEL-GPADC1-4 | AI | N/A | N/A | N/A | VCC18-ADC |
| L29 | DDR-PARA-SEL-GPADC1-5 | AI | N/A | N/A | N/A | VCC18-ADC |
| K27 | GPADC1-6 | AI | N/A | N/A | N/A | VCC18-ADC |
| K26 | GPADC1-7 | AI | N/A | N/A | N/A | VCC18-ADC |
| L28 | GPADC1-8 | AI | N/A | N/A | N/A | VCC18-ADC |
| M24 | GPADC1-9 | AI | N/A | N/A | N/A | VCC18-ADC |
| L27 | GPADC2-0 | AI | N/A | N/A | N/A | VCC18-ADC |
| M25 | GPADC2-1 | AI | N/A | N/A | N/A | VCC18-ADC |
| M26 | GPADC2-2 | AI | N/A | N/A | N/A | VCC18-ADC |
| M27 | GPADC2-3 | AI | N/A | N/A | N/A | VCC18-ADC |
| M28 | GPADC2-4 | AI | N/A | N/A | N/A | VCC18-ADC |
| N27 | GPADC2-5 | AI | N/A | N/A | N/A | VCC18-ADC |
| N28 | GPADC2-6 | AI | N/A | N/A | N/A | VCC18-ADC |
| N29 | GPADC2-7 | AI | N/A | N/A | N/A | VCC18-ADC |
| P26 | GPADC2-8 | AI | N/A | N/A | N/A | VCC18-ADC |
| P27 | GPADC2-9 | AI | N/A | N/A | N/A | VCC18-ADC |
| H27 | GPADC3-0 | AI | N/A | N/A | N/A | VCC18-ADC |
| M23 | VCC18-VCM-ADC | A I/O | N/A | N/A | N/A | VCC18-ADC |
| M21 | VCC18-ADC | P | N/A | N/A | N/A | N/A |
| M22 | AGND-ADC | G | N/A | N/A | N/A | N/A |

(1) The signals of GPADC1-0, GPADC1-1, GPADC 1-2, GPADC1-3 are multiplexed with TPADC signals, see Table 4-21.

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### 4.1.9 TPADC

Table 4-21 TPADC Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| J29 | TP-X1/GPADC1-0 | AI | N/A | N/A | N/A | VCC18-ADC |
| H28 | TP-X2/GPADC1-1 | AI | N/A | N/A | N/A | VCC18-ADC |
| J27 | TP-Y1/GPADC1-2 | AI | N/A | N/A | N/A | VCC18-ADC |
| J28 | TP-Y2/GPADC1-3 | AI | N/A | N/A | N/A | VCC18-ADC |

### 4.1.10 LRADC

Table 4-22 LRADC Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| P28 | LRADC | AI | N/A | N/A | N/A | VCC18-ADC |

### 4.1.11 Power

Table 4-23 Power Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| W20 | VCC18-EFUSE | P | N/A | N/A | N/A | N/A |
| V20 | VCC-IO | P | N/A | N/A | N/A | N/A |
| N9 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| N10 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| P9 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| P11 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| T9 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| T10 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| T11 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| N11 | VDD-CPU | P | N/A | N/A | N/A | N/A |
| T13 | VDD-CPUFB | P | N/A | N/A | N/A | N/A |
| V23 | VDD-CPUS | P | N/A | N/A | N/A | N/A |
| N18 | VDD-NPU | P | N/A | N/A | N/A | N/A |

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| Ball#[1] | Pin Name[2] | Type[3] | Ball Reset State[4] | Pull Up/Down[5] | Default Buffer Strength[6] | I/O Power Supply[7] |
|---|---|---|---|---|---|---|
| N19 | VDD-NPU | P | N/A | N/A | N/A | N/A |
| N20 | VDD-NPU | P | N/A | N/A | N/A | N/A |
| P20 | VDD-NPU | P | N/A | N/A | N/A | N/A |
| R19 | VDD-NPU | P | N/A | N/A | N/A | N/A |
| R20 | VDD-NPU | P | N/A | N/A | N/A | N/A |
| T18 | VDD-NPUFB | P | N/A | N/A | N/A | N/A |
| L14 | VDD-SYS | P | N/A | N/A | N/A | N/A |
| L15 | VDD-SYS | P | N/A | N/A | N/A | N/A |
| N14 | VDD-SYS | P | N/A | N/A | N/A | N/A |
| N15 | VDD-SYS | P | N/A | N/A | N/A | N/A |
| R15 | VDD-SYS | P | N/A | N/A | N/A | N/A |
| T15 | VDD-SYS | P | N/A | N/A | N/A | N/A |
| L13 | VDD-SYSFB | P | N/A | N/A | N/A | N/A |
| L19 | VDD-VE | P | N/A | N/A | N/A | N/A |

### 4.1.12 Ground

Table 4-24 Ground Pin Characteristics

| Ball#[1] | Pin Name[2] | Type[3] |
|---|---|---|
| A1, A2, A28, A29, B1, B2, B29, C3, C5, C11, C13, C15, C17, C19, C21, C23, C25, D3, F3, F15, F17, F19, F23, F25, G15, G21, H3, H7, H9, H11, H17, H21, J15, J17, J19, J20, K7, K9, K13, K18, K19, K24, K25, L2, L11, L16, L17, L18, M8, M12, M16, N13, N16, N21, N22, P12, P16, P22, P25, R16, T1, T8, T17, T20, T22, T23, T28, U14, U15, U16, U18, U19, V22, V24, Y6, Y8, Y9, Y17, Y18, Y19, Y23, Y24, AA8, AA9, AA11, AA13, AA15, AA17, AA19, AB4, AB5, AB6, AB7, AB13, AB19, AB23, AC1, AC7, AC19, AD25, AE1, AE7, AE9, AE11, AE13, AE15, AE17, AE19, AE27, AF5, AF26, AG23, AG27, AH1, AH29, AJ1, AJ2, AJ5, AJ7, AJ9, AJ11, AJ13, AJ15, AJ17, AJ19, AJ21, AJ28, AJ29 | GND | G |

### 4.1.13 Others

Table 4-25 Unconnected Pin List

| Ball#[1] | Pin Name[2] | Type[3] |
|---|---|---|
| G19, AD26 | NC | N/A |

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## 4.2 GPIO Multiplex Function

The following tables provide a description of the GPIO multiplex function.

NOTE

For each GPIO, Function0 is input function; Function1 is output function; Function7 to Function13 are reserved.

### 4.2.1 Port A

Table 4-26 Port A Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PA0 | I/O | GPADC0-0 | | | | | | PA-EINT0 |
| PA1 | I/O | GPADC0-1 | | | | | | PA-EINT1 |
| PA2 | I/O | GPADC0-2 | | | | | | PA-EINT2 |
| PA3 | I/O | GPADC0-3 | | | | | | PA-EINT3 |
| PA4 | I/O | GPADC0-4 | | | | | | PA-EINT4 |
| PA5 | I/O | GPADC0-5 | | | | | | PA-EINT5 |
| PA6 | I/O | GPADC0-6 | | | | | | PA-EINT6 |
| PA7 | I/O | GPADC0-7 | | | | | | PA-EINT7 |
| PA8 | I/O | GPADC0-8 | | | | | | PA-EINT8 |
| PA9 | I/O | GPADC0-9 | | | | | | PA-EINT9 |

### 4.2.2 Port B

Table 4-27 Port B Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PB0 | I/O | UART2-TX | SPI4-CS0 | JTAG-MS | PWM0-6 | | RJTAG-MS | PB-EINT0 |
| PB1 | I/O | UART2-RX | SPI4-CLK | JTAG-CK | PWM0-7 | | RJTAG-CK | PB-EINT1 |
| PB2 | I/O | UART2-RTS | SPI4-MOSI | JTAG-DO | | | RJTAG-DO | PB-EINT2 |
| PB3 | I/O | UART2-CTS | SPI4-MISO | JTAG-DI | | | RJTAG-DI | PB-EINT3 |
| PB4 | I/O | TWI1-SCK | I2S2-MCLK | | PWM0-8 | IR-RX0 | UART8-TX | PB-EINT4 |
| PB5 | I/O | TWI1-SDA | I2S2-BCLK | | PWM0-9 | IR-RX1 | UART8-RX | PB-EINT5 |
| PB6 | I/O | | I2S2-LRCK | | PWM1-0 | IR-TX | UART8-RTS | PB-EINT6 |
| PB7 | I/O | OWA-IN | I2S2-DOUT0 | I2S2-DIN1 | PWM1-1 | CAN1-TX0 | UART8-CTS | PB-EINT7 |
| PB8 | I/O | OWA-OUT | I2S2-DIN0 | I2S2-DOUT1 | PWM0-0 | CAN1-RX0 | UART8-DCD | PB-EINT8 |
| PB9 | I/O | UART0-TX | I2S2-DIN2 | | I2S2-DOUT2 | TWI0-SCK | UART8-DSR | PB-EINT9 |
| PB10 | I/O | UART0-RX | I2S2-DIN3 | PWM0-1 | I2S2-DOUT3 | TWI0-SDA | UART8-DTR | PB-EINT10 |
| PB11 | I/O | TWI5-SCK | UART7-RTS | SPI1-CS0 | PWM0-2 | | UART8-RI | PB-EINT11 |
| PB12 | I/O | TWI5-SDA | UART7-CTS | SPI1-CLK | PWM0-3 | | | PB-EINT12 |
| PB13 | I/O | TWI4-SCK | UART7-TX | SPI1-MOSI | PWM0-4 | IR-RX2 | | PB-EINT13 |
| PB14 | I/O | TWI4-SDA | UART7-RX | SPI1-MISO | PWM0-5 | IR-RX3 | | PB-EINT14 |

### 4.2.3 Port C

Table 4-28 Port C Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PC0 | I/O | NAND-WE | SDC2-DS | | | | | PC-EINT0 |
| PC1 | I/O | NAND-ALE | SDC2-RST | | | | | PC-EINT1 |
| PC2 | I/O | NAND-CLE | | SPI0-MOSI | SPIF-MOSI | | | PC-EINT2 |
| PC3 | I/O | NAND-CE1 | | SPI0-CS0 | SPIF-CS0 | | | PC-EINT3 |

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| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PC4 | I/O | NAND-CE0 | | SPI0-MISO | SPIF-MISO | | | PC-EINT4 |
| PC5 | I/O | NAND-RE | SDC2-CLK | | | | | PC-EINT5 |
| PC6 | I/O | NAND-RB0 | SDC2-CMD | | | | | PC-EINT6 |
| PC7 | I/O | NAND-RB1 | | SPI0-CS1 | SPIF-DQS | | | PC-EINT7 |
| PC8 | I/O | NAND-DQ7 | SDC2-D3 | | SPIF-D7 | | | PC-EINT8 |
| PC9 | I/O | NAND-DQ6 | SDC2-D4 | | SPIF-D6 | | | PC-EINT9 |
| PC10 | I/O | NAND-DQ5 | SDC2-D0 | | SPIF-D5 | | | PC-EINT10 |
| PC11 | I/O | NAND-DQ4 | SDC2-D5 | | SPIF-D4 | | | PC-EINT11 |
| PC12 | I/O | NAND-DQS | | SPI0-CLK | SPIF-CLK | | | PC-EINT12 |
| PC13 | I/O | NAND-DQ3 | SDC2-D1 | | | | | PC-EINT13 |
| PC14 | I/O | NAND-DQ2 | SDC2-D6 | | | | | PC-EINT14 |
| PC15 | I/O | NAND-DQ1 | SDC2-D2 | SPI0-WP | SPIF-WP | | | PC-EINT15 |
| PC16 | I/O | NAND-DQ0 | SDC2-D7 | SPI0-HOLD | SPIF-HOLD | | | PC-EINT16 |

### 4.2.4 Port D

Table 4-29 Port D Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PD0 | I/O | LCD-D2 | LVDS0-D0P | DSI-D0P | PWM0-0 | | UART9-TX | PD-EINT0 |
| PD1 | I/O | LCD-D3 | LVDS0-D0N | DSI-D0N | PWM0-1 | | UART9-RX | PD-EINT1 |
| PD2 | I/O | LCD-D4 | LVDS0-D1P | DSI-D1P | PWM0-2 | | UART9-RTS | PD-EINT2 |
| PD3 | I/O | LCD-D5 | LVDS0-D1N | DSI-D1N | PWM0-3 | | UART9-CTS | PD-EINT3 |
| PD4 | I/O | LCD-D6 | LVDS0-D2P | DSI-CKP | PWM0-4 | | UART10-TX | PD-EINT4 |
| PD5 | I/O | LCD-D7 | LVDS0-D2N | DSI-CKN | PWM0-5 | | UART10-RX | PD-EINT5 |
| PD6 | I/O | LCD-D10 | LVDS0-CKP | DSI-D2P | PWM0-6 | | UART10-RTS | PD-EINT6 |
| PD7 | I/O | LCD-D11 | LVDS0-CKN | DSI-D2N | PWM0-7 | | UART10-CTS | PD-EINT7 |
| PD8 | I/O | LCD-D12 | LVDS0-D3P | DSI-D3P | PWM0-8 | | CAN1-TX0 | PD-EINT8 |
| PD9 | I/O | LCD-D13 | LVDS0-D3N | DSI-D3N | PWM0-9 | | CAN1-RX0 | PD-EINT9 |
| PD10 | I/O | LCD-D14 | LVDS1-D0P | SPI1-CS0/DBI-CSX | PWM1-0 | TWI6-SCK | UART8-TX | PD-EINT10 |
| PD11 | I/O | LCD-D15 | LVDS1-D0N | SPI1-CLK/DBI-SCLK | PWM1-1 | TWI6-SDA | UART8-RX | PD-EINT11 |
| PD12 | I/O | LCD-D18 | LVDS1-D1P | SPI1-MOSI/DBI-SDO | PWM1-2 | | UART8-RTS | PD-EINT12 |
| PD13 | I/O | LCD-D19 | LVDS1-D1N | SPI1-MISO/DBI-SDI/DBI-TE/DBI-DCX | PWM1-3 | | UART8-CTS | PD-EINT13 |
| PD14 | I/O | LCD-D20 | LVDS1-D2P | SPI1-HOLD/DBI-DCX/DBI-WRX | PWM1-4 | UART3-TX | UART8-DCD | PD-EINT14 |
| PD15 | I/O | LCD-D21 | LVDS1-D2N | SPI1-WP/DBI-TE | PWM1-5 | UART3-RX | UART8-DSR | PD-EINT15 |
| PD16 | I/O | LCD-D22 | LVDS1-CKP | CAN2-TX0 | PWM1-6 | UART3-RTS | UART8-DTR | PD-EINT16 |
| PD17 | I/O | LCD-D23 | LVDS1-CKN | CAN2-RX0 | PWM1-7 | UART3-CTS | UART8-RI | PD-EINT17 |
| PD18 | I/O | LCD-CLK | LVDS1-D3P | TWI6-SCK | PWM1-8 | | CAN0-TX0 | PD-EINT18 |
| PD19 | I/O | LCD-DE | LVDS1-D3N | TWI6-SDA | PWM1-9 | | CAN0-RX0 | PD-EINT19 |
| PD20 | I/O | LCD-HSYNC | PWM0-2 | UART2-TX | TWI5-SCK | | | PD-EINT20 |
| PD21 | I/O | LCD-VSYNC | PWM0-3 | UART2-RX | TWI5-SDA | | | PD-EINT21 |
| PD22 | I/O | | PWM0-1 | UART2-RTS | TWI4-SCK | TWI0-SCK | | PD-EINT22 |
| PD23 | I/O | | PWM0-0 | UART2-CTS | TWI4-SDA | TWI0-SDA | | PD-EINT23 |

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### 4.2.5 Port E

Table 4-30 Port E Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PE0 | I/O | MCSI0-MCLK(1) | | | | SPI3-CS0 | LBUS-CS0 | PE-EINT0 |
| PE1 | I/O | TWI2-SCK | UART4-TX | | | SPI3-CLK | LBUS-CS1 | PE-EINT1 |
| PE2 | I/O | TWI2-SDA | UART4-RX | | | SPI3-MOSI | LBUS-CS2 | PE-EINT2 |
| PE3 | I/O | TWI3-SCK | UART4-RTS | CSI0-XHS | | SPI3-MISO | LBUS-CS3 | PE-EINT3 |
| PE4 | I/O | TWI3-SDA | UART4-CTS | CSI1-XHS | | SPI3-WP | LBUS-DP0 | PE-EINT4 |
| PE5 | I/O | MCSI1-MCLK | | | LEDC | SPI3-HOLD | LBUS-DP1 | PE-EINT5 |
| PE6 | I/O | | UART12-TX | TWI2-SCK | TCON-FSYNC0 | | LBUS-DP2 | PE-EINT6 |
| PE7 | I/O | | UART12-RX | TWI2-SDA | CLK-FANOUT0 | | LBUS-DP3 | PE-EINT7 |
| PE8 | I/O | | UART12-RTS | NCSI-D0 | CLK-FANOUT1 | | LBUS-WR | PE-EINT8 |
| PE9 | I/O | | UART12-CTS | NCSI-D1 | | SPI2-CS0 | LBUS-READY | PE-EINT9 |
| PE10 | I/O | MCSI3-MCLK | PWM0-3 | NCSI-D2 | | SPI2-CLK | LBUS-ALE | PE-EINT10 |
| PE11 | I/O | TWI1-SCK | UART5-RTS | NCSI-D3 | UART6-TX | SPI2-MOSI | LBUS-BURST0 | PE-EINT11 |
| PE12 | I/O | TWI1-SDA | UART5-CTS | NCSI-D4 | UART6-RX | SPI2-MISO | LBUS-BURST1 | PE-EINT12 |
| PE13 | I/O | TWI4-SCK | UART5-TX | NCSI-D5 | UART6-RTS | SPI2-WP | LBUS-BURST2 | PE-EINT13 |
| PE14 | I/O | TWI4-SDA | UART5-RX | NCSI-D6 | UART6-CTS | SPI2-HOLD | LBUS-LCLK | PE-EINT14 |
| PE15 | I/O | MCSI2-MCLK | PWM0-2 | NCSI-D7 | | CAN2-TX0 | LBUS-INTR | PE-EINT15 |
| PE16 | I/O | TWI6-SCK | PWM0-4 | CSI0-XVS-FSYNC | TWI0-SCK | CAN2-RX0 | LBUS-DRQ | PE-EINT16 |
| PE17 | I/O | TWI6-SDA | PWM0-5 | CSI1-XVS-FSYNC | TWI0-SDA | | LBUS-LBE | PE-EINT17 |

### 4.2.6 Port F

Table 4-31 Port F Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PF0 | I/O | SDC0-D1 | JTAG-MS | | I2S3-DIN0 | I2S3-DOUT1 | RJTAG-MS | PF-EINT0 |
| PF1 | I/O | SDC0-D0 | JTAG-DI | | I2S3-DOUT0 | I2S3-DIN1 | RJTAG-DI | PF-EINT1 |
| PF2 | I/O | SDC0-CLK | UART0-TX | | I2S3-DIN2 | SPI4-CLK | I2S3-DOUT2 | PF-EINT2 |
| PF3 | I/O | SDC0-CMD | JTAG-DO | | I2S3-LRCK | SPI4-CS0 | RJTAG-DO | PF-EINT3 |
| PF4 | I/O | SDC0-D3 | UART0-RX | | I2S3-DIN3 | SPI4-MOSI | I2S3-DOUT3 | PF-EINT4 |
| PF5 | I/O | SDC0-D2 | JTAG-CK | | I2S3-BCLK | SPI4-MISO | RJTAG-CK | PF-EINT5 |
| PF6 | I/O | | | | I2S3-MCLK | | | PF-EINT6 |

### 4.2.7 Port G

Table 4-32 Port G Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PG0 | I/O | SDC1-CLK | | UART13-TX | | | | PG-EINT0 |
| PG1 | I/O | SDC1-CMD | | UART13-RX | | | | PG-EINT1 |
| PG2 | I/O | SDC1-D0 | PCIE-PERSTN | UART13-RTS | | | | PG-EINT2 |
| PG3 | I/O | SDC1-D1 | PCIE-WAKEN | UART13-CTS | | | | PG-EINT3 |
| PG4 | I/O | SDC1-D2 | PCIE-CLKREQN | | | | | PG-EINT4 |
| PG5 | I/O | SDC1-D3 | | | | | | PG-EINT5 |
| PG6 | I/O | UART1-TX | TWI6-SCK | | | | | PG-EINT6 |
| PG7 | I/O | UART1-RX | TWI6-SDA | | | | | PG-EINT7 |
| PG8 | I/O | UART1-RTS | TWI5-SCK | | | | | PG-EINT8 |
| PG9 | I/O | UART1-CTS | TWI5-SDA | | | | | PG-EINT9 |

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| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PG10 | I/O | UART14-TX | I2S1-MCLK | TWI2-SCK | | | | PG-EINT10 |
| PG11 | I/O | UART14-RX | I2S1-BCLK | TWI2-SDA | | | | PG-EINT11 |
| PG12 | I/O | UART14-RTS | I2S1-LRCK | | CAN3-TX0 | | | PG-EINT12 |
| PG13 | I/O | UART14-CTS | I2S1-DOUT0 | I2S1-DIN1 | CAN3-RX0 | | | PG-EINT13 |
| PG14 | I/O | | I2S1-DIN0 | I2S1-DOUT1 | | | | PG-EINT14 |

### 4.2.8 Port H

Table 4-33 Port H Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PH0 | I/O | TWI0-SCK | PWM2-4 | UART1-TX | CAN1-TX0 | | LBUS-LD24 | PH-EINT0 |
| PH1 | I/O | TWI0-SDA | PWM2-5 | UART1-RX | CAN1-RX0 | | LBUS-LD25 | PH-EINT1 |
| PH2 | I/O | TWI1-SCK | | UART1-RTS | CLK-FANOUT0 | | LBUS-LD26 | PH-EINT2 |
| PH3 | I/O | TWI1-SDA | IR-TX | UART1-CTS | CLK-FANOUT1 | | LBUS-LD27 | PH-EINT3 |
| PH4 | I/O | UART3-TX | SPI3-CS0 | | PWM1-4 | DMIC-CLK | LBUS-LD28 | PH-EINT4 |
| PH5 | I/O | UART3-RX | SPI3-CLK | LEDC | PCIE-PERSTN | DMIC-DATA0 | LBUS-LD29 | PH-EINT5 |
| PH6 | I/O | UART3-RTS | SPI3-MOSI | TWI6-SCK | PCIE-WAKEN | DMIC-DATA1 | LBUS-LD30 | PH-EINT6 |
| PH7 | I/O | UART3-CTS | SPI3-MISO | TWI6-SDA | PCIE-CLKREQN | DMIC-DATA2 | LBUS-LD31 | PH-EINT7 |
| PH8 | I/O | PWM2-6 | SPI3-WP | OWA-IN | LCD-D0 | DMIC-DATA3 | TWI3-SCK | PH-EINT8 |
| PH9 | I/O | PWM2-7 | SPI3-HOLD | I2S3-MCLK | LCD-D1 | | TWI3-SDA | PH-EINT9 |
| PH10 | I/O | PWM2-8 | | I2S3-BCLK | LCD-D8 | CAN3-TX0 | | PH-EINT10 |
| PH11 | I/O | IR-RX3 | PWM2-9 | I2S3-LRCK | LCD-D9 | CAN3-RX0 | | PH-EINT11 |
| PH12 | I/O | IR-RX1 | I2S3-DOUT0 | I2S3-DIN1 | LCD-D16 | CAN2-TX0 | UART11-TX | PH-EINT12 |
| PH13 | I/O | IR-RX2 | I2S3-DOUT1 | I2S3-DIN0 | LCD-D17 | CAN2-RX0 | UART11-RX | PH-EINT13 |
| PH14 | I/O | IR-TX | I2S3-DOUT2 | I2S3-DIN2 | PWM1-5 | | UART11-RTS | PH-EINT14 |
| PH15 | I/O | IR-RX0 | I2S3-DOUT3 | I2S3-DIN3 | LEDC | | UART11-CTS | PH-EINT15 |

### 4.2.9 Port I

Table 4-34 Port I Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PI0 | I/O | I2S0-BCLK | | CAN0-TX0 | | | | PI-EINT0 |
| PI1 | I/O | I2S0-LRCK | | CAN0-RX0 | | | | PI-EINT1 |
| PI2 | I/O | I2S0-DIN0 | UART9-TX | CAN3-TX0 | | | | PI-EINT2 |
| PI3 | I/O | I2S0-MCLK | UART9-RX | CAN3-RX0 | | | | PI-EINT3 |
| PI4 | I/O | TWI5-SCK | UART9-CTS | | | | | PI-EINT4 |
| PI5 | I/O | TWI5-SDA | UART9-RTS | | | | | PI-EINT5 |
| PI6 | I/O | UART6-TX | SPI2-CS0 | PWM2-0 | I2S0-MCLK | LEDC | CLK-FANOUT0 | PI-EINT6 |
| PI7 | I/O | UART6-RX | SPI2-CLK | PWM2-1 | I2S0-DIN2 | I2S0-DOUT2 | CLK-FANOUT1 | PI-EINT7 |
| PI8 | I/O | UART6-CTS | SPI2-MOSI | PWM2-2 | I2S0-BCLK | CAN2-TX0 | | PI-EINT8 |
| PI9 | I/O | UART6-RTS | SPI2-MISO | PWM2-3 | I2S0-LRCK | CAN2-RX0 | IR-TX | PI-EINT9 |
| PI10 | I/O | OWA-OUT | SPI2-WP | PWM2-7 | I2S0-DOUT0 | I2S0-DIN1 | PCIE-PERSTN | PI-EINT10 |
| PI11 | I/O | TWI3-SCK | SPI2-HOLD | PWM2-8 | I2S0-DIN0 | I2S0-DOUT1 | PCIE-WAKEN | PI-EINT11 |
| PI12 | I/O | TWI3-SDA | PWM2-4 | DMIC-CLK | I2S0-DIN3 | I2S0-DOUT3 | PCIE-CLKREQN | PI-EINT12 |
| PI13 | I/O | UART4-TX | PWM2-5 | DMIC-DATA0 | | IR-RX0 | I2S3-BCLK | PI-EINT13 |
| PI14 | I/O | UART4-RX | PWM2-6 | DMIC-DATA1 | | IR-RX1 | I2S3-LRCK | PI-EINT14 |
| PI15 | I/O | UART4-RTS | TWI2-SCK | DMIC-DATA2 | CAN0-TX0 | IR-RX2 | I2S3-DIN0 | PI-EINT15 |
| PI16 | I/O | UART4-CTS | TWI2-SDA | DMIC-DATA3 | CAN0-RX0 | IR-RX3 | I2S3-MCLK | PI-EINT16 |

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### 4.2.10 Port J

Table 4-35 Port J Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PJ0 | I/O | PWM0-0 | I2S2-MCLK | | RGMII0-RXD1/RMII0-RXD1 | | | PJ-EINT0 |
| PJ1 | I/O | PWM0-1 | I2S2-BCLK | | RGMII0-RXD0/RMII0-RXD0 | | | PJ-EINT1 |
| PJ2 | I/O | PWM0-2 | I2S2-LRCK | | RGMII0-RXCTL/RMII0-CRS-DV | | | PJ-EINT2 |
| PJ3 | I/O | PWM0-3 | I2S2-DOUT0 | I2S2-DIN1 | RGMII0-CLKIN/RMII0-RXER | | | PJ-EINT3 |
| PJ4 | I/O | PWM0-4 | I2S2-DIN0 | I2S2-DOUT1 | RGMII0-TXD1/RMII0-TXD1 | | | PJ-EINT4 |
| PJ5 | I/O | PWM0-5 | I2S2-DIN2 | I2S2-DOUT2 | RGMII0-TXD0/RMII0-TXD0 | | | PJ-EINT5 |
| PJ6 | I/O | PWM0-6 | I2S2-DIN3 | I2S2-DOUT3 | RGMII0-TXCK/RMII0-TXCK | | | PJ-EINT6 |
| PJ7 | I/O | PWM0-7 | I2S1-MCLK | | RGMII0-TXCTL/RMII0-TXEN | | | PJ-EINT7 |
| PJ8 | I/O | PWM0-8 | I2S1-BCLK | | RGMII0-MDC | | | PJ-EINT8 |
| PJ9 | I/O | PWM0-9 | I2S1-LRCK | | RGMII0-MDIO | | | PJ-EINT9 |
| PJ10 | I/O | PWM2-0 | I2S1-DOUT0 | I2S1-DIN1 | RGMII0-EPHY-25/50M | | | PJ-EINT10 |
| PJ11 | I/O | PWM2-1 | I2S1-DIN0 | I2S1-DOUT1 | RGMII0-RXD3/RMII0-NULL | | | PJ-EINT11 |
| PJ12 | I/O | PWM2-2 | UART10-TX | | RGMII0-RXD2/RMII0-NULL | | | PJ-EINT12 |
| PJ13 | I/O | PWM2-3 | UART10-RX | | RGMII0-RXCK/RMII0-NULL | | | PJ-EINT13 |
| PJ14 | I/O | PWM2-4 | UART10-RTS | TWI6-SCK | RGMII0-TXD3/RMII0-NULL | | | PJ-EINT14 |
| PJ15 | I/O | PWM2-5 | UART10-CTS | TWI6-SDA | RGMII0-TXD2/RMII0-NULL | | | PJ-EINT15 |
| PJ16 | I/O | PWM2-6 | | IR-RX0 | RGMII1-RXD1/RMII1-RXD1 | | | PJ-EINT16 |
| PJ17 | I/O | PWM2-7 | | IR-RX1 | RGMII1-RXD0/RMII1-RXD0 | | | PJ-EINT17 |
| PJ18 | I/O | PWM2-8 | | IR-RX2 | RGMII1-RXCTL/RMII1-CRS-DV | | | PJ-EINT18 |
| PJ19 | I/O | PWM2-9 | | IR-RX3 | RGMII1-CLKIN/RMII1-RXER | | | PJ-EINT19 |
| PJ20 | I/O | TWI0-SCK | UART14-TX | | RGMII1-TXD1/RMII1-TXD1 | | | PJ-EINT20 |
| PJ21 | I/O | TWI0-SDA | UART14-RX | | RGMII1-TXD0/RMII1-TXD0 | | | PJ-EINT21 |
| PJ22 | I/O | | UART14-RTS | IR-TX | RGMII1-TXCK/RMII1-TXCK | | | PJ-EINT22 |
| PJ23 | I/O | | UART14-CTS | | RGMII1-TXCTL/RMII1-TXEN | | | PJ-EINT23 |
| PJ24 | I/O | PWM2-0 | TWI4-SCK | UART13-RTS | RGMII1-MDC | | | PJ-EINT24 |
| PJ25 | I/O | PWM2-1 | TWI4-SDA | UART13-CTS | RGMII1-MDIO | | | PJ-EINT25 |
| PJ26 | I/O | PWM2-2 | TWI5-SCK | UART13-TX | RGMII1-EPHY-25/50M | | | PJ-EINT26 |
| PJ27 | I/O | PWM2-3 | TWI5-SDA | UART13-RX | RGMII1-RXD3/RMII1-NULL | | | PJ-EINT27 |
| PJ28 | I/O | | TWI1-SCK | | RGMII1-RXD2/RMII1-NULL | | | PJ-EINT28 |
| PJ29 | I/O | | TWI1-SDA | | RGMII1-RXCK/RMII1-NULL | | | PJ-EINT29 |
| PJ30 | I/O | | | | RGMII1-TXD3/RMII1-NULL | | | PJ-EINT30 |
| PJ31 | I/O | | | | RGMII1-TXD2/RMII1-NULL | | | PJ-EINT31 |

### 4.2.11 Port K

Table 4-36 Port K Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PK0 | I/O | MCSIA-D0N | | UART11-TX | | SPI2-CS0 | LBUS-LD0 | PK-EINT0 |
| PK1 | I/O | MCSIA-D0P | | UART11-RX | | SPI2-CLK | LBUS-LD1 | PK-EINT1 |
| PK2 | I/O | MCSIA-D1N | TWI2-SCK | UART11-RTS | | SPI2-MOSI | LBUS-LD2 | PK-EINT2 |
| PK3 | I/O | MCSIA-D1P | TWI2-SDA | UART11-CTS | | SPI2-MISO | LBUS-LD3 | PK-EINT3 |
| PK4 | I/O | MCSIA-CKN | | UART12-TX | NCSI-D0 | SPI2-WP | LBUS-LD4 | PK-EINT4 |
| PK5 | I/O | MCSIA-CKP | | UART12-RX | NCSI-D1 | SPI2-HOLD | LBUS-LD5 | PK-EINT5 |
| PK6 | I/O | MCSIB-D0N | | UART12-RTS | NCSI-D2 | | LBUS-LD6 | PK-EINT6 |

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| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PK7 | I/O | MCSIB-D0P | | UART12-CTS | NCSI-D3 | | LBUS-LD7 | PK-EINT7 |
| PK8 | I/O | MCSIB-D1N | | CAN1-TX0 | NCSI-D4 | | LBUS-LD8 | PK-EINT8 |
| PK9 | I/O | MCSIB-D1P | MCSI3-MCLK | CAN1-RX0 | NCSI-D5 | | LBUS-LD9 | PK-EINT9 |
| PK10 | I/O | MCSIB-CKN | TWI1-SCK | CAN0-TX0 | NCSI-D6 | | LBUS-LD10 | PK-EINT10 |
| PK11 | I/O | MCSIB-CKP | TWI1-SDA | CAN0-RX0 | NCSI-D7 | | LBUS-LD11 | PK-EINT11 |
| PK12 | I/O | MCSIC-D0N | UART7-TX | TWI1-SCK | NCSI-D8 | | LBUS-LD12 | PK-EINT12 |
| PK13 | I/O | MCSIC-D0P | UART7-RX | TWI1-SDA | NCSI-D9 | | LBUS-LD13 | PK-EINT13 |
| PK14 | I/O | MCSIC-D1N | UART7-RTS | UART5-RTS | NCSI-D10 | SPI3-CS0 | LBUS-LD14 | PK-EINT14 |
| PK15 | I/O | MCSIC-D1P | UART7-CTS | UART5-CTS | NCSI-D11 | SPI3-CLK | LBUS-LD15 | PK-EINT15 |
| PK16 | I/O | MCSIC-CKN | TWI4-SCK | UART5-TX | NCSI-D12 | SPI3-MOSI | LBUS-LD16 | PK-EINT16 |
| PK17 | I/O | MCSIC-CKP | TWI4-SDA | UART5-RX | NCSI-D13 | SPI3-MISO | LBUS-LD17 | PK-EINT17 |
| PK18 | I/O | MCSID-D0N | MCSI0-MCLK | UART6-TX | NCSI-D14 | SPI3-WP | LBUS-LD18 | PK-EINT18 |
| PK19 | I/O | MCSID-D0P | TWI2-SCK | UART6-RX | NCSI-D15 | SPI3-HOLD | LBUS-LD19 | PK-EINT19 |
| PK20 | I/O | MCSID-D1N | TWI2-SDA | UART6-RTS | NCSI-PCLK | | LBUS-LD20 | PK-EINT20 |
| PK21 | I/O | MCSID-D1P | MCSI1-MCLK | UART6-CTS | NCSI-MCLK | | LBUS-LD21 | PK-EINT21 |
| PK22 | I/O | MCSID-CKN | TWI3-SCK | PWM0-6 | NCSI-HSYNC | CAN3-TX0 | LBUS-LD22 | PK-EINT22 |
| PK23 | I/O | MCSID-CKP | TWI3-SDA | PWM0-7 | NCSI-VSYNC | CAN3-RX0 | LBUS-LD23 | PK-EINT23 |

### 4.2.12 Port L

Table 4-37 Port L Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PL0 | I/O | S-TWI0-SCK | | | | | | PL-EINT0 |
| PL1 | I/O | S-TWI0-SDA | | | | | | PL-EINT1 |
| PL2 | I/O | S-UART0-TX | S-IR-RX | S-PWM0-2 | | | | PL-EINT2 |
| PL3 | I/O | S-UART0-RX | | S-PWM0-3 | | | | PL-EINT3 |
| PL4 | I/O | | S-UART1-TX | S-IR-RX | | | | PL-EINT4 |
| PL5 | I/O | | S-UART1-RX | | | | | PL-EINT5 |
| PL6 | I/O | | S-UART1-RTS | | | | | PL-EINT6 |
| PL7 | I/O | | S-UART1-CTS | | | | | PL-EINT7 |
| PL8 | I/O | S-TWI1-SCK | | S-RJTAG-MS | | | | PL-EINT8 |
| PL9 | I/O | S-TWI1-SDA | | S-RJTAG-CK | S-PWM0-1 | | | PL-EINT9 |
| PL10 | I/O | S-PWM0-0 | S-UART0-TX | | | S-SPI0-CS0 | | PL-EINT10 |
| PL11 | I/O | S-IR-RX | S-UART0-RX | | | S-SPI0-CLK | | PL-EINT11 |
| PL12 | I/O | S-PWM0-1 | S-UART0-RTS | | | S-SPI0-MOSI | | PL-EINT12 |
| PL13 | I/O | S-PWM0-3 | S-UART0-CTS | | | S-SPI0-MISO | | PL-EINT13 |

### 4.2.13 Port M

Table 4-38 Port M Multiplex Function

| Pin Name | IO Type | Function2 | Function3 | Function4 | Function5 | Function6 | Function7 | Function14 |
|---|---|---|---|---|---|---|---|---|
| PM0 | I/O | S-TWI0-SCK | S-UART1-TX | S-PWM0-0 | | | | PM-EINT0 |
| PM1 | I/O | S-TWI0-SDA | S-UART1-RX | S-PWM0-1 | | | | PM-EINT1 |
| PM2 | I/O | S-TWI1-SCK | S-UART1-RTS | S-UART0-TX | | | | PM-EINT2 |
| PM3 | I/O | S-TWI1-SDA | S-UART1-CTS | S-UART0-RX | | | | PM-EINT3 |
| PM4 | I/O | | S-PWM0-2 | S-UART0-RTS | | | | PM-EINT4 |
| PM5 | I/O | S-IR-RX | S-PWM0-3 | S-UART0-CTS | | | | PM-EINT5 |

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## 4.3 Detailed Signal Description

The following tables show the detailed function description of every signal based on the different interfaces.

[1] **Signal Name**: The name of every signal.

[2] **Description**: The detailed function description of every signal.

[3] **Type**: Denotes the signal direction:

I (Input),

O (Output),

I/O (Input/Output),

OD (Open-Drain),

A (Analog),

AI (Analog Input),

AO (Analog Output),

A I/O (Analog Input/Output),

P (Power),

G (Ground)

### 4.3.1 SDRAM

**Table 4-39 SDRAM Signal Description**

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SRST | DRAM Reset Signal | O |
| SZQ | DRAM ZQ Calibration(the signal connects to an external reference resistor which is used to calibrate DRAM input/output buffer) | AI |
| SA0 | DRAM Address Signal 0 | O |
| SA1 | DRAM Address Signal 1 | O |
| SA2 | DRAM Address Signal 2 | O |
| SA3 | DRAM Address Signal 3 | O |
| SA4 | DRAM Address Signal 4 | O |
| SA5 | DRAM Address Signal 5 | O |
| SA6 | DRAM Address Signal 6 | O |
| SA7 | DRAM Address Signal 7 | O |
| SA8 | DRAM Address Signal 8 | O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SA9 | DRAM Address Signal 9 | O |
| SA10 | DRAM Address Signal 10 | O |
| SA11 | DRAM Address Signal 11 | O |
| SA12 | DRAM Address Signal 12 | O |
| SA13 | DRAM Address Signal 13 | O |
| SA14 | DRAM Address Signal 14 | O |
| SA15 | DRAM Address Signal 15 | O |
| SA16 | DRAM Address Signal 16 | O |
| SA17 | DRAM Address Signal 17 | O |
| SACT | DRAM Activation Command Output | O |
| SBA0 | DRAM Bank Address Signal 0 | O |
| SBA1 | DRAM Bank Address Signal 1 | O |
| SBG0 | DRAM Bank Group Address Signal 0 | O |
| SBG1 | DRAM Bank Group Address Signal 1 | O |
| SCKE0 | DRAM Clock Enable Signal 0 | O |
| SCKN | DRAM Differential Clock (Negative) | O |
| SCKP | DRAM Differential Clock (Positive) | O |
| SCS0 | DRAM Chip Select Signal 0 | O |
| SCS1 | DRAM Chip Select Signal 1 | O |
| SDQM0 | DRAM Data Mask Signal | I/O |
| SDQM1 | DRAM Data Mask Signal | I/O |
| SDQM2 | DRAM Data Mask Signal | I/O |
| SDQM3 | DRAM Data Mask Signal | I/O |
| SDQ0 | DRAM Bidirectional Data Line 0 | I/O |
| SDQ1 | DRAM Bidirectional Data Line 1 | I/O |
| SDQ2 | DRAM Bidirectional Data Line 2 | I/O |
| SDQ3 | DRAM Bidirectional Data Line 3 | I/O |
| SDQ4 | DRAM Bidirectional Data Line 4 | I/O |
| SDQ5 | DRAM Bidirectional Data Line 5 | I/O |
| SDQ6 | DRAM Bidirectional Data Line 6 | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SDQ7 | DRAM Bidirectional Data Line 7 | I/O |
| SDQ8 | DRAM Bidirectional Data Line 8 | I/O |
| SDQ9 | DRAM Bidirectional Data Line 9 | I/O |
| SDQ10 | DRAM Bidirectional Data Line 10 | I/O |
| SDQ11 | DRAM Bidirectional Data Line 11 | I/O |
| SDQ12 | DRAM Bidirectional Data Line 12 | I/O |
| SDQ13 | DRAM Bidirectional Data Line 13 | I/O |
| SDQ14 | DRAM Bidirectional Data Line 14 | I/O |
| SDQ15 | DRAM Bidirectional Data Line 15 | I/O |
| SDQ16 | DRAM Bidirectional Data Line 16 | I/O |
| SDQ17 | DRAM Bidirectional Data Line 17 | I/O |
| SDQ18 | DRAM Bidirectional Data Line 18 | I/O |
| SDQ19 | DRAM Bidirectional Data Line 19 | I/O |
| SDQ20 | DRAM Bidirectional Data Line 20 | I/O |
| SDQ21 | DRAM Bidirectional Data Line 21 | I/O |
| SDQ22 | DRAM Bidirectional Data Line 22 | I/O |
| SDQ23 | DRAM Bidirectional Data Line 23 | I/O |
| SDQ24 | DRAM Bidirectional Data Line 24 | I/O |
| SDQ25 | DRAM Bidirectional Data Line 25 | I/O |
| SDQ26 | DRAM Bidirectional Data Line 26 | I/O |
| SDQ27 | DRAM Bidirectional Data Line 27 | I/O |
| SDQ28 | DRAM Bidirectional Data Line 28 | I/O |
| SDQ29 | DRAM Bidirectional Data Line 29 | I/O |
| SDQ30 | DRAM Bidirectional Data Line 30 | I/O |
| SDQ31 | DRAM Bidirectional Data Line 31 | I/O |
| SDQS0N | DRAM Bidirectional Data Strobe 0 (Active Low) | I/O |
| SDQS0P | DRAM Bidirectional Data Strobe 0 (Active-High) | I/O |
| SDQS1N | DRAM Bidirectional Data Strobe 1 (Active Low) | I/O |
| SDQS1P | DRAM Bidirectional Data Strobe 1 (Active-High) | I/O |
| SDQS2N | DRAM Bidirectional Data Strobe 2 (Active Low) | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SDQS2P | DRAM Bidirectional Data Strobe 2 (Active-High) | I/O |
| SDQS3N | DRAM Bidirectional Data Strobe 3 (Active Low) | I/O |
| SDQS3P | DRAM Bidirectional Data Strobe 3 (Active-High) | I/O |
| SODT0 | DRAM On-Die Termination Output Signal 0 | O |
| SODT1 | DRAM On-Die Termination Output Signal 1 | O |
| VCC-DRAM | DRAM IO Power Supply | P |
| VCC-DRAML | LPDDR4x Power Supply | P |
| VDD18-DRAM | SDRAM Controller Power Supply | P |

### 4.3.2 Raw NAND Flash

Table 4-40 Raw NAND Flash Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| NAND-WE | NAND Flash Write Enable | O |
| NAND-ALE | NAND Flash Address Latch Enable | O |
| NAND-CLE | NAND Flash Command latch Enable | O |
| NAND-CE0 | NAND Flash Chip Select 0 | O |
| NAND-CE1 | NAND Flash Chip Select 1 | O |
| NAND-RE | NAND Flash Read Enable | O |
| NAND-RB0 | NAND Flash Ready/Busy Status Indicator Signal 0 | I |
| NAND-RB1 | NAND Flash Ready/Busy Status Indicator Signal 1 | I |
| NAND-DQ0 | NAND Flash Data Bit 0 | I/O |
| NAND-DQ1 | NAND Flash Data Bit 1 | I/O |
| NAND-DQ2 | NAND Flash Data Bit 2 | I/O |
| NAND-DQ3 | NAND Flash Data Bit 3 | I/O |
| NAND-DQ4 | NAND Flash Data Bit 4 | I/O |
| NAND-DQ5 | NAND Flash Data Bit 5 | I/O |
| NAND-DQ6 | NAND Flash Data Bit 6 | I/O |
| NAND-DQ7 | NAND Flash Data Bit 7 | I/O |
| NAND-DQS | NAND Flash Data Strobe | I/O |

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### 4.3.3 System Control

Table 4-41 System Control Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| FEL | The signal is used to force this chip into download mode (Active Low). | I |
| JTAG-SEL | JTAG mode select<br>The signal is used to select the port from which JTAG function outputs. | I |
| NMI | Non-Maskable Interrupt | I/O, OD |
| RESET | SoC Reset Signal (Active Low) | I/O, OD |

### 4.3.4 RTC&PLL

Table 4-42 RTC&PLL Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| X32KFOUT | 32.768 kHz Clock Fanout<br>Provides low frequency clock for external devices | AO,OD |
| X32KIN | 32.768 kHz Crystal Clock Input | AI |
| X32KOUT | 32.768 kHz Crystal Clock Output | AO |
| PLLTEST | PLL Test Signal | AO, OD |
| CPU-PLLTEST | CPU PLL Test Signal | AO, OD |
| VCC-CPU-PLL | CPU PLL Power Supply | P |
| VCC-RTC | RTC Power Supply | P |
| VCC-PLL | System PLL Supply | P |

### 4.3.5 DCXO

Table 4-43 DCXO Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| DXIN | Digital Compensated Crystal Oscillator Input | AI |
| DXOUT | Digital Compensated Crystal Oscillator Output | AO |
| REFCLK-OUT | Digital Compensated Crystal Oscillator Clock Fanout | AO |

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### 4.3.6 Clock Fanout

Table 4-44 Clock Fanout Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| CLK-FANOUT0 | Internal Clock Fanout<br>Optional Frequency: 32 kHz, 12 MHz, 16 MHz, 24 MHz, 25 MHz, 27 MHz, 50 MHz | O |
| CLK-FANOUT1 | Internal Clock Fanout<br>Optional Frequency: 32 kHz, 12 MHz, 16 MHz, 24 MHz, 25 MHz, 27 MHz, 50 MHz | O |

### 4.3.7 SD Card/SDIO/eMMC

Table 4-45 SD Card/SDIO/eMMC Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SDC0-CMD | SD Card Command Output/Response Input | I/O, OD |
| SDC0-CLK | SD Card Clock Output | O |
| SDC0-D0 | SD Card Data Input/ Output 0 | I/O |
| SDC0-D1 | SD Card Data Input/ Output 1 | I/O |
| SDC0-D2 | SD Card Data Input/ Output 2 | I/O |
| SDC0-D3 | SD Card Data Input/ Output 3 | I/O |
| SDC1-CMD | SDIO WIFI Command Output/Response Input | I/O, OD |
| SDC1-CLK | SDIO WIFI Clock Output | O |
| SDC1-D0 | SDIO WIFI Data Input/ Output 0 | I/O |
| SDC1-D1 | SDIO WIFI Data Input/ Output 1 | I/O |
| SDC1-D2 | SDIO WIFI Data Input/ Output 2 | I/O |
| SDC1-D3 | SDIO WIFI Data Input/ Output 3 | I/O |
| SDC2-CMD | eMMC Command Output/Response Input | I/O, OD |
| SDC2-CLK | eMMC Clock Output | O |
| SDC2-D0 | eMMC Data Input/ Output 0 | I/O |
| SDC2-D1 | eMMC Data Input/ Output 1 | I/O |
| SDC2-D2 | eMMC Data Input/ Output 2 | I/O |
| SDC2-D3 | eMMC Data Input/ Output 3 | I/O |
| SDC2-D4 | eMMC Data Input/ Output 4 | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SDC2-D5 | eMMC Data Input/ Output 5 | I/O |
| SDC2-D6 | eMMC Data Input/ Output 6 | I/O |
| SDC2-D7 | eMMC Data Input/ Output 7 | I/O |
| SDC2-RST | eMMC Reset | O |
| SDC2-DS | eMMC Data Strobe | I |

### 4.3.8 SerDes

Table 4-46 SerDes Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| USB0-DM | USB0/USB2_U2 Differential Data (Negative) | A I/O |
| USB0-DP | USB0/USB2_U2 Differential Data (Positive) | A I/O |
| USB1-DM | USB1/USB2_U2 Differential Data (Negative) | A I/O |
| USB1-DP | USB1/USB2_U2 Differential Data (Positive) | A I/O |
| PCIE-REF-CLKN | SerDes Reference Clock Input/Output (Negative) | A I/O |
| PCIE-REF-CLKP | SerDes Reference Clock Input/Output (Positive) | A I/O |
| PCIE-RX0-DN/USB2-U3-RXN | SerDes Differential Receive Signal (Negative) | AI |
| PCIE-RX0-DP/USB2-U3-RXP | SerDes Differential Receive Signal (Positive) | AI |
| PCIE-TX0-DN/USB2-U3-TXN | SerDes Differential Transmit Signal (Negative) | AO |
| PCIE-TX0-DP/USB2-U3-TXP | SerDes Differential Transmit Signal (Positive) | AO |
| PCIE-REXT | Pin to Connect to External Calibration Resistor | A I/O |
| PCIE-PERSTN | PCIe2.1 Fundamental Reset | I/O |
| PCIE-WAKEN | PCIe2.1 Wake Up | I/O |
| PCIE-CLKREQN | PCIe2.1 Clock Request | I/O |
| VCC18-COMB | PCIe2.1 & USB3.1 Combo PHY 1.8V Power Supply | P |
| VCC33-USB0-USB1 | USB2.0 DRD and USB 2.0 Host 3.3V Power Supply | P |
| VDD09-COMB | PCIe2.1 & USB3.1 Combo PHY 0.9V Power Supply | P |
| VDD09-USB0-USB1 | USB2.0 DRD and USB 2.0 Host 0.9V Power Supply | P |

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### 4.3.9 I2S/PCM

Table 4-47 I2S/PCM Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| I2S0-MCLK | I2S0 Master Clock | O |
| I2S0-LRCK | I2S0/PCM0 Sample Rate Clock/Sync | I/O |
| I2S0-BCLK | I2S0/PCM0 Bit Rate Clock | I/O |
| I2S0-DIN0 | I2S0/PCM0 Serial Data Input 0 | I |
| I2S0-DIN1 | I2S0/PCM0 Serial Data Input 1 | I |
| I2S0-DIN2 | I2S0/PCM0 Serial Data Input 2 | I |
| I2S0-DIN3 | I2S0/PCM0 Serial Data Input 3 | I |
| I2S0-DOUT0 | I2S0/PCM0 Serial Data Output 0 | O |
| I2S0-DOUT1 | I2S0/PCM0 Serial Data Output 1 | O |
| I2S0-DOUT2 | I2S0/PCM0 Serial Data Output 2 | O |
| I2S0-DOUT3 | I2S0/PCM0 Serial Data Output 3 | O |
| I2S1-MCLK | I2S1 Master Clock | O |
| I2S1-LRCK | I2S1/PCM1 Sample Rate Clock/Sync | I/O |
| I2S1-BCLK | I2S1/PCM1 Bit Rate Clock | I/O |
| I2S1-DIN0 | I2S1/PCM1 Serial Data Input 0 | I |
| I2S1-DIN1 | I2S1/PCM1 Serial Data Input 1 | I |
| I2S1-DOUT0 | I2S1/PCM1 Serial Data Output 0 | O |
| I2S1-DOUT1 | I2S1/PCM1 Serial Data Output 1 | O |
| I2S2-MCLK | I2S2 Master Clock | O |
| I2S2-LRCK | I2S2/PCM2 Sample Rate Clock/Sync | I/O |
| I2S2-BCLK | I2S2/PCM2 Bit Rate Clock | I/O |
| I2S2-DIN0 | I2S2/PCM2 Serial Data Input 0 | I |
| I2S2-DIN1 | I2S2/PCM2 Serial Data Input 1 | I |
| I2S2-DIN2 | I2S2/PCM2 Serial Data Input 2 | I |
| I2S2-DIN3 | I2S2/PCM2 Serial Data Input 3 | I |
| I2S2-DOUT0 | I2S2/PCM2 Serial Data Output 0 | O |
| I2S2-DOUT1 | I2S2/PCM2 Serial Data Output 1 | O |
| I2S2-DOUT2 | I2S2/PCM2 Serial Data Output 2 | O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| I2S2-DOUT3 | I2S2/PCM2 Serial Data Output 3 | O |
| I2S3-MCLK | I2S3 Master Clock | O |
| I2S3-LRCK | I2S3/PCM3 Sample Rate Clock/Sync | I/O |
| I2S3-BCLK | I2S3/PCM3 Bit Rate Clock | I/O |
| I2S3-DIN0 | I2S3/PCM3 Serial Data Input 0 | I |
| I2S3-DIN1 | I2S3/PCM3 Serial Data Input 1 | I |
| I2S3-DIN2 | I2S3/PCM3 Serial Data Input 2 | I |
| I2S3-DIN3 | I2S3/PCM3 Serial Data Input 3 | I |
| I2S3-DOUT0 | I2S3/PCM3 Serial Data Output 0 | O |
| I2S3-DOUT1 | I2S3/PCM3 Serial Data Output 1 | O |
| I2S3-DOUT2 | I2S3/PCM3 Serial Data Output 2 | O |
| I2S3-DOUT3 | I2S3/PCM3 Serial Data Output 3 | O |

### 4.3.10 Audio Codec

Table 4-48 Audio Codec Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LINEOUTN | Lineout Differential Output (Negative) | AO |
| LINEOUTP | Lineout Differential Output (Positive) | AO |
| AVCC | Power Supply for Analog Part | P |
| VRA1 | Internal Reference Voltage | AO |
| AGND | Analog Ground | G |

### 4.3.11 DMIC

Table 4-49 DMIC Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| DMIC-CLK | Digital Microphone Clock Output | O |
| DMIC-DATA0 | Digital Microphone Data Input 0 | I |
| DMIC-DATA1 | Digital Microphone Data Input 1 | I |
| DMIC-DATA2 | Digital Microphone Data Input 2 | I |
| DMIC-DATA3 | Digital Microphone Data Input 3 | I |

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### 4.3.12 OWA

Table 4-50 OWA Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| OWA-IN | One Wire Audio Input | I |
| OWA-OUT | One Wire Audio Output | O |

### 4.3.13 GPADC

Table 4-51 GPADC Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| VCC18-VCM-ADC | Pin to Connect to External Capacitor | A I/O |
| GPADC0-0 | General Purpose ADC0 Input 0 | AI |
| GPADC0-1 | General Purpose ADC0 Input 1 | AI |
| GPADC0-2 | General Purpose ADC0 Input 2 | AI |
| GPADC0-3 | General Purpose ADC0 Input 3 | AI |
| GPADC0-4 | General Purpose ADC0 Input 4 | AI |
| GPADC0-5 | General Purpose ADC0 Input 5 | AI |
| GPADC0-6 | General Purpose ADC0 Input 6 | AI |
| GPADC0-7 | General Purpose ADC0 Input 7 | AI |
| GPADC0-8 | General Purpose ADC0 Input 8 | AI |
| GPADC0-9 | General Purpose ADC0 Input 9 | AI |
| GPADC1-0 | General Purpose ADC1 Input 0 | AI |
| GPADC1-1 | General Purpose ADC1 Input 1 | AI |
| GPADC1-2 | General Purpose ADC1 Input 2 | AI |
| GPADC1-3 | General Purpose ADC1 Input 3 | AI |
| BOOT-SEL-GPADC1-4 | General Purpose ADC1 Input 4/Boot Media Select | AI |
| DDR-PARA-SEL-GPADC1-5 | General Purpose ADC1 Input 5/DDR Parameter Select | AI |
| GPADC1-6 | General Purpose ADC1 Input 6 | AI |
| GPADC1-7 | General Purpose ADC1 Input 7 | AI |
| GPADC1-8 | General Purpose ADC1 Input 8 | AI |
| GPADC1-9 | General Purpose ADC1 Input 9 | AI |
| GPADC3-0 | General Purpose ADC3 Input 0 | AI |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| VCC18-ADC | GPADC and LRADC Power Supply | P |
| AGND-ADC | Analog Ground | G |

### 4.3.14 LRADC

Table 4-52 LRADC Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LRADC | Low Rate ADC Input | AI |

### 4.3.15 TPADC

Table 4-53 TPADC Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| TP-X1 | Touch Panel X1 Input | AI |
| TP-X2 | Touch Panel X2 Input | AI |
| TP-Y1 | Touch Panel Y1 Input | AI |
| TP-Y2 | Touch Panel Y2 Input | AI |

### 4.3.16 MIPI CSI

Table 4-54 MIPI CSI Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| MCSI0-MCLK | MIPI CSI A Master Clock | O |
| MCSIA-CKN | MIPI CSI A Differential Receive Clock Signal (Negative) | AI |
| MCSIA-CKP | MIPI CSI A Differential Receive Clock Signal (Positive) | AI |
| MCSIA-D0N | MIPI CSI A Data Line 0 Input (Negative) | AI |
| MCSIA-D0P | MIPI CSI A Data Line 0 Input (Positive) | AI |
| MCSIA-D1N | MIPI CSI A Data Line 1 Input (Negative) | AI |
| MCSIA-D1P | MIPI CSI A Data Line 1 Input (Positive) | AI |
| MCSI1-MCLK | MIPI CSI B Master Clock | O |
| MCSIB-CKN | MIPI CSI B Differential Receive Clock Signal (Negative) | AI |
| MCSIB-CKP | MIPI CSI B Differential Receive Clock Signal (Positive) | AI |
| MCSIB-D0N | MIPI CSI B Data Line 0 Input (Negative) | AI |
| MCSIB-D0P | MIPI CSI B Data Line 0 Input (Positive) | AI |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| MCSIB-D1N | MIPI CSI B Data Line 1 Input (Negative) | AI |
| MCSIB-D1P | MIPI CSI B Data Line 1 Input (Positive) | AI |
| MCSI2-MCLK | MIPI CSI C Master Clock | O |
| MCSIC-CKN | MIPI CSI C Differential Receive Clock Signal (Negative) | AI |
| MCSIC-CKP | MIPI CSI C Differential Receive Clock Signal (Positive) | AI |
| MCSIC-D0N | MIPI CSI C Data Line 0 Input (Negative) | AI |
| MCSIC-D0P | MIPI CSI C Data Line 0 Input (Positive) | AI |
| MCSIC-D1N | MIPI CSI C Data Line 1 Input (Negative) | AI |
| MCSIC-D1P | MIPI CSI C Data Line 1 Input (Positive) | AI |
| MCSI3-MCLK | MIPI CSI D Master Clock | O |
| MCSID-CKN | MIPI CSI D Differential Receive Clock Signal (Negative) | AI |
| MCSID-CKP | MIPI CSI D Differential Receive Clock Signal (Positive) | AI |
| MCSID-D0N | MIPI CSI D Data Line 0 Input (Negative) | AI |
| MCSID-D0P | MIPI CSI D Data Line 0 Input (Positive) | AI |
| MCSID-D1N | MIPI CSI D Data Line 1 Input (Negative) | AI |
| MCSID-D1P | MIPI CSI D Data Line 1 Input (Positive) | AI |
| CSI0-XHS | CSI Horizontal SYNC 0 | O |
| CSI1-XHS | CSI Horizontal SYNC 1 | O |
| CSI0-XVS-FSYNC | CSI Vertical SYNC/Frame SYNC 0 | I/O |
| CSI1-XVS-FSYNC | CSI Vertical SYNC/Frame SYNC 1 | I/O |
| VCC18-MCSI | MIPI CSI Power Supply | P |

### 4.3.17 Parallel CSI

Table 4-55 Parallel CSI Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| NCSI-PCLK | Parallel CSI Pixel Clock Input | I |
| NCSI-MCLK | Parallel CSI Master Clock Output | O |
| NCSI-HSYNC | Parallel CSI Horizontal Synchronous Input | I |
| NCSI-VSYNC | Parallel CSI Vertical Synchronous Input | I |
| NCSI-D0 | Parallel CSI Pixel Data 0 | I |
| NCSI-D1 | Parallel CSI Pixel Data 1 | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| NCSI-D2 | Parallel CSI Pixel Data 2 | I |
| NCSI-D3 | Parallel CSI Pixel Data 3 | I |
| NCSI-D4 | Parallel CSI Pixel Data 4 | I |
| NCSI-D5 | Parallel CSI Pixel Data 5 | I |
| NCSI-D6 | Parallel CSI Pixel Data 6 | I |
| NCSI-D7 | Parallel CSI Pixel Data 7 | I |
| NCSI-D8 | Parallel CSI Pixel Data 8 | I |
| NCSI-D9 | Parallel CSI Pixel Data 9 | I |
| NCSI-D10 | Parallel CSI Pixel Data 10 | I |
| NCSI-D11 | Parallel CSI Pixel Data 11 | I |
| NCSI-D12 | Parallel CSI Pixel Data 12 | I |
| NCSI-D13 | Parallel CSI Pixel Data 13 | I |
| NCSI-D14 | Parallel CSI Pixel Data 14 | I |
| NCSI-D15 | Parallel CSI Pixel Data 15 | I |

### 4.3.18 LCD

Table 4-56 LCD Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LCD-CLK | LCD Clock | O |
| LCD-VSYNC | LCD Vertical Synchronization | O |
| LCD-HSYNC | LCD Horizontal Synchronization | O |
| LCD-DE | LCD Data Enable | O |
| TCON-FSYNC0 | Frame Synchronization Signal for TCON and Sensor | O |
| LCD-D0 | LCD Data Input/Output 0 | I/O |
| LCD-D1 | LCD Data Input/Output 1 | I/O |
| LCD-D2 | LCD Data Input/Output 2 | I/O |
| LCD-D3 | LCD Data Input/Output 3 | I/O |
| LCD-D4 | LCD Data Input/Output 4 | I/O |
| LCD-D5 | LCD Data Input/Output 5 | I/O |
| LCD-D6 | LCD Data Input/Output 6 | I/O |
| LCD-D7 | LCD Data Input/Output 7 | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LCD-D8 | LCD Data Input/Output 8 | I/O |
| LCD-D9 | LCD Data Input/Output 9 | I/O |
| LCD-D10 | LCD Data Input/Output 10 | I/O |
| LCD-D11 | LCD Data Input/Output 11 | I/O |
| LCD-D12 | LCD Data Input/Output 12 | I/O |
| LCD-D13 | LCD Data Input/Output 13 | I/O |
| LCD-D14 | LCD Data Input/Output 14 | I/O |
| LCD-D15 | LCD Data Input/Output 15 | I/O |
| LCD-D16 | LCD Data Input/Output 16 | I/O |
| LCD-D17 | LCD Data Input/Output 17 | I/O |
| LCD-D18 | LCD Data Input/Output 18 | I/O |
| LCD-D19 | LCD Data Input/Output 19 | I/O |
| LCD-D20 | LCD Data Input/Output 20 | I/O |
| LCD-D21 | LCD Data Input/Output 21 | I/O |
| LCD-D22 | LCD Data Input/Output 22 | I/O |
| LCD-D23 | LCD Data Input/Output 23 | I/O |

### 4.3.19 LVDS

Table 4-57 LVDS Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LVDS0-CKP | LVDS0 Differential Clock (Positive) | AO |
| LVDS0-CKN | LVDS0 Differential Clock (Negative) | AO |
| LVDS0-D0P | LVDS0 Differential Data 0 (Positive) | AO |
| LVDS0-D0N | LVDS0 Differential Data 0 (Negative) | AO |
| LVDS0-D1P | LVDS0 Differential Data 1 (Positive) | AO |
| LVDS0-D1N | LVDS0 Differential Data 1 (Negative) | AO |
| LVDS0-D2P | LVDS0 Differential Data 2 (Positive) | AO |
| LVDS0-D2N | LVDS0 Differential Data 2 (Negative) | AO |
| LVDS0-D3P | LVDS0 Differential Data 3 (Positive) | AO |
| LVDS0-D3N | LVDS0 Differential Data 3 (Negative) | AO |
| LVDS1-CKP | LVDS1 Differential Clock (Positive) | AO |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LVDS1-CKN | LVDS1 Differential Clock (Negative) | AO |
| LVDS1-D0P | LVDS1 Differential Data 0 (Positive) | AO |
| LVDS1-D0N | LVDS1 Differential Data 0 (Negative) | AO |
| LVDS1-D1P | LVDS1 Differential Data 1 (Positive) | AO |
| LVDS1-D1N | LVDS1 Differential Data 1 (Negative) | AO |
| LVDS1-D2P | LVDS1 Differential Data 2 (Positive) | AO |
| LVDS1-D2N | LVDS1 Differential Data 2 (Negative) | AO |
| LVDS1-D3P | LVDS1 Differential Data 3 (Positive) | AO |
| LVDS1-D3N | LVDS1 Differential Data 3 (Negative) | AO |
| VCC18-LVDS | LVDS0/1 Power Supply | P |

### 4.3.20 MIPI DSI

Table 4-58 MIPI DSI Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| DSI-CKP | DSI Differential Clock (Positive) | AO |
| DSI-CKN | DSI Differential Clock (Negative) | AO |
| DSI-D0P | DSI Differential Data Line 0 (Positive) | A I/O |
| DSI-D0N | DSI Differential Data Line 0 (Negative) | A I/O |
| DSI-D1P | DSI Differential Data Line 1 (Positive) | AO |
| DSI-D1N | DSI Differential Data Line 1 (Negative) | AO |
| DSI-D2P | DSI Differential Data Line 2 (Positive) | AO |
| DSI-D2N | DSI Differential Data Line 2 (Negative) | AO |
| DSI-D3P | DSI Differential Data Line 3 (Positive) | AO |
| DSI-D3N | DSI Differential Data Line 3 (Negative) | AO |

### 4.3.21 CAN-FD

Table 4-59 CAN-FD Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| CAN0-RX0 | CAN0 Receive Data 0 | I |
| CAN0-TX0 | CAN0 Transmit Data 0 | O |
| CAN1-RX0 | CAN1 Receive Data 0 | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| CAN1-TX0 | CAN1 Transmit Data 0 | O |
| CAN2-RX0 | CAN2 Receive Data 0 | I |
| CAN2-TX0 | CAN2 Transmit Data 0 | O |
| CAN3-RX0 | CAN3 Receive Data 0 | I |
| CAN3-TX0 | CAN3 Transmit Data 0 | O |

### 4.3.22 GMAC

Table 4-60 GMAC Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| RGMII0-RXCK/ RMII0-NULL | RGMII0 Receive Clock | I |
| RGMII0-TXCK/RMII0-TXCK | RGMII0/RMII0 Transmit Clock<br>For RGMII, IO type is output;<br>For RMII, IO type is input. | I/O |
| RGMII0-CLKIN/RMII0-RXER | RGMII0 Transmit Clock from External/RMII0 Receive Error | I |
| RGMII0-RXCTL/RMII0-CRS-DV | RGMII0 Receive Control/RMII0 Carrier Sense Receive Data Valid | I |
| RGMII0-TXCTL/RMII0-TXEN | RGMII0 Transmit Control/RMII0 Transmit Enable | O |
| RGMII0-MDC | RGMII0 Management Data Clock | O |
| RGMII0-MDIO | RGMII0 Management Data Input/ Output | I/O |
| RGMII0-EPHY-25/50M | GMAC PHY 25 MHz or 50 MHz Clock Output | O |
| RGMII0-RXD0/RMII0-RXD0 | RGMII0/RMII0 Receive Data 0 | I |
| RGMII0-RXD1/RMII0-RXD1 | RGMII0/RMII0 Receive Data 1 | I |
| RGMII0-RXD2/RMII0-NULL | RGMII0 Receive Data 2 | I |
| RGMII0-RXD3/RMII0-NULL | RGMII0 Receive Data 3 | I |
| RGMII0-TXD0/RMII0-TXD0 | RGMII0/RMII0 Transmit Data 0 | O |
| RGMII0-TXD1/RMII0-TXD1 | RGMII0/RMII0 Transmit Data 1 | O |
| RGMII0-TXD2/RMII0-NULL | RGMII0 Transmit Data 2 | O |
| RGMII0-TXD3/RMII0-NULL | RGMII0 Transmit Data 3 | O |
| RGMII1-RXCK/ RMII1-NULL | RGMII1 Receive Clock | I |
| RGMII1-TXCK/RMII1-TXCK | RGMII1/RMII1 Transmit Clock<br>For RGMII, IO type is output; | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| | For RMII, IO type is input. | |
| RGMII1-CLKIN/RMII1-RXER | RGMII1 Transmit Clock from External/RMII1 Receive Error | I |
| RGMII1-RXCTL/RMII1-CRS-DV | RGMII1 Receive Control/RMII1 Carrier Sense Receive Data Valid | I |
| RGMII1-TXCTL/RMII1-TXEN | RGMII1 Transmit Control/RMII1 Transmit Enable | O |
| RGMII1-MDC | RGMII1 Management Data Clock | O |
| RGMII1-MDIO | RGMII1 Management Data Input/ Output | I/O |
| RGMII1-EPHY-25/50M | GMAC PHY 25 MHz or 50 MHz Clock Output | O |
| RGMII1-RXD0/RMII1-RXD0 | RGMII1/RMII1 Receive Data 0 | I |
| RGMII1-RXD1/RMII1-RXD1 | RGMII1/RMII1 Receive Data 1 | I |
| RGMII1-RXD2/RMII1-NULL | RGMII1 Receive Data 2 | I |
| RGMII1-RXD3/RMII1-NULL | RGMII1 Receive Data 3 | I |
| RGMII1-TXD0/RMII1-TXD0 | RGMII1/RMII1 Transmit Data 0 | O |
| RGMII1-TXD1/RMII1-TXD1 | RGMII1/RMII1 Transmit Data 1 | O |
| RGMII1-TXD2/RMII1-NULL | RGMII1 Transmit Data 2 | O |
| RGMII1-TXD3/RMII1-NULL | RGMII1 Transmit Data 3 | O |

### 4.3.23 Local Bus Controller

**Table 4-61 Local Bus Controller Signal Description**

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LBUS-CS0 | Local Bus Chip Select Signal 0 | O |
| LBUS-CS1 | Local Bus Chip Select Signal 1 | O |
| LBUS-CS2 | Local Bus Chip Select Signal 2 | O |
| LBUS-CS3 | Local Bus Chip Select Signal 3 | O |
| LBUS-DP0 | Local Bus Data Parity Signal 0 | I/O |
| LBUS-DP1 | Local Bus Data Parity Signal 1 | I/O |
| LBUS-DP2 | Local Bus Data Parity Signal 2 | I/O |
| LBUS-DP3 | Local Bus Data Parity Signal 3 | I/O |
| LBUS-WR | Local Bus Write Ready | O |
| LBUS-READY | Local Bus Input Ready | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LBUS-ALE | Local Bus Address Lock Enable | O |
| LBUS-BURST0 | Local Bus Burst Command 0 | O |
| LBUS-BURST1 | Local Bus Burst Command 1 | O |
| LBUS-BURST2 | Local Bus Burst Command 2 | O |
| LBUS-LCLK | Local Bus Clock Output | O |
| LBUS-INTR | Local Bus Interrupt | I |
| LBUS-DRQ | Local Bus Asynchronous Data Request | I |
| LBUS-LBE | Local Bus Byte Enable | O |
| LBUS-LD0 | Local Bus Data 0 | I/O |
| LBUS-LD1 | Local Bus Data 1 | I/O |
| LBUS-LD2 | Local Bus Data 2 | I/O |
| LBUS-LD3 | Local Bus Data 3 | I/O |
| LBUS-LD4 | Local Bus Data 4 | I/O |
| LBUS-LD5 | Local Bus Data 5 | I/O |
| LBUS-LD6 | Local Bus Data 6 | I/O |
| LBUS-LD7 | Local Bus Data 7 | I/O |
| LBUS-LD8 | Local Bus Data 8 | I/O |
| LBUS-LD9 | Local Bus Data 9 | I/O |
| LBUS-LD10 | Local Bus Data 10 | I/O |
| LBUS-LD11 | Local Bus Data 11 | I/O |
| LBUS-LD12 | Local Bus Data 12 | I/O |
| LBUS-LD13 | Local Bus Data 13 | I/O |
| LBUS-LD14 | Local Bus Data 14 | I/O |
| LBUS-LD15 | Local Bus Data 15 | I/O |
| LBUS-LD16 | Local Bus Data 16 | I/O |
| LBUS-LD17 | Local Bus Data 17 | I/O |
| LBUS-LD18 | Local Bus Data 18 | I/O |
| LBUS-LD19 | Local Bus Data 19 | I/O |
| LBUS-LD20 | Local Bus Data 20 | I/O |
| LBUS-LD21 | Local Bus Data 21 | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LBUS-LD22 | Local Bus Data 22 | I/O |
| LBUS-LD23 | Local Bus Data 23 | I/O |
| LBUS-LD24 | Local Bus Data 24 | I/O |
| LBUS-LD25 | Local Bus Data 25 | I/O |
| LBUS-LD26 | Local Bus Data 26 | I/O |
| LBUS-LD27 | Local Bus Data 27 | I/O |
| LBUS-LD28 | Local Bus Data 28 | I/O |
| LBUS-LD29 | Local Bus Data 29 | I/O |
| LBUS-LD30 | Local Bus Data 30 | I/O |
| LBUS-LD31 | Local Bus Data 31 | I/O |

### 4.3.24 SPI&SPI DBI

Table 4-62 SPI&SPI DBI Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SPI0-CS0 | SPI0 Chip Select 0 (Active Low) | I/O |
| SPI0-CS1 | SPI0 Chip Select 1 (Active Low) | I/O |
| SPI0-CLK | SPI0 Clock | I/O |
| SPI0-MOSI | SPI0 Master Data Out, Slave Data In | I/O |
| SPI0-MISO | SPI0 Master Data In, Slave Data Out | I/O |
| SPI0-WP | SPI0 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI0-HOLD | SPI0 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI1-CS0 | SPI1 Chip Select 0 (Active Low) | I/O |
| SPI1-CLK | SPI1 Clock | I/O |
| SPI1-MOSI | SPI1 Master Data Out, Slave Data In | I/O |
| SPI1-MISO | SPI1 Master Data In, Slave Data Out | I/O |
| SPI1-WP | SPI1 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI1-HOLD | SPI1 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI2-CS0 | SPI2 Chip Select 0 (Active Low) | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SPI2-CLK | SPI2 Clock | I/O |
| SPI2-MOSI | SPI2 Master Data Out, Slave Data In | I/O |
| SPI2-MISO | SPI2 Master Data In, Slave Data Out | I/O |
| SPI2-WP | SPI2 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI2-HOLD | SPI2 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI3-CS0 | SPI3 Chip Select 0 (Active Low) | I/O |
| SPI3-CLK | SPI3 Clock | I/O |
| SPI3-MOSI | SPI3 Master Data Out, Slave Data In | I/O |
| SPI3-MISO | SPI3 Master Data In, Slave Data Out | I/O |
| SPI3-WP | SPI3 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI3-HOLD | SPI3 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output | I/O |
| SPI4-CS0 | SPI4 Chip Select 0 (Active Low) | I/O |
| SPI4-CLK | SPI4 Clock | I/O |
| SPI4-MOSI | SPI4 Master Data Out, Slave Data In | I/O |
| SPI4-MISO | SPI4 Master Data In, Slave Data Out | I/O |
| S-SPI0-CS0 | S-SPI0 Chip Select 0 (Active Low) | I/O |
| S-SPI0-CLK | S-SPI0 Clock | I/O |
| S-SPI0-MOSI | S-SPI0 Master Data Out, Slave Data In | I/O |
| S-SPI0-MISO | S-SPI0 Master Data In, Slave Data Out | I/O |
| DBI-CSX | Chip Select Signal (Active Low) | I/O |
| DBI-SCLK | Serial Clock Signal | I/O |
| DBI-SDO | Data Output Signal | I/O |
| DBI-SDI | Data Input Signal<br>The data is sampled on the rising edge and the falling edge. | I/O |
| DBI-TE | Tearing Effect Input<br>It is used to capture the external TE signal edge. The rising and falling edge is configurable. | I/O |
| DBI-DCX | DCX pin is the select output signal of data and command. | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| | DCX = 0: register command;<br>DCX = 1: data or parameter. | |
| DBI-WRX | When DBI operates in dual data lane format, the RGB666 format 2 can use WRX to transfer data | I/O |

### 4.3.25 SPIF

Table 4-63 SPIF Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| SPIF-CS0 | SPI Peripheral Chip Select Signal, Low Active | O |
| SPIF-CLK | SPI Master Mode Clock Output | O |
| SPIF-MOSI | SPI Master Data Out, Slave Data In | I/O |
| SPIF-MISO | SPI Master Data In, Slave Data Out | I/O |
| SPIF-DQS | SPI Data Strobe Signal | I |
| SPIF-WP | SPI Write Protection (Active Low) | I/O |
| SPIF-HOLD | SPI Hold Signal | I/O |
| SPIF-D4 | SPI Master Mode Data 4 in Octal Mode. | I/O |
| SPIF-D5 | SPI Master Mode Data 5 in Octal Mode. | I/O |
| SPIF-D6 | SPI Master Mode Data 6 in Octal Mode. | I/O |
| SPIF-D7 | SPI Master Mode Data 7 in Octal Mode. | I/O |

### 4.3.26 UART

Table 4-64 UART Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| UART0-RX | UART0 Data Receiver | I |
| UART0-TX | UART0 Data Transmitter | O |
| UART1-CTS | UART1 Clear to Send | I |
| UART1-RTS | UART1 Request to Send | O |
| UART1-RX | UART1 Data Receiver | I |
| UART1-TX | UART1 Data Transmitter | O |
| UART2-CTS | UART2 Clear to Send | I |
| UART2-RTS | UART2 Request to Send | O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| UART2-RX | UART2 Data Receiver | I |
| UART2-TX | UART2 Data Transmitter | O |
| UART3-CTS | UART3 Clear to Send | I |
| UART3-RTS | UART3 Request to Send | O |
| UART3-RX | UART3 Data Receiver | I |
| UART3-TX | UART3 Data Transmitter | O |
| UART4-CTS | UART4 Clear to Send | I |
| UART4-RTS | UART4 Request to Send | O |
| UART4-RX | UART4 Data Receiver | I |
| UART4-TX | UART4 Data Transmitter | O |
| UART5-CTS | UART5 Clear to Send | I |
| UART5-RTS | UART5 Request to Send | O |
| UART5-RX | UART5 Data Receiver | I |
| UART5-TX | UART5 Data Transmitter | O |
| UART6-CTS | UART6 Clear to Send | I |
| UART6-RTS | UART6 Request to Send | O |
| UART6-RX | UART6 Data Receiver | I |
| UART6-TX | UART6 Data Transmitter | O |
| UART7-CTS | UART7 Clear to Send | I |
| UART7-RTS | UART7 Request to Send | O |
| UART7-RX | UART7 Data Receiver | I |
| UART7-TX | UART7 Data Transmitter | O |
| UART8-CTS | UART8 Clear to Send | I |
| UART8-RTS | UART8 Request to Send | O |
| UART8-DCD | UART8 Data Carrier Detect | I |
| UART8-DSR | UART8 Data Set Ready | I |
| UART8-DTR | UART8 Data Terminal Ready | O |
| UART8-RI | UART8 Ring Information | I |
| UART8-RX | UART8 Data Receiver | I |
| UART8-TX | UART8 Data Transmitter | O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| UART9-CTS | UART9 Clear to Send | I |
| UART9-RTS | UART9 Request to Send | O |
| UART9-RX | UART9 Data Receiver | I |
| UART9-TX | UART9 Data Transmitter | O |
| UART10-CTS | UART10 Clear to Send | I |
| UART10-RTS | UART10 Request to Send | O |
| UART10-RX | UART10 Data Receiver | I |
| UART10-TX | UART10 Data Transmitter | O |
| UART11-CTS | UART11 Clear to Send | I |
| UART11-RTS | UART11 Request to Send | O |
| UART11-RX | UART11 Data Receiver | I |
| UART11-TX | UART11 Data Transmitter | O |
| UART12-CTS | UART12 Clear to Send | I |
| UART12-RTS | UART12 Request to Send | O |
| UART12-RX | UART12 Data Receiver | I |
| UART12-TX | UART12 Data Transmitter | O |
| UART13-CTS | UART13 Clear to Send | I |
| UART13-RTS | UART13 Request to Send | O |
| UART13-RX | UART13 Data Receiver | I |
| UART13-TX | UART13 Data Transmitter | O |
| UART14-CTS | UART14 Clear to Send | I |
| UART14-RTS | UART14 Request to Send | O |
| UART14-RX | UART14 Data Receiver | I |
| UART14-TX | UART14 Data Transmitter | O |
| S-UART0-CTS | S_UART0 Clear to Send | I |
| S-UART0-RTS | S_UART0 Request to Send | O |
| S-UART0-RX | S_UART0 Data Receiver | I |
| S-UART0-TX | S_UART0 Data Transmitter | O |
| S-UART1-CTS | S_UART1 Clear to Send | I |
| S-UART1-RTS | S_UART1 Request to Send | O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| S-UART1-RX | S_UART1 Data Receiver | I |
| S-UART1-TX | S_UART1 Data Transmitter | O |

### 4.3.27 PWM

Table 4-65 PWM Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PWM0-0 | PWM0 Wave Output /Capture Wave Input 0 | I/O |
| PWM0-1 | PWM0 Wave Output /Capture Wave Input 1 | I/O |
| PWM0-2 | PWM0 Wave Output /Capture Wave Input 2 | I/O |
| PWM0-3 | PWM0 Wave Output /Capture Wave Input 3 | I/O |
| PWM0-4 | PWM0 Wave Output /Capture Wave Input 4 | I/O |
| PWM0-5 | PWM0 Wave Output /Capture Wave Input 5 | I/O |
| PWM0-6 | PWM0 Wave Output /Capture Wave Input 6 | I/O |
| PWM0-7 | PWM0 Wave Output /Capture Wave Input 7 | I/O |
| PWM0-8 | PWM0 Wave Output /Capture Wave Input 8 | I/O |
| PWM0-9 | PWM0 Wave Output /Capture Wave Input 9 | I/O |
| PWM1-0 | PWM1 Wave Output /Capture Wave Input 0 | I/O |
| PWM1-1 | PWM1 Wave Output /Capture Wave Input 1 | I/O |
| PWM1-2 | PWM1 Wave Output /Capture Wave Input 2 | I/O |
| PWM1-3 | PWM1 Wave Output /Capture Wave Input 3 | I/O |
| PWM1-4 | PWM1 Wave Output /Capture Wave Input 4 | I/O |
| PWM1-5 | PWM1 Wave Output /Capture Wave Input 5 | I/O |
| PWM1-6 | PWM1 Wave Output /Capture Wave Input 6 | I/O |
| PWM1-7 | PWM1 Wave Output /Capture Wave Input 7 | I/O |
| PWM1-8 | PWM1 Wave Output /Capture Wave Input 8 | I/O |
| PWM1-9 | PWM1 Wave Output /Capture Wave Input 9 | I/O |
| PWM2-0 | PWM2 Wave Output /Capture Wave Input 0 | I/O |
| PWM2-1 | PWM2 Wave Output /Capture Wave Input 1 | I/O |
| PWM2-2 | PWM2 Wave Output /Capture Wave Input 2 | I/O |
| PWM2-3 | PWM2 Wave Output /Capture Wave Input 3 | I/O |
| PWM2-4 | PWM2 Wave Output /Capture Wave Input 4 | I/O |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PWM2-5 | PWM2 Wave Output /Capture Wave Input 5 | I/O |
| PWM2-6 | PWM2 Wave Output /Capture Wave Input 6 | I/O |
| PWM2-7 | PWM2 Wave Output /Capture Wave Input 7 | I/O |
| PWM2-8 | PWM2 Wave Output /Capture Wave Input 8 | I/O |
| PWM2-9 | PWM2 Wave Output /Capture Wave Input 9 | I/O |
| S-PWM0-0 | S_PWM0 Wave Output /Capture Wave Input 0 | I/O |
| S-PWM0-1 | S_PWM0 Wave Output /Capture Wave Input 1 | I/O |
| S-PWM0-2 | S_PWM0 Wave Output /Capture Wave Input 2 | I/O |
| S-PWM0-3 | S_PWM0 Wave Output /Capture Wave Input 3 | I/O |

### 4.3.28 TWI

Table 4-66 TWI Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| TWI0-SCK | TWI0 Serial Clock Signal | I/O |
| TWI0-SDA | TWI0 Serial Data Signal | I/O |
| TWI1-SCK | TWI1 Serial Clock Signal | I/O |
| TWI1-SDA | TWI1 Serial Data Signal | I/O |
| TWI2-SCK | TWI2 Serial Clock Signal | I/O |
| TWI2-SDA | TWI2 Serial Data Signal | I/O |
| TWI3-SCK | TWI3 Serial Clock Signal | I/O |
| TWI3-SDA | TWI3 Serial Data Signal | I/O |
| TWI4-SCK | TWI4 Serial Clock Signal | I/O |
| TWI4-SDA | TWI4 Serial Data Signal | I/O |
| TWI5-SCK | TWI5 Serial Clock Signal | I/O |
| TWI5-SDA | TWI5 Serial Data Signal | I/O |
| TWI6-SCK | TWI6 Serial Clock Signal | I/O |
| TWI6-SDA | TWI6 Serial Data Signal | I/O |
| S-TWI0-SCK | S_TWI0 Serial Clock Signal | I/O |
| S-TWI0-SDA | S_TWI0 Serial Data Signal | I/O |
| S-TWI1-SCK | S_TWI1 Serial Clock Signal | I/O |
| S-TWI1-SDA | S_TWI1 Serial Data Signal | I/O |

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### 4.3.29 IR RX

Table 4-67 IR RX Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| IR-RX0 | Consumer Infrared Receiver | I |
| IR-RX1 | Consumer Infrared Receiver | I |
| IR-RX2 | Consumer Infrared Receiver | I |
| IR-RX3 | Consumer Infrared Receiver | I |
| S-IR-RX | Consumer infrared Receiver | I |

### 4.3.30 IR TX

Table 4-68 IR TX Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| IR-TX | Consumer Infrared Transmitter | O |

### 4.3.31 LEDC

Table 4-69 LEDC Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| LEDC | Intelligent Control LED Signal Output | O |

### 4.3.32 JTAG

Table 4-70 JTAG Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| JTAG-MS | A55 JTAG Mode Selection | I |
| JTAG-CK | A55JTAG Clock Signal | I |
| JTAG-DI | A55 JTAG Data Input | I |
| JTAG-DO | A55 JTAG Data Output | O |
| RJTAG-MS | E907 JTAG Mode Selection | I |
| RJTAG-CK | E907JTAG Clock Signal | I |
| RJTAG-DI | E907 JTAG Data Input | I |
| RJTAG-DO | E907 JTAG Data Output | O |
| S-RJTAG-MS | E902 JTAG Mode Select | I |
| S-RJTAG-CK | E902 JTAG Clock Signal | I |

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### 4.3.33 Interrupt

Table 4-71 Interrupt Signal Description

| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PA-EINT0 | Port A Interrupt | I |
| PA-EINT1 | Port A Interrupt | I |
| PA-EINT2 | Port A Interrupt | I |
| PA-EINT3 | Port A Interrupt | I |
| PA-EINT4 | Port A Interrupt | I |
| PA-EINT5 | Port A Interrupt | I |
| PA-EINT6 | Port A Interrupt | I |
| PA-EINT7 | Port A Interrupt | I |
| PA-EINT8 | Port A Interrupt | I |
| PA-EINT9 | Port A Interrupt | I |
| PB-EINT0 | Port B Interrupt | I |
| PB-EINT1 | Port B Interrupt | I |
| PB-EINT2 | Port B Interrupt | I |
| PB-EINT3 | Port B Interrupt | I |
| PB-EINT4 | Port B Interrupt | I |
| PB-EINT5 | Port B Interrupt | I |
| PB-EINT6 | Port B Interrupt | I |
| PB-EINT7 | Port B Interrupt | I |
| PB-EINT8 | Port B Interrupt | I |
| PB-EINT9 | Port B Interrupt | I |
| PB-EINT10 | Port B Interrupt | I |
| PB-EINT11 | Port B Interrupt | I |
| PB-EINT12 | Port B Interrupt | I |
| PB-EINT13 | Port B Interrupt | I |
| PB-EINT14 | Port B Interrupt | I |
| PC-EINT0 | Port C Interrupt | I |
| PC-EINT1 | Port C Interrupt | I |
| PC-EINT2 | Port C Interrupt | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PC-EINT3 | Port C Interrupt | I |
| PC-EINT4 | Port C Interrupt | I |
| PC-EINT5 | Port C Interrupt | I |
| PC-EINT6 | Port C Interrupt | I |
| PC-EINT7 | Port C Interrupt | I |
| PC-EINT8 | Port C Interrupt | I |
| PC-EINT9 | Port C Interrupt | I |
| PC-EINT10 | Port C Interrupt | I |
| PC-EINT11 | Port C Interrupt | I |
| PC-EINT12 | Port C Interrupt | I |
| PC-EINT13 | Port C Interrupt | I |
| PC-EINT14 | Port C Interrupt | I |
| PC-EINT15 | Port C Interrupt | I |
| PC-EINT16 | Port C Interrupt | I |
| PD-EINT0 | Port D Interrupt | I |
| PD-EINT1 | Port D Interrupt | I |
| PD-EINT2 | Port D Interrupt | I |
| PD-EINT3 | Port D Interrupt | I |
| PD-EINT4 | Port D Interrupt | I |
| PD-EINT5 | Port D Interrupt | I |
| PD-EINT6 | Port D Interrupt | I |
| PD-EINT7 | Port D Interrupt | I |
| PD-EINT8 | Port D Interrupt | I |
| PD-EINT9 | Port D Interrupt | I |
| PD-EINT10 | Port D Interrupt | I |
| PD-EINT11 | Port D Interrupt | I |
| PD-EINT12 | Port D Interrupt | I |
| PD-EINT13 | Port D Interrupt | I |
| PD-EINT14 | Port D Interrupt | I |
| PD-EINT15 | Port D Interrupt | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PD-EINT16 | Port D Interrupt | I |
| PD-EINT17 | Port D Interrupt | I |
| PD-EINT18 | Port D Interrupt | I |
| PD-EINT19 | Port D Interrupt | I |
| PD-EINT20 | Port D Interrupt | I |
| PD-EINT21 | Port D Interrupt | I |
| PD-EINT22 | Port D Interrupt | I |
| PD-EINT23 | Port D Interrupt | I |
| PE-EINT0 | Port E Interrupt | I |
| PE-EINT1 | Port E Interrupt | I |
| PE-EINT2 | Port E Interrupt | I |
| PE-EINT3 | Port E Interrupt | I |
| PE-EINT4 | Port E Interrupt | I |
| PE-EINT5 | Port E Interrupt | I |
| PE-EINT6 | Port E Interrupt | I |
| PE-EINT7 | Port E Interrupt | I |
| PE-EINT8 | Port E Interrupt | I |
| PE-EINT9 | Port E Interrupt | I |
| PE-EINT10 | Port E Interrupt | I |
| PE-EINT11 | Port E Interrupt | I |
| PE-EINT12 | Port E Interrupt | I |
| PE-EINT13 | Port E Interrupt | I |
| PE-EINT14 | Port E Interrupt | I |
| PE-EINT15 | Port E Interrupt | I |
| PE-EINT16 | Port E Interrupt | I |
| PE-EINT17 | Port E Interrupt | I |
| PF-EINT0 | Port F Interrupt | I |
| PF-EINT1 | Port F Interrupt | I |
| PF-EINT2 | Port F Interrupt | I |
| PF-EINT3 | Port F Interrupt | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PF-EINT4 | Port F Interrupt | I |
| PF-EINT5 | Port F Interrupt | I |
| PF-EINT6 | Port F Interrupt | I |
| PG-EINT0 | Port G Interrupt | I |
| PG-EINT1 | Port G Interrupt | I |
| PG-EINT2 | Port G Interrupt | I |
| PG-EINT3 | Port G Interrupt | I |
| PG-EINT4 | Port G Interrupt | I |
| PG-EINT5 | Port G Interrupt | I |
| PG-EINT6 | Port G Interrupt | I |
| PG-EINT7 | Port G Interrupt | I |
| PG-EINT8 | Port G Interrupt | I |
| PG-EINT9 | Port G Interrupt | I |
| PG-EINT10 | Port G Interrupt | I |
| PG-EINT11 | Port G Interrupt | I |
| PG-EINT12 | Port G Interrupt | I |
| PG-EINT13 | Port G Interrupt | I |
| PG-EINT14 | Port G Interrupt | I |
| PH-EINT0 | Port H Interrupt | I |
| PH-EINT1 | Port H Interrupt | I |
| PH-EINT2 | Port H Interrupt | I |
| PH-EINT3 | Port H Interrupt | I |
| PH-EINT4 | Port H Interrupt | I |
| PH-EINT5 | Port H Interrupt | I |
| PH-EINT6 | Port H Interrupt | I |
| PH-EINT7 | Port H Interrupt | I |
| PH-EINT8 | Port H Interrupt | I |
| PH-EINT9 | Port H Interrupt | I |
| PH-EINT10 | Port H Interrupt | I |
| PH-EINT11 | Port H Interrupt | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PH-EINT12 | Port H Interrupt | I |
| PH-EINT13 | Port H Interrupt | I |
| PH-EINT14 | Port H Interrupt | I |
| PH-EINT15 | Port H Interrupt | I |
| PI-EINT0 | Port I Interrupt | I |
| PI-EINT1 | Port I Interrupt | I |
| PI-EINT2 | Port I Interrupt | I |
| PI-EINT3 | Port I Interrupt | I |
| PI-EINT4 | Port I Interrupt | I |
| PI-EINT5 | Port I Interrupt | I |
| PI-EINT6 | Port I Interrupt | I |
| PI-EINT7 | Port I Interrupt | I |
| PI-EINT8 | Port I Interrupt | I |
| PI-EINT9 | Port I Interrupt | I |
| PI-EINT10 | Port I Interrupt | I |
| PI-EINT11 | Port I Interrupt | I |
| PI-EINT12 | Port I Interrupt | I |
| PI-EINT13 | Port I Interrupt | I |
| PI-EINT14 | Port I Interrupt | I |
| PI-EINT15 | Port I Interrupt | I |
| PI-EINT16 | Port I Interrupt | I |
| PJ-EINT0 | Port J Interrupt | I |
| PJ-EINT1 | Port J Interrupt | I |
| PJ-EINT2 | Port J Interrupt | I |
| PJ-EINT3 | Port J Interrupt | I |
| PJ-EINT4 | Port J Interrupt | I |
| PJ-EINT5 | Port J Interrupt | I |
| PJ-EINT6 | Port J Interrupt | I |
| PJ-EINT7 | Port J Interrupt | I |
| PJ-EINT8 | Port J Interrupt | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PJ-EINT9 | Port J Interrupt | I |
| PJ-EINT10 | Port J Interrupt | I |
| PJ-EINT11 | Port J Interrupt | I |
| PJ-EINT12 | Port J Interrupt | I |
| PJ-EINT13 | Port J Interrupt | I |
| PJ-EINT14 | Port J Interrupt | I |
| PJ-EINT15 | Port J Interrupt | I |
| PJ-EINT16 | Port J Interrupt | I |
| PJ-EINT17 | Port J Interrupt | I |
| PJ-EINT18 | Port J Interrupt | I |
| PJ-EINT19 | Port J Interrupt | I |
| PJ-EINT20 | Port J Interrupt | I |
| PJ-EINT21 | Port J Interrupt | I |
| PJ-EINT22 | Port J Interrupt | I |
| PJ-EINT23 | Port J Interrupt | I |
| PJ-EINT24 | Port J Interrupt | I |
| PJ-EINT25 | Port J Interrupt | I |
| PJ-EINT26 | Port J Interrupt | I |
| PJ-EINT27 | Port J Interrupt | I |
| PJ-EINT28 | Port J Interrupt | I |
| PJ-EINT29 | Port J Interrupt | I |
| PJ-EINT30 | Port J Interrupt | I |
| PJ-EINT31 | Port J Interrupt | I |
| PK-EINT0 | Port K Interrupt | I |
| PK-EINT1 | Port K Interrupt | I |
| PK-EINT2 | Port K Interrupt | I |
| PK-EINT3 | Port K Interrupt | I |
| PK-EINT4 | Port K Interrupt | I |
| PK-EINT5 | Port K Interrupt | I |
| PK-EINT6 | Port K Interrupt | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PK-EINT7 | Port K Interrupt | I |
| PK-EINT8 | Port K Interrupt | I |
| PK-EINT9 | Port K Interrupt | I |
| PK-EINT10 | Port K Interrupt | I |
| PK-EINT11 | Port K Interrupt | I |
| PK-EINT12 | Port K Interrupt | I |
| PK-EINT13 | Port K Interrupt | I |
| PK-EINT14 | Port K Interrupt | I |
| PK-EINT15 | Port K Interrupt | I |
| PK-EINT16 | Port K Interrupt | I |
| PK-EINT17 | Port K Interrupt | I |
| PK-EINT18 | Port K Interrupt | I |
| PK-EINT19 | Port K Interrupt | I |
| PK-EINT20 | Port K Interrupt | I |
| PK-EINT21 | Port K Interrupt | I |
| PK-EINT22 | Port K Interrupt | I |
| PK-EINT23 | Port K Interrupt | I |
| PL-EINT0 | Port L Interrupt | I |
| PL-EINT1 | Port L Interrupt | I |
| PL-EINT2 | Port L Interrupt | I |
| PL-EINT3 | Port L Interrupt | I |
| PL-EINT4 | Port L Interrupt | I |
| PL-EINT5 | Port L Interrupt | I |
| PL-EINT6 | Port L Interrupt | I |
| PL-EINT7 | Port L Interrupt | I |
| PL-EINT8 | Port L Interrupt | I |
| PL-EINT9 | Port L Interrupt | I |
| PL-EINT10 | Port L Interrupt | I |
| PL-EINT11 | Port L Interrupt | I |
| PL-EINT12 | Port L Interrupt | I |

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| Signal Name[1] | Description[2] | Type[3] |
|---|---|---|
| PL-EINT13 | Port L Interrupt | I |
| PM-EINT0 | Port M Interrupt | I |
| PM-EINT1 | Port M Interrupt | I |
| PM-EINT2 | Port M Interrupt | I |
| PM-EINT3 | Port M Interrupt | I |
| PM-EINT4 | Port M Interrupt | I |
| PM-EINT5 | Port M Interrupt | I |

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# 5 Electrical characteristics

## 5.1 Parameter Conditions

### 5.1.1 Minimum and Maximum Values

Unless otherwise specified the minimum and maximum values are guaranteed in the worst conditions of ambient temperature, supply voltage, and frequencies by tests in production on 100% of the devices with ambient temperature at Ta = 25 °C and Ta = Ta max.

Data based on characterization results, design simulation, and/or technology characteristics are indicated in the table footnotes and are not tested in production.

### 5.1.2 Typical Values

Unless otherwise specified, the typical data are based on Ta = 25 °C. They are given only as design guidelines.

### 5.1.3 Temperature Definitions

- Ambient Temperature— the temperature of the surrounding environment.
- Junction Temperature— the hottest temperature of the silicon chip inside the package.
- Absolute Maximum Junction Temperature— the temperature beyond which damage occurs to the device. The device may not function or meet expected performance at this temperature.
- Recommended Operating Temperature— the junction temperature at which the device operates continuously at the designated performance over the designed lifetime. The reliability of the device may be degraded if the device operates above this temperature. Some devices will not function electrically above this temperature.

## 5.2 Absolute Maximum Ratings

Absolute Maximum Ratings are those values beyond which damage to the device may occur. The following table specifies the absolute maximum ratings.



CAUTION

Stresses beyond those listed under Table 5-1 may affect reliability or cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated under section 5.4 Recommended Operating Conditions is not implied. Exposure to absolute maximum rated conditions for extended periods may affect device reliability.

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Table 5-1 Absolute Maximum Ratings

| Supply Name | Description | Min(1) | Max(1) | Unit |
|---|---|---|---|---|
| AVCC | Power Supply for Analog Part | -0.3 | 2.16 | V |
| VCC-CPU-PLL | CPU PLL Power Supply | -0.3 | 2.16 | V |
| VCC-PLL | System PLL Supply | -0.3 | 2.16 | V |
| VCC-RTC | RTC Power Supply | -0.3 | 2.16 | V |
| VCC-DRAM | DRAM IO Power Supply | -0.3 | VDDQ+0.3 | V |
| VCC-DRAML | LPDDR4x Power Supply | -0.3 | VDDQ+0.3 | V |
| VCC18-ADC | GPADC and LRADC Power Supply | -0.3 | 2.16 | V |
| VCC18-EFUSE | eFuse Power Supply | -0.3 | 2.16 | V |
| VCC18-LVDS | LVDS and MIPI DSI Power Supply | -0.3 | 2.16 | V |
| VCC18-MCSI | MIPI CSI Power Supply | -0.3 | 2.16 | V |
| VCC-IO | GPIO/FEL/BOOT_SEL/JTAG_SEL Power Supply | -0.3 | 3.96 | V |
| VCC-PA | Digital Port A Power | -0.3 | 3.96 | V |
| VCC-PC | Digital Port C Power | -0.3 | 3.96 | V |
| VCC-PD | Digital Port D Power | -0.3 | 3.96 | V |
| VCC-PE | Digital Port E Power | -0.3 | 3.96 | V |
| VCC18-PF | Digital Port F Power | -0.3 | 2.16 | V |
| VCC-PG | Digital Port G Power | -0.3 | 3.96 | V |
| VCC-PH | Digital Port H Power | -0.3 | 3.96 | V |
| VCC-PI | Digital Port I Power | -0.3 | 3.96 | V |
| VCC-PJ | Digital Port J Power | -0.3 | 3.96 | V |
| VCC-PK | Digital Port K Power | -0.3 | 3.96 | V |
| VCC-PL | Digital Port L Power | -0.3 | 3.96 | V |
| VCC-PM | Digital Port M Power | -0.3 | 3.96 | V |
| VCC18-COMB | PCIe2.1 & USB3.1 Combo PHY 1.8V Power Supply | -0.3 | 2.16 | V |
| VCC33-USB0-USB1 | USB2.0 DRD and USB 2.0 Host 3.3V Power Supply | -0.3 | 3.63 | V |
| VDD18-DRAM | SDRAM Controller Power Supply | -0.3 | 2.16 | V |

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| Supply Name | Description | Min(1) | Max(1) | Unit |
|---|---|---|---|---|
| VDD09-USB0-USB1 | USB2.0 DRD and USB 2.0 Host 0.9V Power Supply | -0.3 | 1.08 | V |
| VDD09-COMB | PCIe2.1 & USB3.1 Combo PHY 0.9V Power Supply | -0.3 | 1.05 | V |
| VDD-VE | VE Power Supply | -0.3 | 1.05 | V |
| VDD-CPUS | CPUS Power Supply | -0.3 | 1.05 | V |
| VDD-CPU | ARM Cortex™-A55 Power Supply | -0.3 | 1.25 | V |
| VDD-NPU | NPU Power Supply | -0.3 | 1.25 | V |
| VDD-SYS | System Power Supply | -0.3 | 1.05 | V |
| $V_{ESD}$ | Human Body Model (HBM)(3) | ±2000 | | V |
| | Charged Device Model (CDM)(4) | ±500 | | V |
| $I_{Latch-up}$ | Latch-up I-test performance current-pulse injection on each IO pin(5) | PASS | | |
| | Latch-up over-voltage performance voltage injection on each IO pin(6) | PASS | | |

(1) The min/max voltages of power rails are guaranteed by design, not tested in production.

(2) Electrostatic discharge (ESD) to measure device sensitivity/immunity to damage caused by electrostatic discharges into the devices.

(3) Level listed above is the passing level per ESDA/JEDEC JS-001-2023.

(4) Level listed above is the passing level per ESDA/JEDEC JS-002-2022.

(5) Based on JESD78F; each device is tested with IO pin injection of ±200 mA at room temperature.

(6) Based on JESD78F; each device is tested with a stress voltage of 1.5 x Vddmax at room temperature.

## 5.3 Power-On Hours (POH)

**Table 5-2 Reliability Data**

| Junction Temperature ($T_j$) | 1.608GHz/1.15 V | 1.560 GHz/1.1V | 1.512 GHz/1.06V | 1.416 GHz/1.0V |
|---|---|---|---|---|
| 85°C | 20.8K | 83.2K | 1664.K | 1408K |
| 95°C | 10.4K | 41.6K | 83.2K | 704K |
| 105°C | 5.2K | 20.8K | 41.6K | 352K |
| 115°C | 2.6K | 10.4K | 20.8K | 176K |

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| Junction Temperature ($T_j$) | 1.608GHz/1.15 V | 1.560 GHz/1.1V | 1.512 GHz/1.06V | 1.416 GHz/1.0V |
|---|---|---|---|---|
| 125°C | 1.3K | 5.2K | 10.4K | 88K |

(1) The power-on hours (POH) information in this table is provided solely for your convenience and does not extend or modify the warranty provided under Allwinner's standard terms and conditions for Allwinner products.

(2) To avoid significant degradation, the device power-on hours (POH) must be limited as described in this table.

(3) Logic functions and parameter values are not assured out of the range specified in the recommended operating conditions.

(4) The previous notations cannot be deemed a warranty or deemed to extend or modify the warranty under Allwinner's standard terms and conditions for Allwinner products.

(5) For more detailed description about chip lifetime, see *T536_Chip_Lifetime_Evaluation_Guide.*

## 5.4 Recommended Operating Conditions

The following table describes operating conditions of the device.

NOTE

Logic functions and parameter values are not assured out of the range specified in the recommended operating conditions.

**Table 5-3 Recommended Operating Conditions**

| Supply Name | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| AVCC | Power Supply for Analog Part | 1.782 | 1.8 | 1.818 | V |
| VCC-CPU-PLL | CPU PLL Power Supply | 1.71 | 1.8 | 1.89 | V |
| VCC-PLL | System PLL Supply | 1.71 | 1.8 | 1.89 | V |
| VCC-RTC | RTC Power Supply | 1.62 | 1.8 | 1.98 | V |
| VCC-DRAM | DDR3 Power Supply | 1.425 | 1.5 | 1.575 | V |
| | DDR3L Power Supply | 1.283 | 1.35 | 1.45 | |
| | LPDDR3 Power Supply | 1.14 | 1.2 | 1.30 | |
| | DDR4 Power Supply | 1.14 | 1.2 | 1.26 | |
| | LPDDR4 and LPDDR4x Power Supply | 1.06 | 1.1 | 1.17 | |
| VCC-DRAML | LPDDR4x Power Supply | 0.57 | 0.6(1) | 0.65 | V |
| VCC18-ADC | GPADC and LRADC Power Supply | 1.782 | 1.8 | 1.818 | V |
| VCC18-EFUSE | eFuse Power Supply | 1.71 | 1.8 | 1.98 | V |

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| Supply Name | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VCC18-LVDS | LVDS and MIPI DSI Power Supply | 1.71 | 1.8 | 1.89 | V |
| VCC18-MCSI | MIPI CSI Power Supply | 1.71 | 1.8 | 1.89 | V |
| VCC-IO | GPIO/FEL/BOOT_SEL/JTAG_SEL Power Supply | 2.97 | 3.3 | 3.63 | V |
| VCC-PA | Digital Port A Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PC | Digital Port C Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PD | Digital Port D Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PE | Digital Port E Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC18-PF | Digital Port F Power<br>1.8 V Voltage | 1.62 | 1.8 | 1.98 | V |
| VCC-PG | Digital Port G Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PH | Digital Port H Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PI | Digital Port I Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PJ | Digital Port J Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PK | Digital Port K Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |

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| Supply Name | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VCC-PL | Digital Port L Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC-PM | Digital Port M Power<br>1.8 V Voltage<br>3.3 V Voltage | 1.62<br>2.97 | 1.8<br>3.3 | 1.98<br>3.63 | V |
| VCC18-COMB | PCIe & USB3.1 Combo PHY 1.8V Power Supply | 1.71 | 1.8 | 1.89 | V |
| VCC33-USB0-USB1 | USB2.0 DRD and USB 2.0 Host 3.3V Power Supply | 2.97 | 3.3 | 3.63 | V |
| VDD18-DRAM | SDRAM Controller Power Supply | 1.71 | 1.8 | 1.89 | V |
| VDD09-USB0-USB1 | USB2.0 DRD and USB 2.0 Host 0.9V Power Supply | 0.837 | 0.9 | 1.00 | V |
| VDD09-COMB | PCIe & USB3.1 Combo PHY 0.9V Power Supply | 0.9 | - | 1.00 | V |
| VDD-VE | VE Power Supply | 0.9 | - | 1.00 | V |
| VDD-CPUS | CPUS Power Supply | 0.88 | - | 0.99 | V |
| VDD-CPU | ARM Cortex™-A55 Power Supply | 0.9 | - | 1.15 | V |
| VDD-NPU | NPU Power Supply | 0.9 | - | 1.15 | V |
| VDD-SYS | System Power Supply | 0.9 | - | 1.00 | V |

(1) When only LPDDR4x is selected, VCC-DRAML is 0.6 V. Otherwise, VCC-DRAM and VCC-DRAML are the same.

## 5.5 GPIO DC Electrical Characteristics

Table 5-4 summarizes the GPIO DC electrical characteristics of the device.

**Table 5-4 GPIO DC Electrical Characteristics(1)**

**(VCC: VCC-IO/VCC-PA/VCC-PC/ VCC-PD/VCC18-PF/VCC-PH/VCC-PG/ VCC-PI/VCC-PJ/VCC-PK/ VCC-PL)**

| Symbol | Parameter | | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| $V_{IH}$ | High-Level Input Voltage | | 0.7 * VCC | - | 1.1*VCC | V |
| $V_{IL}$ | Low-Level Input Voltage | | -0.3 | - | 0.3 * VCC | V |
| $R_{PU}$ | | PI4, PI5, PL0, PL1 | 3.29 | 4.7 | 6.11 | kΩ |

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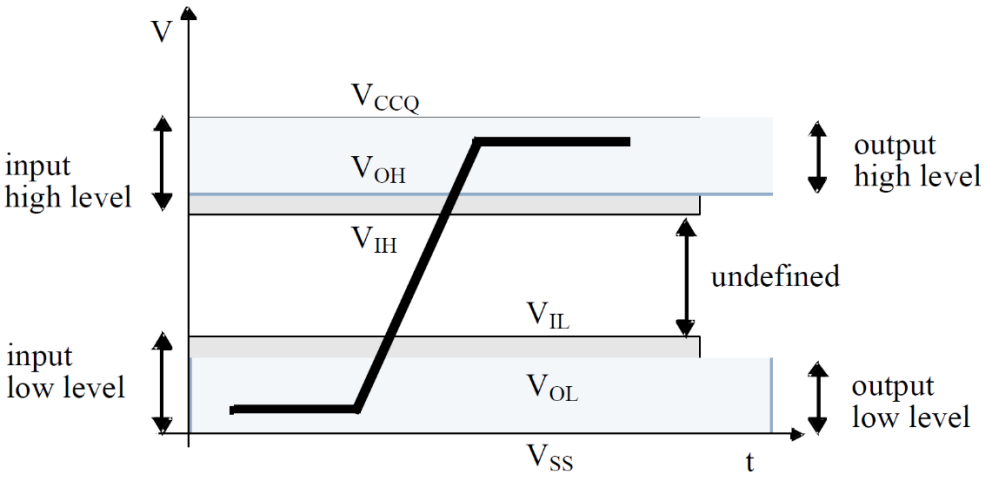
| Symbol | Parameter | | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| | Output Pull-up Resistance | PC0, PC1, PC3, PC6, PC7, PF3, PF6 | 10.5 | 15 | 19.5 | kΩ |
| | | PG1, PG2, PG3, PG4, PG5 | 23.1 | 33 | 42.9 | kΩ |
| | | Other GPIOs | 70 | 100 | 130 | kΩ |
| $R_{PD}$ | Output Pull-down Resistance | PI4, PI5, PL0, PL1 | 3.29 | 4.7 | 6.11 | kΩ |
| | | PC0, PC1, PC3, PC6, PC7, PF3, PF6 | 10.5 | 15 | 19.5 | kΩ |
| | | PG1, PG2, PG3, PG4, PG5 | 23.1 | 33 | 42.9 | kΩ |
| | | Other GPIOs | 70 | 100 | 130 | kΩ |
| $I_{IH}$ | High-Level Input Current | | - | - | 10 | uA |
| $I_{IL}$ | Low-Level Input Current | | - | - | 10 | uA |
| $V_{OH}$ | High-Level Output Voltage | | VCC – 0.3 | - | VCC | V |
| $V_{OL}$ | Low-Level Output Voltage | | 0 | - | 0.2 | V |
| $I_{OZ}$ | Tri-State Output Leakage Current | | -10 | - | 10 | uA |
| $C_{IN}$ | Input Capacitance | | - | - | 5 | pF |
| $C_{OUT}$ | Output Capacitance | | - | - | 5 | pF |

(1) Guaranteed by design.

## 5.6 SMHC Electrical Characteristics

The SMHC electrical parameters are related to different supply voltage.

**Figure 5-1 SMHC Voltage Waveform**




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Table 5-5 shows 3.3 V SMHC electrical parameters.

**Table 5-5 3.3 V SMHC Electrical Parameters**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VDD | Power voltage | 2.7 | - | 3.6 | V |
| $V_{CCQ}$ | I/O voltage | 2.7 | | 3.6 | V |
| $V_{OH}$ | Output high-level voltage | 0.75 * $V_{CCQ}$ | - | - | V |
| $V_{OL}$ | Output low-level voltage | - | - | 0.125 * $V_{CCQ}$ | V |
| $V_{IH}$ | Input high-level voltage | 0.625 * $V_{CCQ}$ | - | $V_{CCQ}$ + 0.3 | V |
| $V_{IL}$ | Input low-level voltage | $V_{SS}$ – 0.3 | - | 0.25 * $V_{CCQ}$ | V |

Table 5-6 shows 1.8 V SMHC electrical parameters.

**Table 5-6 1.8 V SMHC Electrical Parameters**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| VDD | Power voltage | 2.7 | - | 3.6 | V |
| $V_{CCQ}$ | I/O voltage | 1.7 | | 1.95 | V |
| $V_{OH}$ | Output high-level voltage | $V_{CCQ}$ – 0.45 | - | - | V |
| $V_{OL}$ | Output low-level voltage | - | - | 0.45 | V |
| $V_{IH}$ | Input high-level voltage | 0.65 * $V_{CCQ}$ (1) | - | $V_{CCQ}$ + 0.3 | V |
| $V_{IL}$ | Input low-level voltage | $V_{SS}$ – 0.3 | - | 0.35 * $V_{CCQ}$ (2) | V |
| (1).0.7 * $V_{CCQ}$ for MMC4.3 or lower.<br>(2).0.3 * $V_{CCQ}$ for MMC4.3 or lower. | | | | | |

## 5.7 GPADC Electrical Characteristics

Table 5-7 lists the GPADC electrical characteristics.

**Table 5-7 GPADC Electrical Characteristics**

| Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|
| ADC Resolution | - | 12 | - | bits |
| Full-scale Input Range | 0 | - | VCC18-ADC | V |
| Sampling Rate | - | - | 1 | MHz |
| Conversion Time | 0.5 | - | - | us |

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## 5.8 LRADC Electrical Characteristics

The following table lists the LRADC electrical characteristics.

**Table 5-8 LRADC Electrical Characteristics**

| Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|
| ADC Resolution | - | 6 | - | bits |
| Full-scale Input Range | | | LEVELB(1) | V |
| Effective Precision | - | 5 | - | bits |
| Sampling Rate | - | - | 2 | kHz |
| Conversion Time | - | 6 | - | ADC Clock Cycles |

(1) The maximum value of LEVELB is 1.286 V. For details, see the register description of LRADC in *T536_User_Manual*.

## 5.9 Audio Codec Electrical Characteristics

**Test Conditions**

VDD-SYS = 0.9 V, AVCC = 1.8 V, Ta = 25 °C, 1 kHz sinusoid signal, DAC fs = 48 kHz.

**Table 5-9 Audio Codec Typical Performance Parameters**

| Symbol | Parameter | Test Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| DAC Path | DAC to LINEOUTP/N(R=100K) | | | | | |
| | Full-scale | 0dBFS 1kHz | - | 1.08 | - | Vrms |
| | SNR(A-weighted) | 0data | - | 100 | - | dB |
| | THD+N | 0dBFS 1kHz | - | -89 | - | dB |

## 5.10 MIPI DPHY Electrical Characteristics

This section describes some test parameters for MIPI DPHY based on D-PHY Physical Layer Conformance Test Suite Version 1.00.

**Table 5-10 MIPI DPHY CTS Reference**

| Symbol | Parameters | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Data Lane HS-TX Parameters | | | | | |
| $V_{OD(0)}$ | Data Lane HS-TX Differential Voltages | - | 247 | - | mV |

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| Symbol | Parameters | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| $V_{OD(1)}$ | Data Lane HS-TX Differential Voltages | - | 249 | - | mV |
| $t_R$ | Data Lane HS-TX 20%-80% Rise Time | - | 0.36 | - | UI |
| $t_F$ | Data Lane HS-TX 80%-20% Fall Time | - | 0.36 | - | UI |
| Clock Lane HS-TX Parameters | | | | | |
| $V_{OD(0)}$ | Clock Lane HS-TX Differential Voltages | - | 243 | - | mV |
| $V_{OD(1)}$ | Clock Lane HS-TX Differential Voltages | - | 216 | - | mV |
| $t_R$ | Clock Lane HS-TX 20%-80% Rise Time | - | 0.36 | - | UI |
| $t_F$ | Clock Lane HS-TX 80%-20% Fall Time | - | 0.36 | - | UI |

# 5.11 External Clock Source Electrical Characteristics

## 5.11.1 High-speed Crystal/Ceramic Resonator Characteristics

The high-speed external clock can be supplied with a 24 MHz crystal resonator (oscillation mode). The 24 MHz crystal resonator provides 24 MHz reference clock which is connected to the DXIN and DXOUT terminals.

**Table 5-11 High-speed 24 MHz Crystal Requirements**

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| $f_{X24M_IN}$ | Crystal parallel resonance frequency | - | 24 | - | MHz |
| | Crystal frequency stability and tolerance at 25 °C (1) | -50 | - | +50 | ppm |
| | Oscillation mode | Fundamental | | | - |
| $C_0$ | Shunt capacitance (2) | - | 6.5 | - | pF |

1. The 50 ppm frequency stability and tolerance can meet the requirement of the device. We recommend selecting 20 ppm crystal devices. If the REFCLK-OUT (24 MHz fanout) is used for Wi-Fi chip, the crystal uses the recommended specification or the specified model for Wi-Fi chip.
2. The 6.5 pF is only a simulation value. The crystal shunt capacitance ($C_0$) is given by the crystal manufacturer.

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Table 5-12 Crystal Circuit Parameters

| Symbol | Parameter |
|---|---|
| $C_1$ | $C_1$ capacitance |
| $C_2$ | $C_2$ capacitance |
| $C_L$ | Equivalent load capacitance, specified by the crystal manufacturer |
| $C_0$ | Crystal shunt capacitance, specified by the crystal manufacturer |
| $C_{shunt}$ | Total shunt capacitance |

Frequency stability mainly requires that the total load capacitance ($C_L$) be constant. The crystal manufacturer typically specifies a total load capacitance which is the series combination of $C_1$, $C_2$, and $C_{shunt}$.

The total load capacitance is $C_L = [(C_1 * C_2)/ (C_1 + C_2)] + C_{shunt}$.

- $C_1$ and $C_2$ represent the total capacitance of the respective PCB trace, load capacitor, and other components (excluding the crystal) connected to each crystal terminal. $C_1$ and $C_2$ are usually the same size.
- $C_{shunt}$ is the crystal shunt capacitance ($C_0$) plus any mutual capacitance ($C_{pkg} + C_{PCB}$) seen across the DXIN and DXOUT signals.

In the application, the crystal resonator and the load capacitors must be placed close to the oscillator pins in order to minimize output distortion and the startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics.

### 5.11.2 Low-speed Crystal/Ceramic Resonator Characteristics

This device contains an RC oscillation circuit that generates a 32.768 kHz clock, meanwhile, the DCXO module can calibrate the RC oscillation circuit regularly. If the product does not have a high requirement for the accuracy of the system clock, the external 32.768 kHz crystal circuit can be omitted and the internal RC oscillation circuit can be adopted, meanwhile, the relevant clock configuration needs to be turned on by the software.

This device also can connect to a 32.768 kHz crystal resonator (oscillation mode). The 32.768 kHz crystal resonator provides 32.768 kHz reference clock which is connected to the X32KIN and X32KOUT terminals. In the application, the crystal resonator and the load capacitors must be placed close to the oscillator pins to minimize output distortion and the startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics.

Table 5-13 Low-speed 32.768 kHz Crystal Circuit Characteristics

| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| $f_{X32K_IN}$ | Crystal parallel resonance frequency | - | 32.768 | - | kHz |
| | Crystal frequency stability and tolerance at 25 °C (1) | - | - | - | ppm |
| | Oscillation mode | Fundamental | | | - |

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| Symbol | Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| $C_0$ | Shunt capacitance (2) | - | 1.1 | - | pF |

This device has no requirement for the frequency stability and tolerance of 32.768 kHz crystal. If the actual product has requirement for the accuracy of timing function, the 20 ppm stability and tolerance is recommended.

The 1.1 pF is only a simulation value. The crystal shunt capacitance ($C_0$) is given by the crystal manufacturer.

# 5.12 Interface Timing Characteristics

## 5.12.1 Raw NAND Flash Interface Timing

Figure 5-2 Conventional Serial Access Cycle Timing (SAM0)

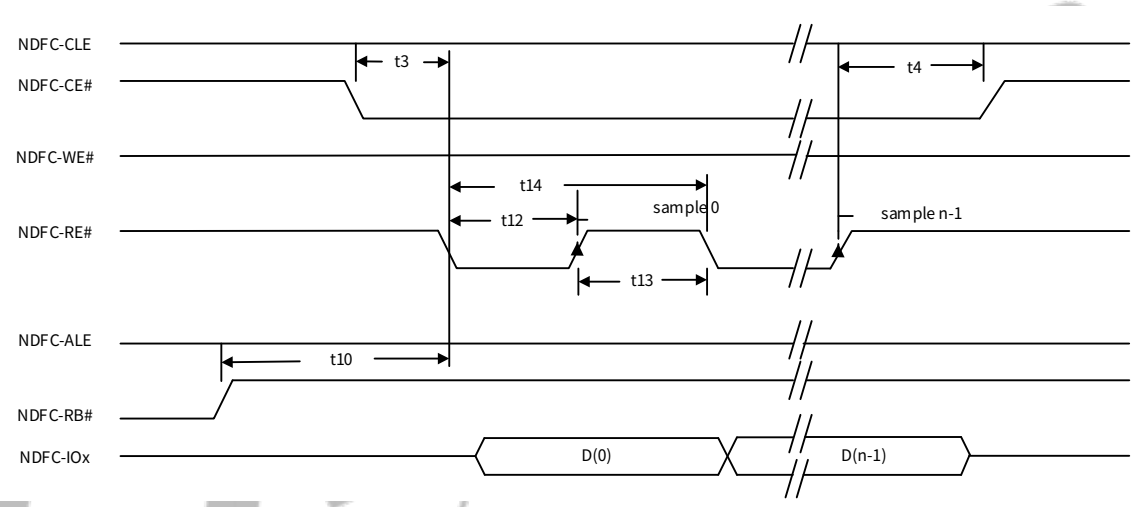
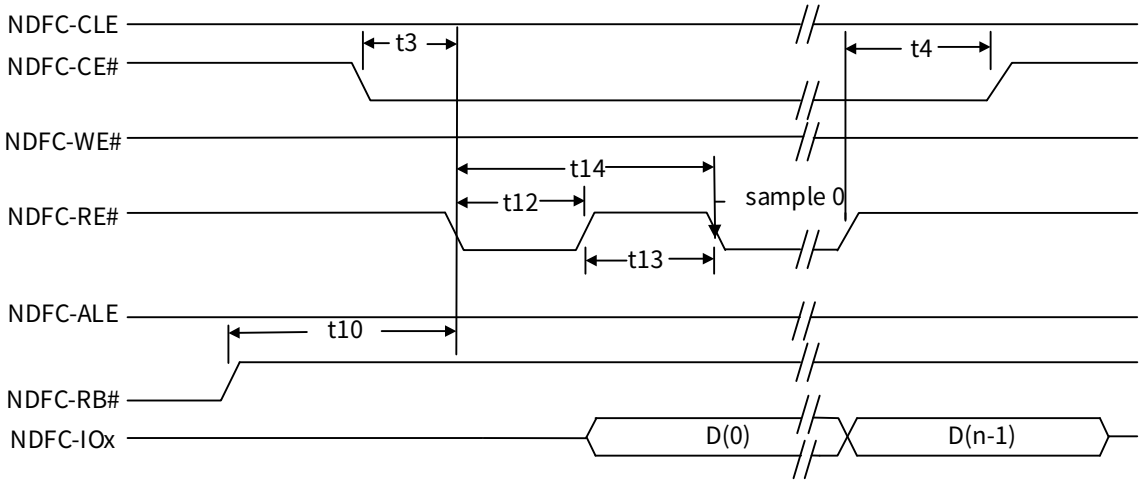



Figure 5-3 EDO Type Serial Access after Read Cycle Timing (SAM1)




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Figure 5-4 Extending EDO Type Serial Access Mode Timing (SAM2)

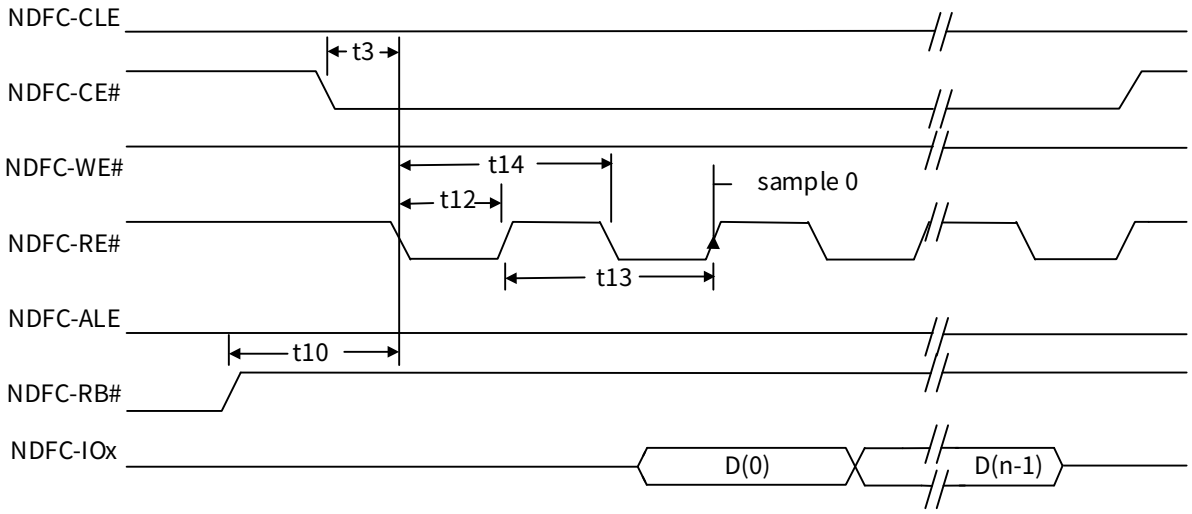



Figure 5-5 Command Latch Cycle Timing

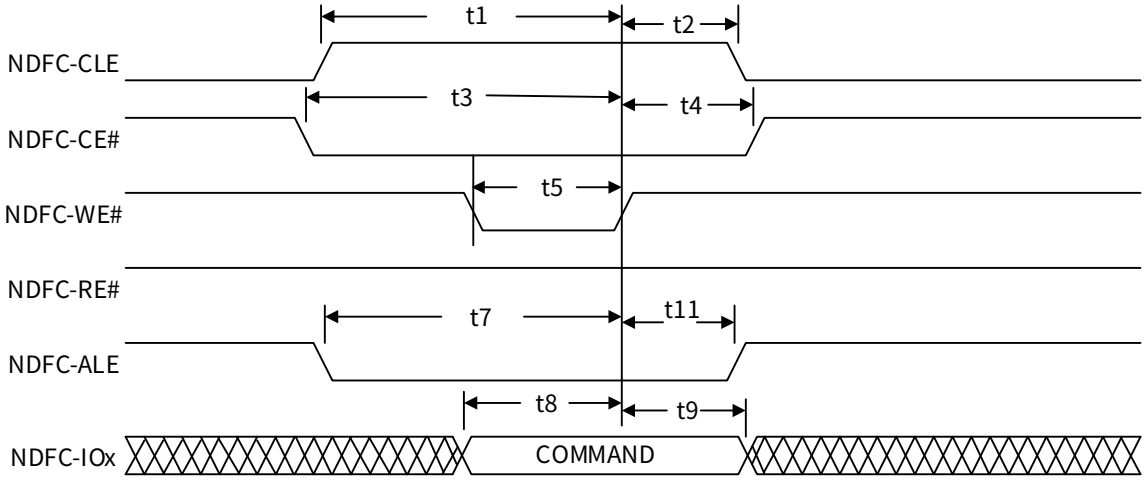
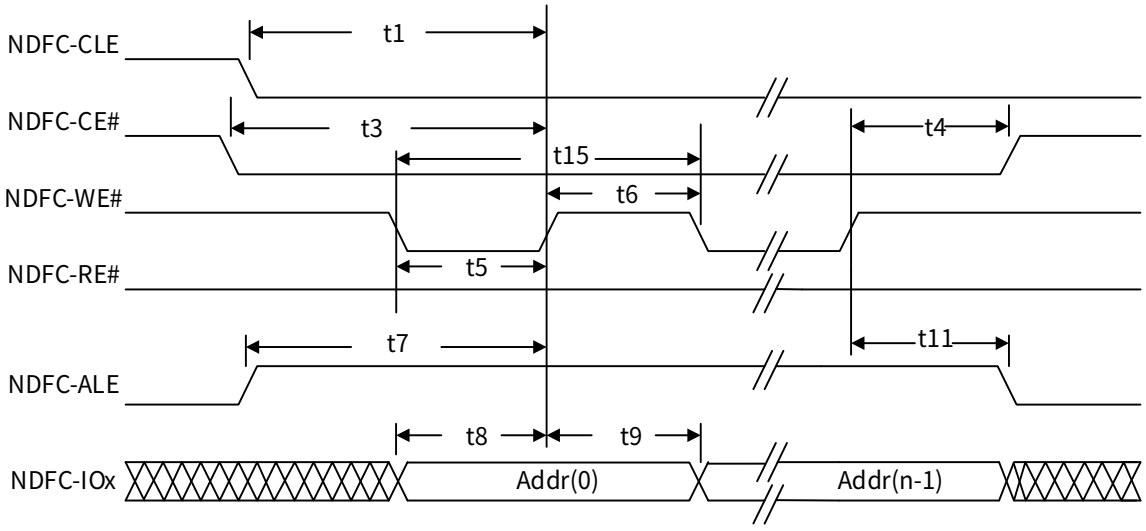



Figure 5-6 Address Latch Cycle Timing




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Figure 5-7 Write Data to Flash Cycle Timing

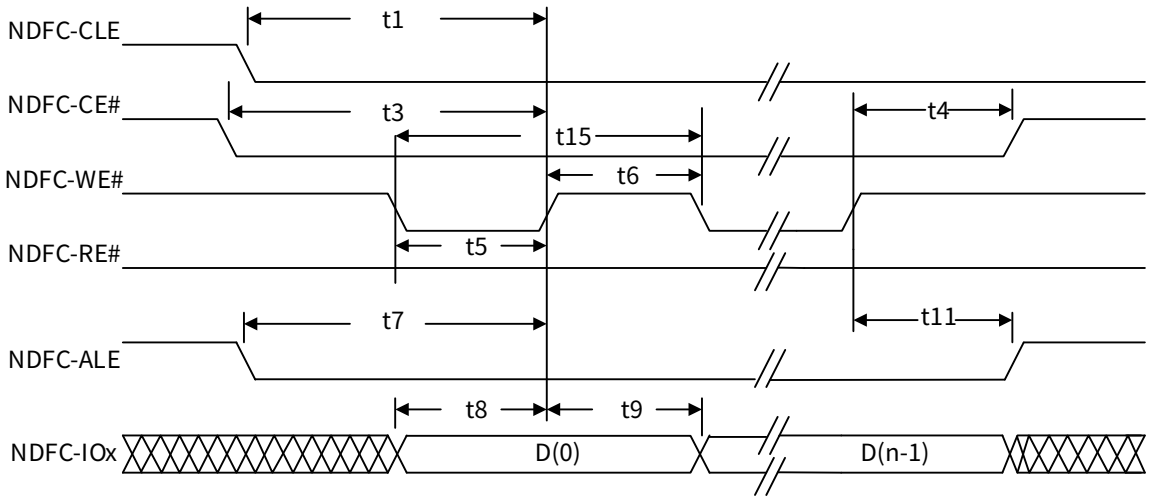



Figure 5-8 Waiting R/B# Ready Timing

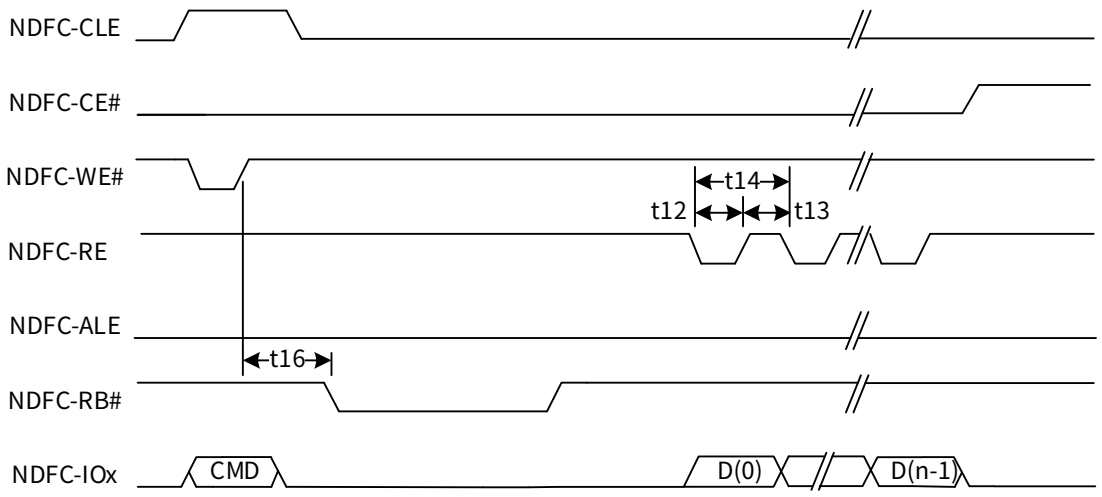
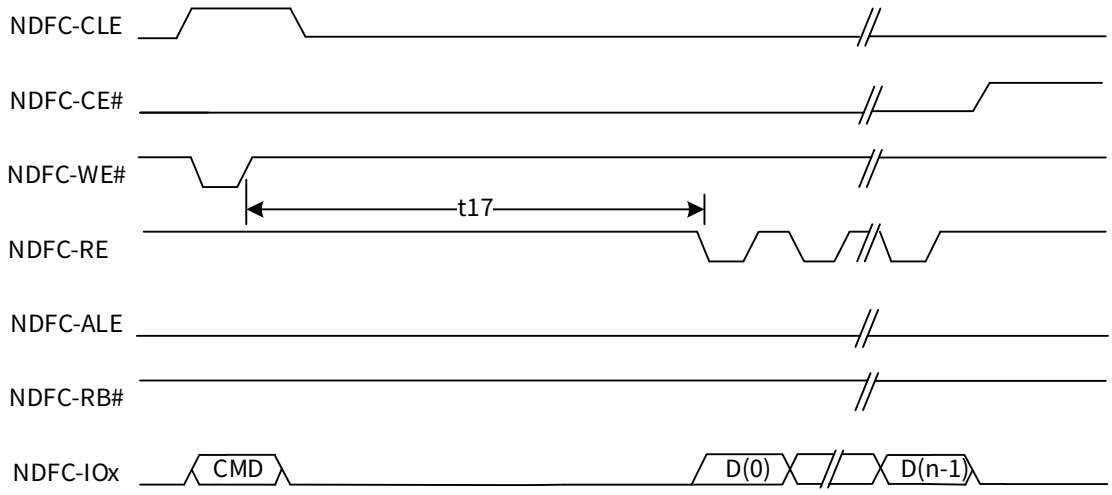



Figure 5-9 WE# High to RE# Low Timing




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Figure 5-10 RE# High to WE# Low Timing

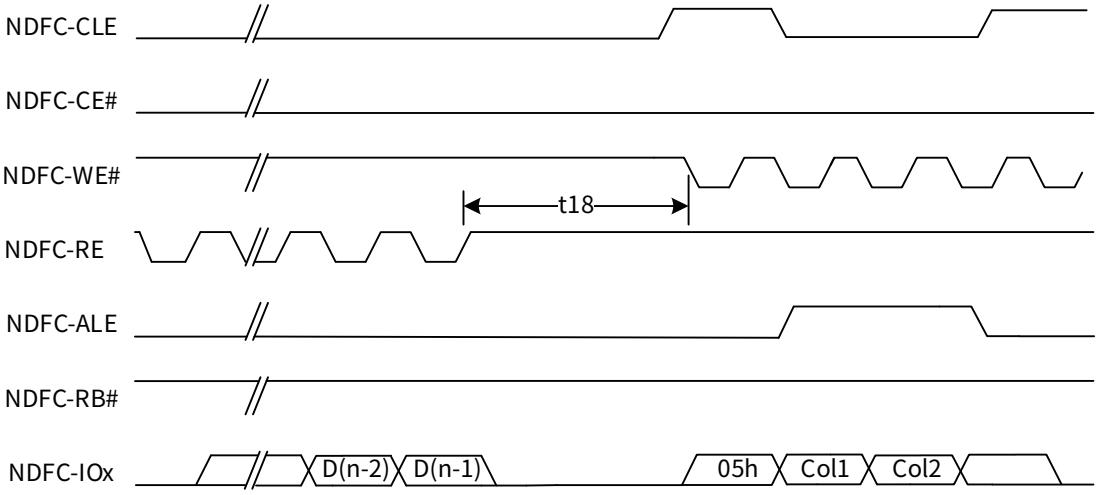



Figure 5-11 Address to Data Loading Timing

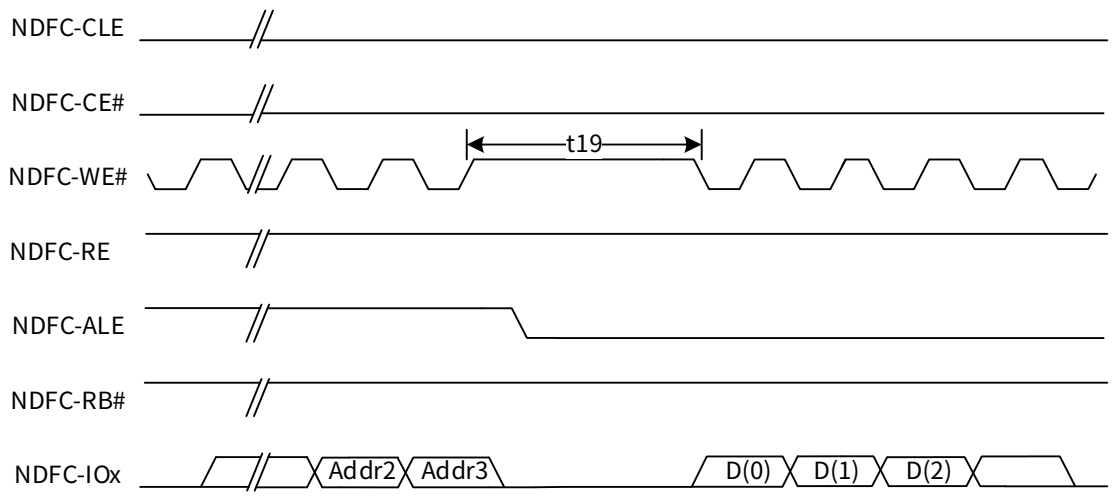



Table 5-14 Raw NAND Flash Timing Constants

| Parameter | Symbol | Timing | Unit |
|---|---|---|---|
| NDFC-CLE setup time | t1 | 2T | ns |
| NDFC-CLE hold time | t2 | 2T | ns |
| NDFC-CE setup time | t3 | 2T | ns |
| NDFC-CE hold time | t4 | 2T | ns |
| NDFC-WE# pulse width | t5 | T(1) | ns |
| NDFC-WE# hold time | t6 | T | ns |
| NDFC-ALE setup time | t7 | 2T | ns |
| Data setup time | t8 | T | ns |
| Data hold time | t9 | T | ns |

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| Parameter | Symbol | Timing | Unit |
|---|---|---|---|
| Ready to NDFC-RE# low | t10 | 3T | ns |
| NDFC-ALE hold time | t11 | 2T | ns |
| NDFC-RE# pulse width | t12 | T | ns |
| NDFC-RE# hold time | t13 | T | ns |
| Read cycle time | t14 | 2T | ns |
| Write cycle time | t15 | 2T | ns |
| NDFC-WE# high to R/B# busy | t16 | T_WB(2) | ns |
| NDFC-WE# high to NDFC-RE# low | t17 | T_WHR(3) | ns |
| NDFC-RE# high to NDFC-WE# low | t18 | T_RHW(4) | ns |
| Address to Data Loading time | t19 | T_ADL(5) | ns |
| (1) T is the cycle of internal clock.<br>(2), (3), (4), (5) This values are configurable in NAND Flash controller. The value of T_WB could be 14*2T/22*2T/30*2T/38*2T, the value of T_WHR could be 8*2T/16*2T/24*2T/32*2T, the value of T_RHW could be 4*2T/8*2T/12*2T/20*2T, the value of T_ADL could be 0*2T/8*2T/16*2T/24*2T. | | | |

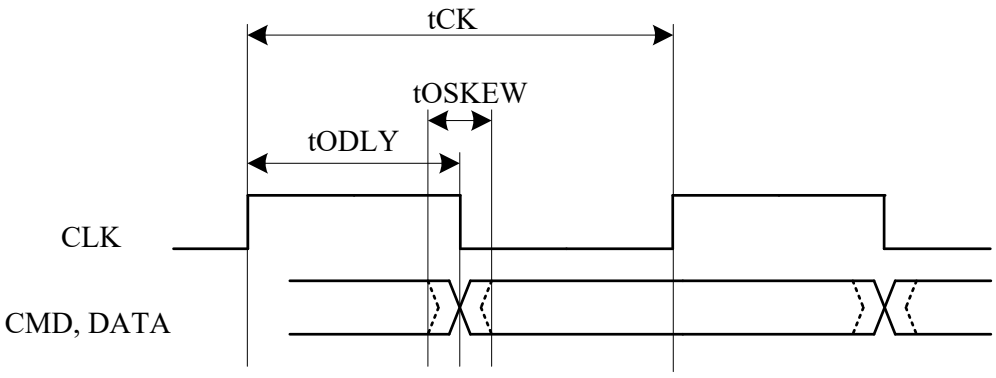
## 5.12.2 SMHC Interface Timing

### 5.12.2.1 HS-SDR Mode

NOTE

IO voltage is 1.8V or 3.3V.

Figure 5-12 SMHC HS-SDR Mode Output Timing Diagram




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Table 5-15 SMHC HS-SDR Mode Output Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **CLK** | | | | | |
| Clock frequency | tCK | 0 | 50 | 50 | MHz |
| Duty Cycle | DC | 45 | 50 | 55 | % |
| **Output CMD, DATA(referenced to CLK)** | | | | | |
| CMD, Data output delay time | tODLY | - | 0.25 | 0.5 | UI |
| Data output delay skew time | tOSKEW | - | - | 0.884 | ns |
| (1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=20 ns at 50 MHz.<br>(2) The driver strength level of GPIO is 2 for test. | | | | | |

Figure 5-13 SMHC HS-SDR Mode Input Timing Diagram

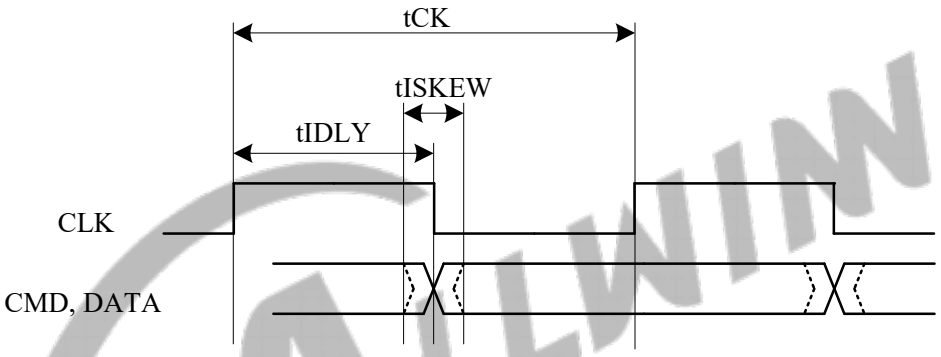



Table 5-16 SMHC HS-SDR Mode Input Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **CLK** | | | | | |
| Clock frequency | tCK | 0 | 50 | 50 | MHz |
| Duty Cycle | DC | 45 | 50 | 55 | % |
| **Input CMD, DATA(referenced to CLK 50MHz)** | | | | | |
| Data input delay in SDR mode. It includes the PCB delay time of Clock, the PCB delay time of Data and the data output delay of Device | tIDLY | - | - | 20 | ns |
| Data input skew time in SDR mode | tISKEW | - | - | 0.858 | ns |
| The driver strength level of GPIO is 2 for test. | | | | | |

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### 5.12.2.2 HS-DDR Mode

Figure 5-14 SMHC HS-DDR Mode Output Timing Diagram

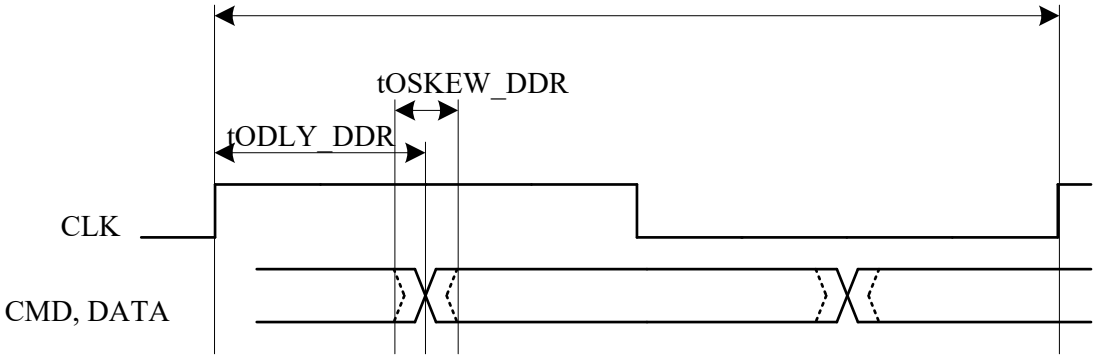



Table 5-17 SMHC HS-DDR Mode Output Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **CLK** | | | | | |
| Clock frequency | tCK | 0 | 50 | 50 | MHz |
| Duty Cycle | DC | 45 | 50 | 55 | % |
| **Output CMD, DATA(referenced to CLK)** | | | | | |
| CMD, Data output delay time in DDR mode | tODLY-DDR | - | 0.25 | 0.25 | UI |
| Data output delay skew time | tOSKEW | - | - | 0.884 | ns |
| (1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=20 ns at 50 MHz.<br>(2) The driver strength level of GPIO is 2 for test. | | | | | |

Figure 5-15 SMHC HS-DDR Mode Input Timing Diagram

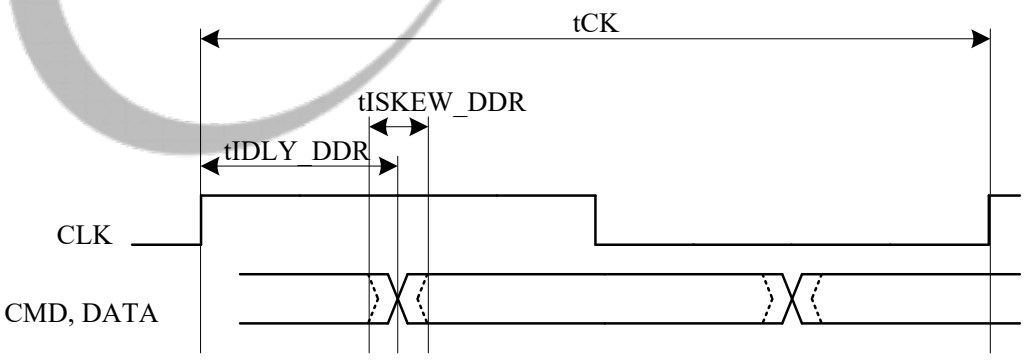



Table 5-18 SMHC HS-DDR Mode Input Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **CLK** | | | | | |
| Clock frequency | tCK | 0 | 50 | 50 | MHz |

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| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Duty Cycle | DC | 45 | 50 | 55 | % |
| **Input CMD, DATA(referenced to CLK 50MHz)** | | | | | |
| Data input delay in DDR mode. It includes the PCB delay time of Clock, the PCB delay time of Data and the data output delay of Device | tIDLY-DDR | - | - | 8.3 | ns |
| Data input skew time in DDR mode | tISKEW-DDR | - | - | 0.858 | ns |
| The driver strength level of GPIO is 2 for test. | | | | | |

#### 5.12.2.3 HS200/SDR104 Mode

Figure 5-16 SMHC HS200/SDR104 Mode Host Output and Device InputTiming Diagram

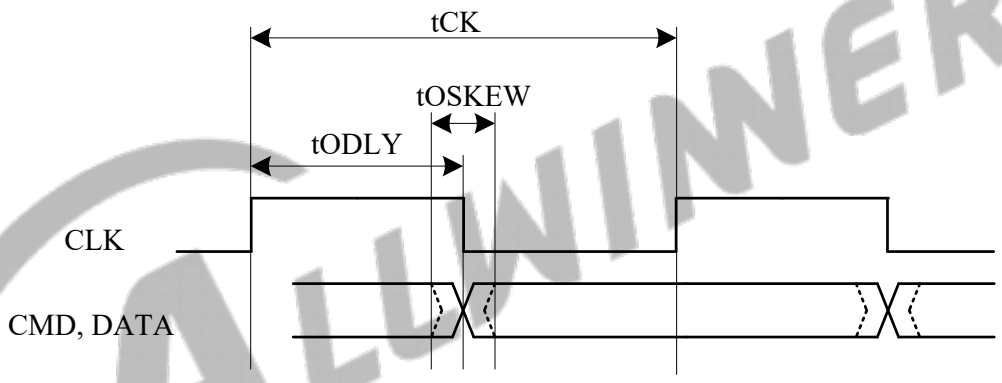



Table 5-19 SMHC HS200/SDR104 Mode Host Output and Device InputTiming Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| **CLK** | | | | | |
| Clock frequency | tCK | - | - | 200 | MHz |
| Duty Cycle | DC | 45 | 50 | 55 | % |
| Rise time, fall time | tTLH, tTHL | - | - | 0.2 | UI |
| **Host Output CMD, DATA (referenced to CLK)** | | | | | |
| Host CMD, Data output delay time | tODLY | - | 0.25 | 0.5 | UI |
| Host Data output delay skew time | tOSKEW | - | - | 0.884 | ns |
| (1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=10 ns at 100 MHz.<br>(2) The driver strength level of GPIO is 3 for test. | | | | | |

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Figure 5-17 SMHC HS200/SDR104 Mode Host Input and Device Output Timing Diagram

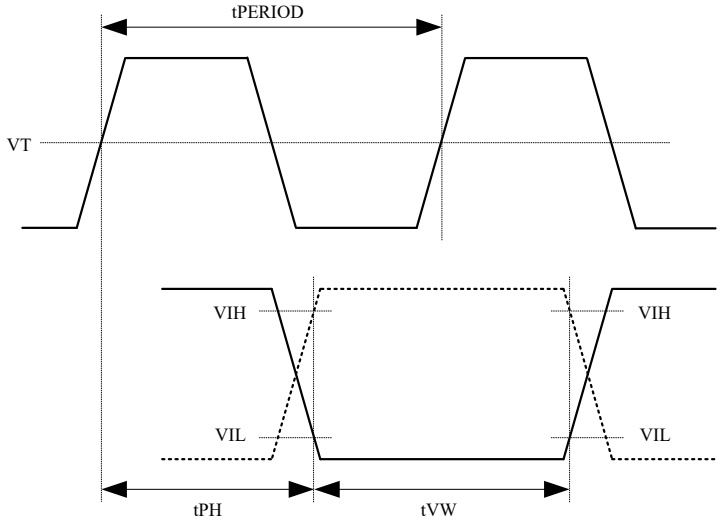



Table 5-20 SMHC HS200/SDR104 Mode Host Input and Device Output Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit | Remark |
|---|---|---|---|---|---|---|
| **CLK** | | | | | | |
| Clock Period | tPERIOD | 5 | - | - | ns | Max: 200 MHz |
| Duty Cycle | DC | 45 | 50 | 55 | % | |
| Rise time, fall time | tTLH, tTHL | - | - | 0.2 | UI | |
| **Host Input CMD, DATA (referenced to CLK)** | | | | | | |
| Device Output delay | tPH | 0 | - | 2 | UI | |
| Device Output delay variation due to temperature change after tuning | dPH | -350(3) | - | 1550(4) | ps | |
| Device CMD, Data valid window | tVW | 0.575 | - | - | UI | |

(1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=10 ns at 100 MHz.

(2) The driver strength level of GPIO is 3 for test.

(3) Temperature variation: -20 °C.

(4) Temperature variation: 90 °C.

### 5.12.2.4 HS400 Mode

The CMD output timing for HS400 mode is the same as CMD output timing for HS200 mode.

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Figure 5-18 SMHC HS400 Mode Host Output Timing and Device Input Timing Diagram

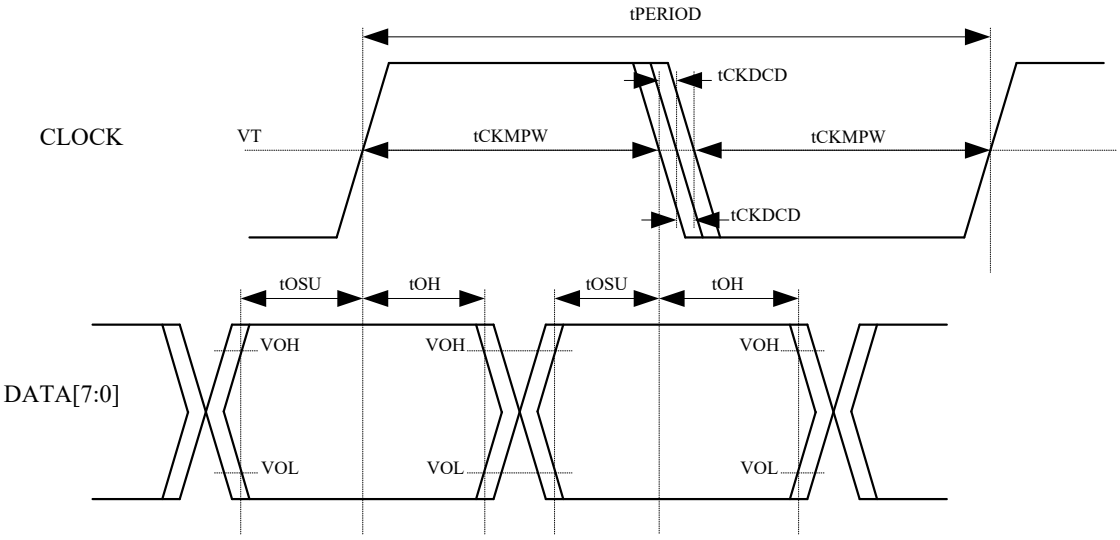



Table 5-21 SMHC HS400 Mode Host Output Timing and Device Input Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit | Remark |
|---|---|---|---|---|---|---|
| **CLK** | | | | | | |
| Clock period | tPERIOD | 5 | - | - | ns | Max: 200MHz |
| Clock slew rate | SR | 1.125 | - | - | V/ns | |
| Clock duty cycle distortion | tCKDCD | 0 | - | 0.5 | ns | |
| Clock minimum pulse width | tCKMPW | 2.2 | - | - | ns | |
| **Output DATA (referenced to CLK)** | | | | | | |
| Data output setup time | tOSU | 0.4 | - | - | ns | |
| Data output hold time | tOH | 0.4 | - | - | ns | |
| Data output slew rate | SR | 0.9 | - | - | V/ns | |
| (1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=10 ns at 100 MHz.<br>(2) The driver strength level of GPIO is 3 for test. | | | | | | |

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Figure 5-19 SMHC HS400 Mode Host Input Timing and Device Output Timing Diagram

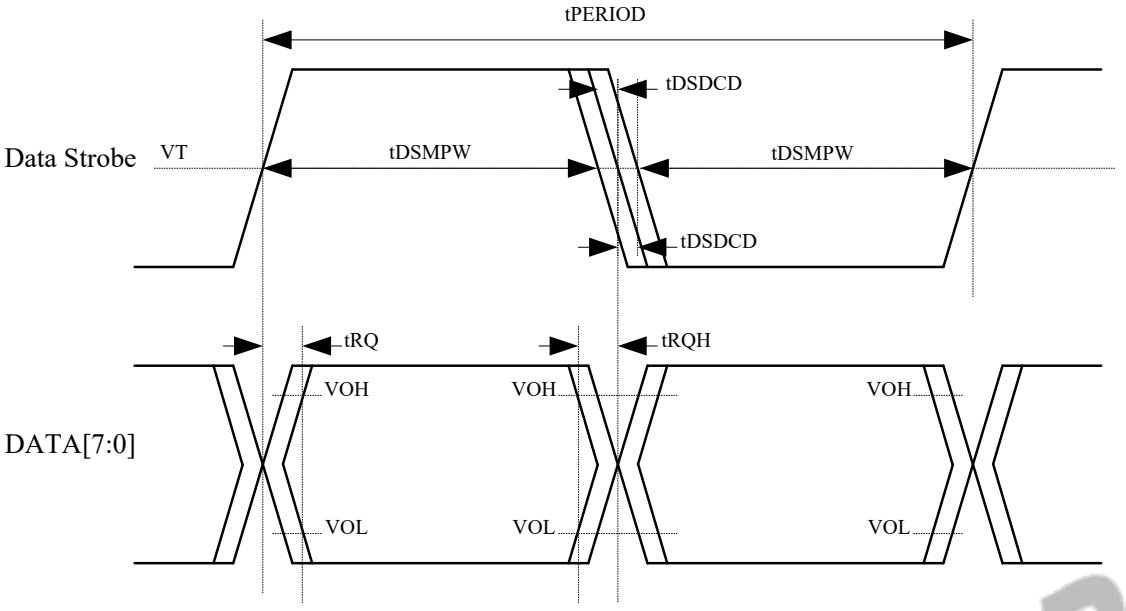


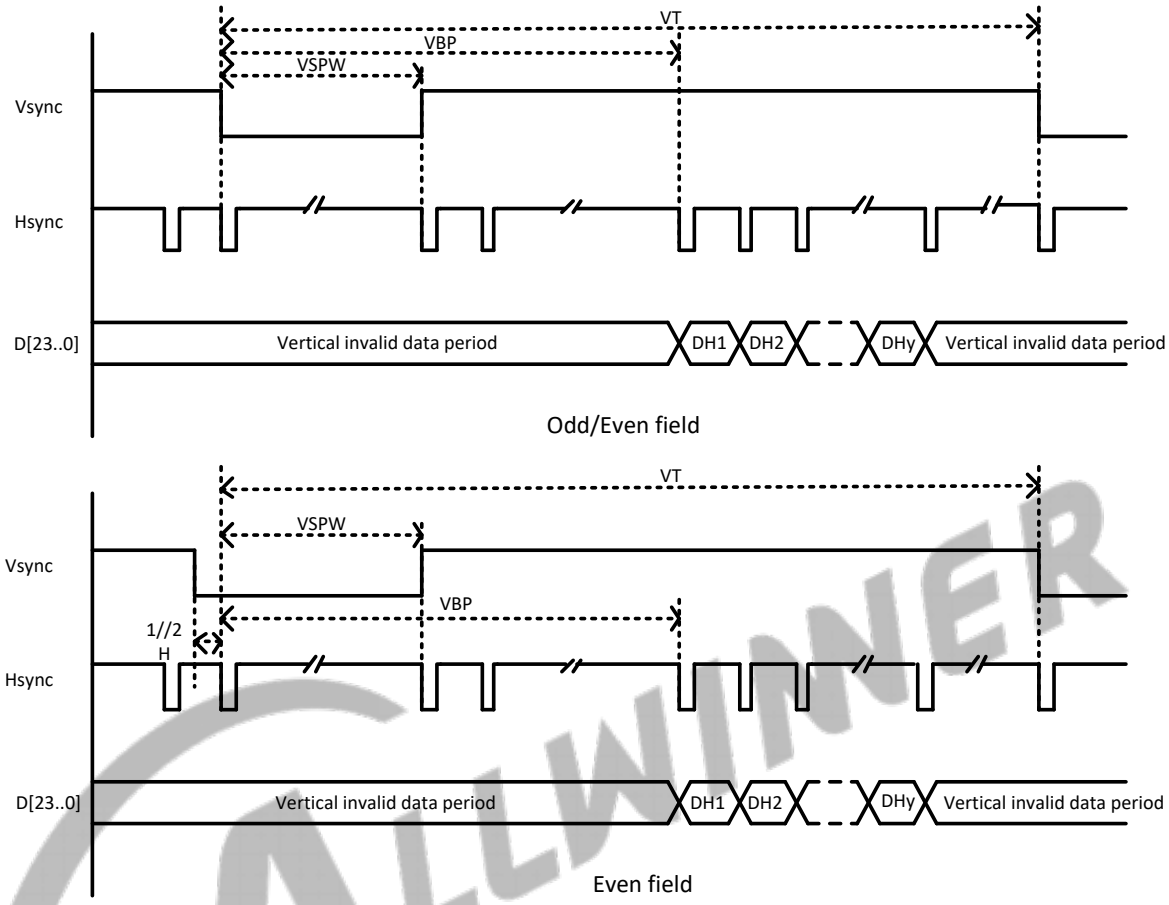

Table 5-22 SMHC HS400 Mode Input Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit | Remark |
|---|---|---|---|---|---|---|
| **DS (Data Strobe)** | | | | | | |
| DS period | tPERIOD | 5 | - | - | ns | Max: 200 MHz |
| DS slew rate | SR | 1.125 | - | - | V/ns | |
| DS duty cycle distortion | tDSDCD | 0.0 | - | 0.4 | ns | |
| DS minimum pulse width | tDSMPW | 2.0 | - | - | ns | |
| **Input DATA (referenced to DS)** | | | | | | |
| Data input skew | tRQ | - | - | 0.4 | ns | |
| Data input hold skew | tRQH | - | - | 0.4 | ns | |
| Data input slew rate | SR | 0.85 | - | - | V/ns | |
| (1) The Unit Interval (UI) is 1-bit nominal time. For example, UI=10 ns at 100 MHz.<br>(2) The driver strength level of GPIO is 3 for test. | | | | | | |

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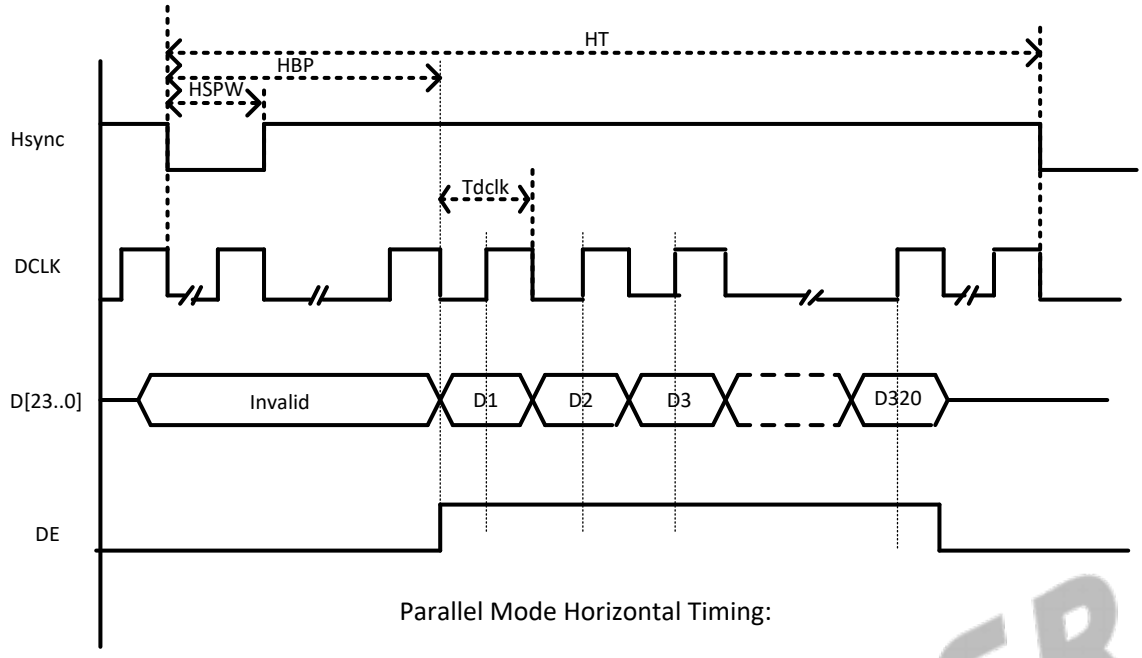
## 5.12.3 LCD Interface Timing

Figure 5-20 HV_IF Vertical Timing

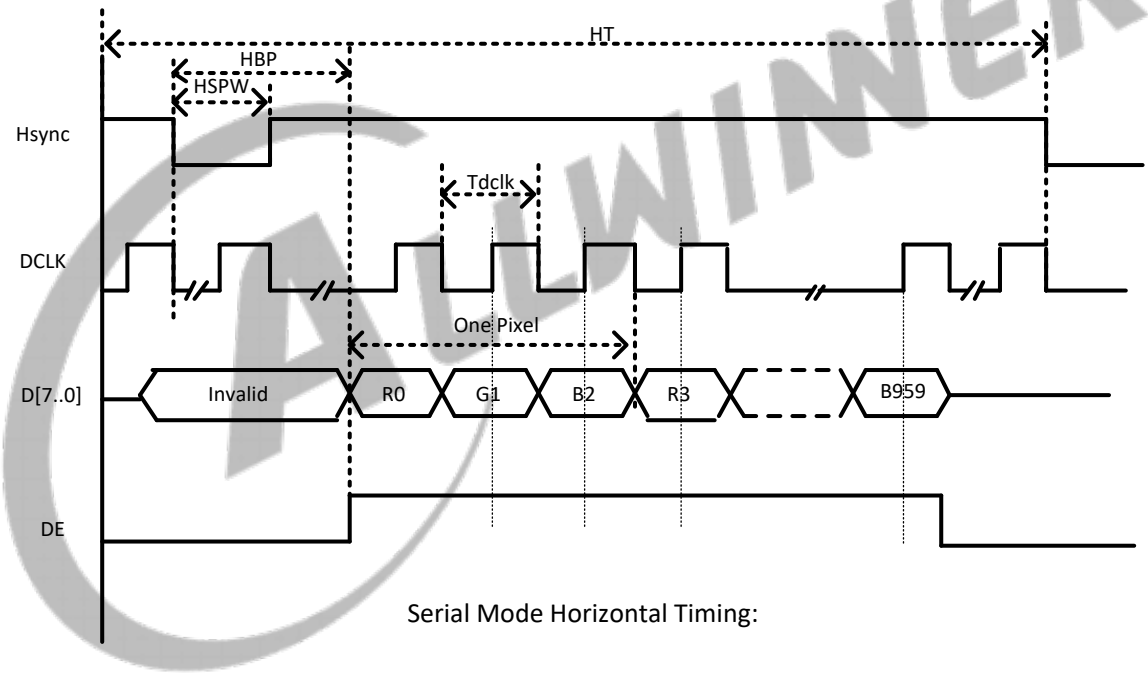



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Figure 5-21 HV_IF Horizontal Timing




Parallel Mode Horizontal Timing:




Serial Mode Horizontal Timing:

Table 5-23 LCD HV_IF Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| DCLK Cycle Time | tDCLK | 5.9 | - | - | ns |
| Hsync Period Time | tHT | - | HT+1 | - | tDCLK |
| Hsync Width | tHSPW | - | HSPW+1 | - | tDCLK |
| Hsync Back Porch | tHBP | - | HBP+1 | - | tDCLK |
| Vsync Period Time | tVT | - | VT/2 | - | tHT |
| Vsync Width | tVSPW | - | VSPW+1 | - | tHT |

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| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Vsync Back Porch | tVBP | - | VBP+1 | - | tHT |
| (1) Vsync: Vertical sync, indicates one new frame.<br>(2) Hsync: Horizontal sync, indicate one new scan line.<br>(3) DCLK: Dot clock, pixel data are sync by this clock.<br>(4) LDE: LCD data enable.<br>(5) LD[23..0]: 24Bit RGB/YUV output from input FIFO for panel. | | | | | |

### 5.12.4 Parallel CSI Interface Timing

Figure 5-22 Parallel CSI Data Sample Timing

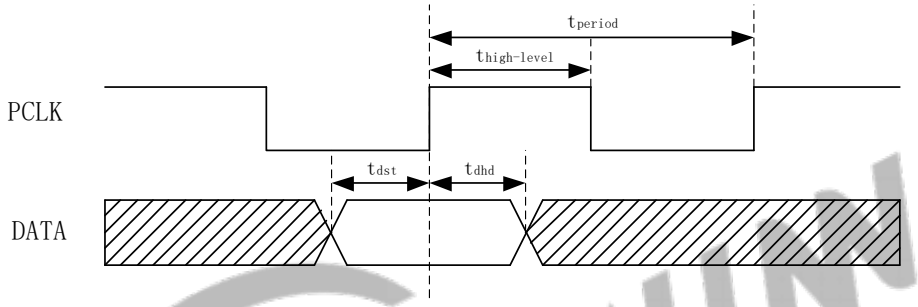
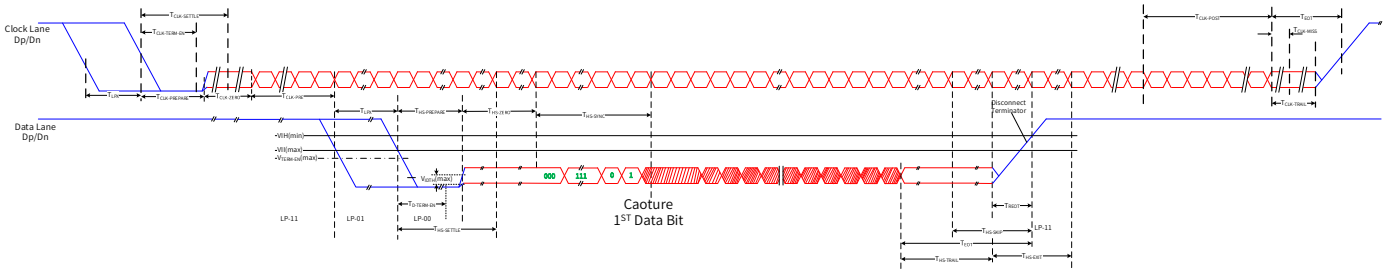



Table 5-24 Parallel CSI Interface Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Pclk period | $t_{period}$ | 6.73 | - | - | ns |
| Pclk frequency | $1/t_{period}$ | - | - | 148.5 | MHz |
| Pclk duty | $t_{high-level}/t_{period}$ | 40 | 50 | 60 | % |
| Data input setup time | $t_{dst}$ | 0.6 | - | - | ns |
| Data input hold time | $t_{dhd}$ | 0.6 | - | - | ns |

### 5.12.5 MIPI CSI Interface Timing

Figure 5-23 MIPI CSI Interface Timing




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Table 5-25 MIPI CSI Interface Timing Constants

| Parameter | Description | Min | Typ | Max | Unit | Notes |
|---|---|---|---|---|---|---|
| $T_{CLK-MISS}$ | Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX. | | | 60 | ns | (6), (11) |
| $T_{CLK-POST}$ | Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of THS-TRAIL to the beginning of $T_{CLK-TRAIL}$. | 60ns + 52*UI | | | ns | (10) |
| $T_{CLK-PRE}$ | Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode. | 8 | | | UI | (10) |
| $T_{CLK-PREPARE}$ | Time that the transmitter drives the Clock Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission. | 38 | | 95 | ns | (10) |
| $T_{CLK-SETTLE}$ | Time interval during which the HS receiver should ignore any Clock Lane HS transitions, starting from the beginning of TCLK-PREPARE. | 95 | | 300 | ns | (11), (12) |
| $T_{CLK-TERM-EN}$ | Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$. | Time for Dn to reach $V_{TERM-EN}$ | | 38 | ns | (11) |
| $T_{CLK-TRAIL}$ | Time that the transmitter drives the HS-0 state after the last payload clock bit of a HS transmission burst. | 60 | | | ns | (10) |
| $T_{CLK-PREPARE}$ + $T_{CLK-ZERO}$ | TCLK-PREPARE + time that the transmitter drives the HS-0 state prior to starting the Clock. | 300 | | | ns | (10) |
| $T_{D-TERM-EN}$ | Time for the Data Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$. | Time for Dn to reach $V_{TERM-EN}$ | | 35ns + 4*UI | | (11) |
| $T_{EOT}$ | Transmitted time interval from the start of $T_{HS-TRAIL}$ or $T_{CLK-TRAIL}$, to the start | | | 105ns + n*12*UI | | (8), (10) |

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| Parameter | Description | Min | Typ | Max | Unit | Notes |
|---|---|---|---|---|---|---|
| | of the LP-11 state following a HS burst. | | | | | |
| $T_{HS-EXIT}$ | Time that the transmitter drives LP-11 following a HS burst. | 100 | | | ns | (10) |
| $T_{HS-PREPARE}$ | Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission | 40ns + 4*UI | | 85ns + 6*UI | ns | (10) |
| $T_{HS-PREPARE}$ + $T_{HS-ZERO}$ | $T_{HS-PREPARE}$ + time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence. | 145ns + 10*UI | | | ns | (10) |
| $T_{HS-SETTLE}$ | Time interval during which the HS receiver shall ignore any Data Lane HS transitions, starting from the beginning of $T_{HS-PREPARE}$.<br>The HS receiver shall ignore any Data Lane transitions before the minimum value, and the HS receiver shall respond to any Data Lane transitions after the maximum value. | 85ns + 6*UI | | 145ns + 10*UI | ns | (11) |
| $T_{HS-SKIP}$ | Time interval during which the HS-RX should ignore any transitions on the Data Lane, following a HS burst. The end point of the interval is defined as the beginning of the LP-11 state following the HS burst. | 40 | | | ns | (11) |
| $T_{HS-TRAIL}$ | Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst | max(n*8*UI, 60ns + n*4*UI) | | | ns | (7), (8), (10) |
| $T_{INIT}$ | | 100 | | | ns | (10) |
| $T_{LPX}$ | Transmitted length of any Low-Power state period | 50 | | | ns | (9), (10) |
| Ratio $T_{LPX}$ | Ratio of $T_{LPX(MASTER)}/T_{LPX(SLAVE)}$ between Master and Slave side | 2/3 | | 3/2 | ns | |
| $T_{TA-GET}$ | Time that the new transmitter drives the Bridge state (LP-00) after accepting control during a Link Turnaround. | $5*T_{LPX}$ | | | ns | (10) |

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| Parameter | Description | Min | Typ | Max | Unit | Notes |
|---|---|---|---|---|---|---|
| $T_{TA-GO}$ | Time that the transmitter drives the Bridge state (LP-00) before releasing control during a Link Turnaround. | $4*T_{LPX}$ | | | ns | (10) |
| $T_{TA-SURE}$ | Time that the new transmitter waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround. | $T_{LPX}$ | | $2*T_{LPX}$ | ns | (10) |
| $T_{WAKEUP}$ | Time that a transmitter drives a Mark-1 state prior to a Stop state in order to initiate an exit from ULPS. | 1 | | | ms | (10) |

NOTE

(6) The minimum value depends on the bit rate. Implementations should ensure proper operation for all the supported bit rates.

(7) If a > b then max (a, b) = a otherwise max (a, b) = b.

(8) Where n = 1 for Forward-direction HS mode and n = 4 for Reverse-direction HS mode.

(9) TLPX is an internal state machine timing reference. Externally measured values may differ slightly from the specified values due to asymmetrical rise and fall times.

(10) Transmitter-specific parameter.

(11) Receiver-specific parameter.

(12) The stated values are considered informative guidelines rather than normative requirements since this parameter is untestable in typical applications.

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### 5.12.6 MIPI DPHY Interface Timing

Figure 5-24 MIPI DPHY Timing

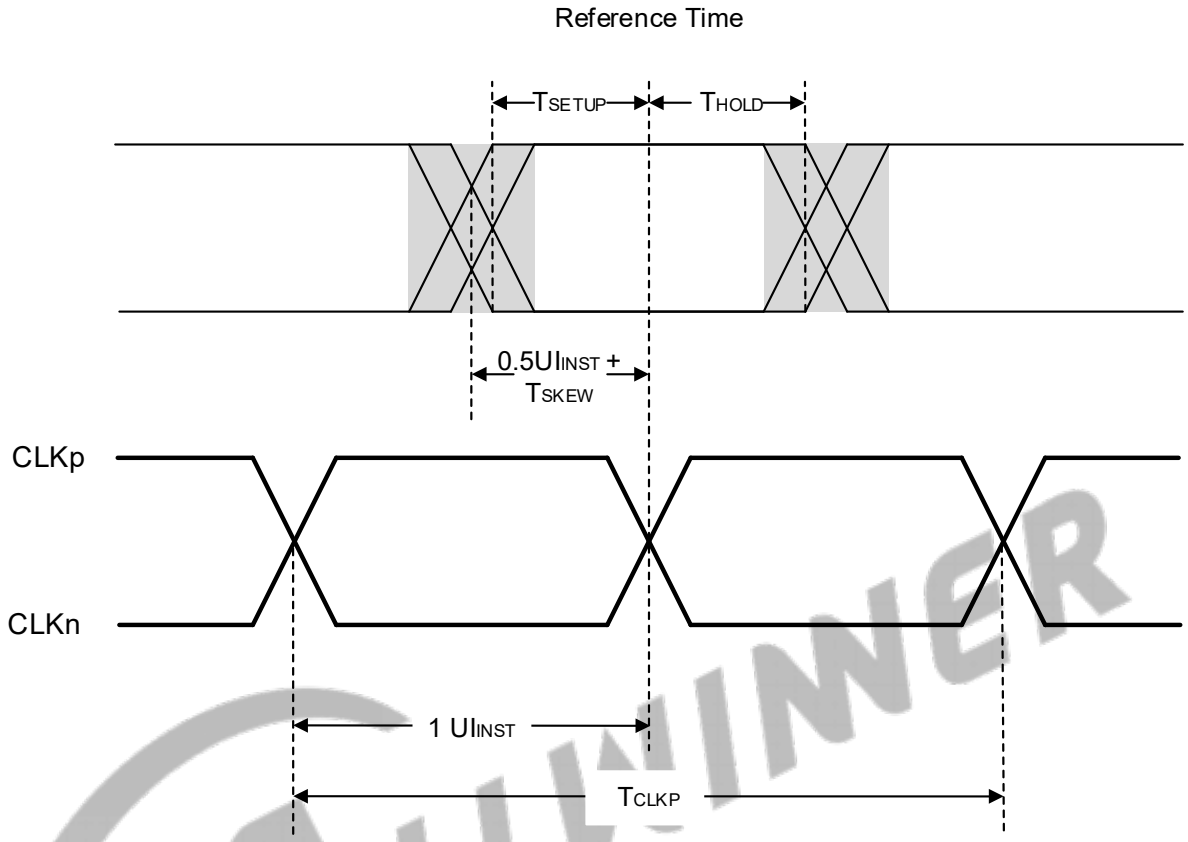



Table 5-26 MIPI DPHY Timing Constants

| Parameter | Symbol | Units in Ulinst | | | Oprational Frequency in Gbps | |
|---|---|---|---|---|---|---|
| | | Min | Max | Total | Min | Max |
| Data to Clock Skew | $T_{skew[tx]}$ | -0.15 | 0.15 | 0.3 | 0.08 | 1.0 |
| Data to Clock Setup Time | $T_{setup[rx]}$ | 0.15 | - | - | 0.08 | 1.0 |
| Clock to Data Hold Time | $T_{hold[rx]}$ | 0.15 | - | - | 0.08 | 1.0 |

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## 5.12.7 GMAC Interface Timing

### 5.12.7.1 RGMII

Figure 5-25 RGMII Receive Timing

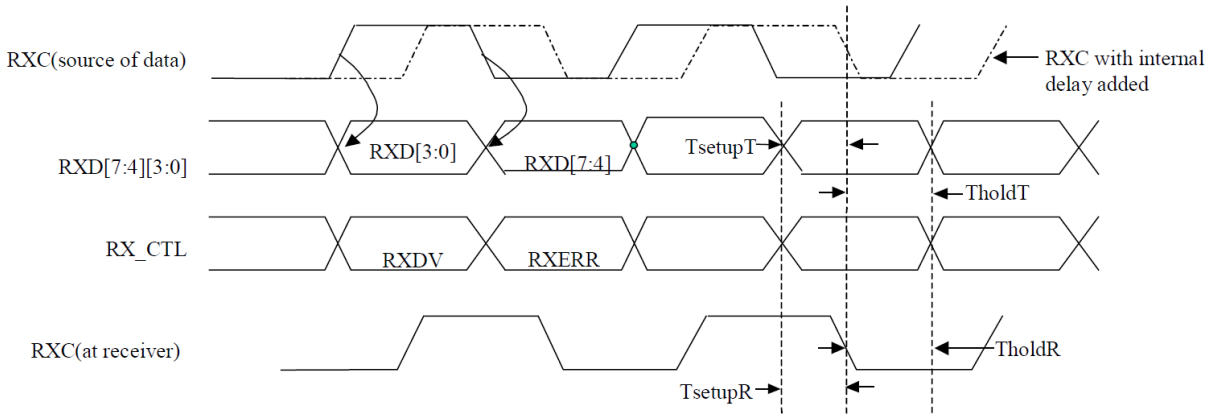



Figure 5-26 RGMII Transmit Timing

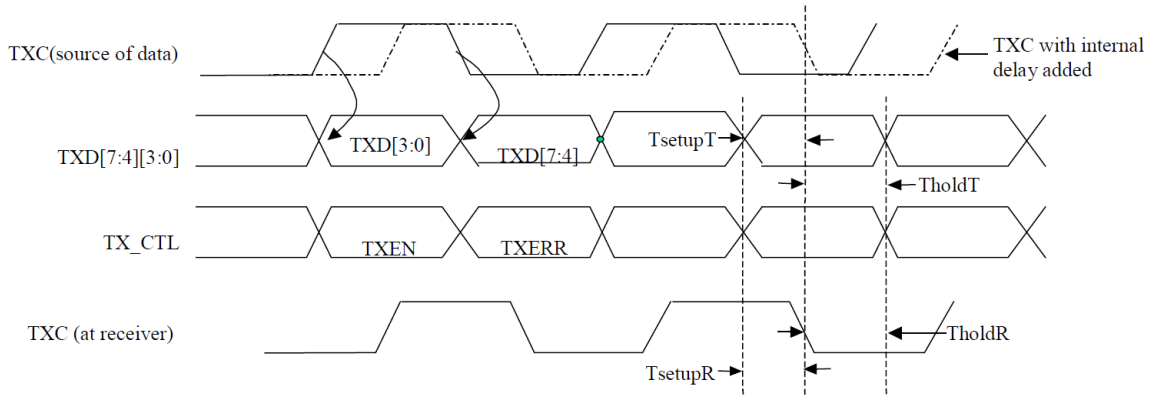



Table 5-27 RGMII Timing Parameters

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Tcyc | Clock Cycle Duration *note1 | 7.2 | 8 | 8.8 | ns |
| Duty_G | Duty Cycle Duration for Gigabit | 45 | 50 | 55 | % |
| Duty_T | Duty Cycle for 10/100T | 40 | 50 | 60 | % |
| TsetupT | Data to clock output setup(at Transmitter integrated delay) | 1.2 | 2.0 | | ns |
| TholdT | Data to clock output hold(at Transmitter integrated delay) | 1.2 | 2.0 | | ns |
| TsetupR | Data to clock input setup(at Receiver integrated delay) | 1.0 | 2.0 | | ns |

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| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| TholdR | Data to clock input hold(at Receiver integrated delay) | 1.0 | 2.0 | | ns |
| For 10Mbps and 100Mbps, Tcyc will scale 400ns±40ns and 40ns±4ns. | | | | | |

### 5.12.7.2 RMII

Figure 5-27 RMII Receive Timing

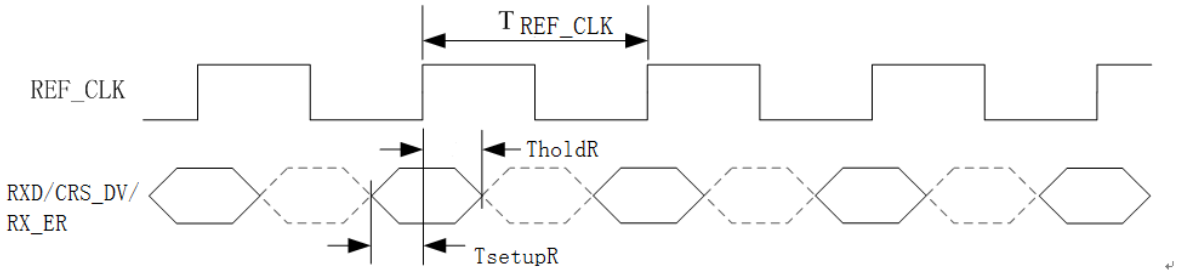



Figure 5-28 RMII Transmit Timing

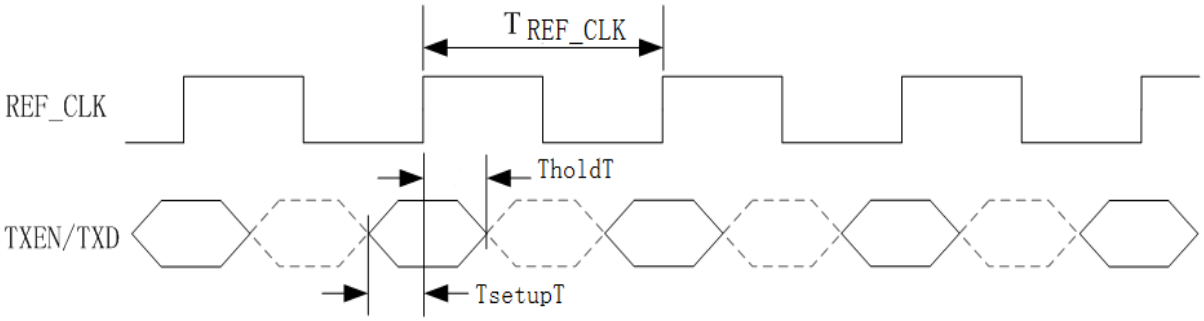



Table 5-28 RMII Timing Parameters

| Parameter | Description | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| $T_{REF_CLK}$ | Reference Clock Period | - | 20 | - | ns |
| Tduty | REF_CLK duty cycle | 35 | | 65 | % |
| TsetupT | TXD/TXEN to REF_CLK setup time | 4 | | | ns |
| TholdT | TXD/TXEN to REF_CLK hold time | 2 | | | ns |
| TsetupR | RXD/CRS_DV/RX_ER to REF_CLK setup time | 4 | | | ns |
| TholdR | RXD/CRS_DV/RX_ER to REF_CLK hold time | 2 | | | ns |

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## 5.12.8 Local Bus Timing

Figure 5-29 Local Bus Timing

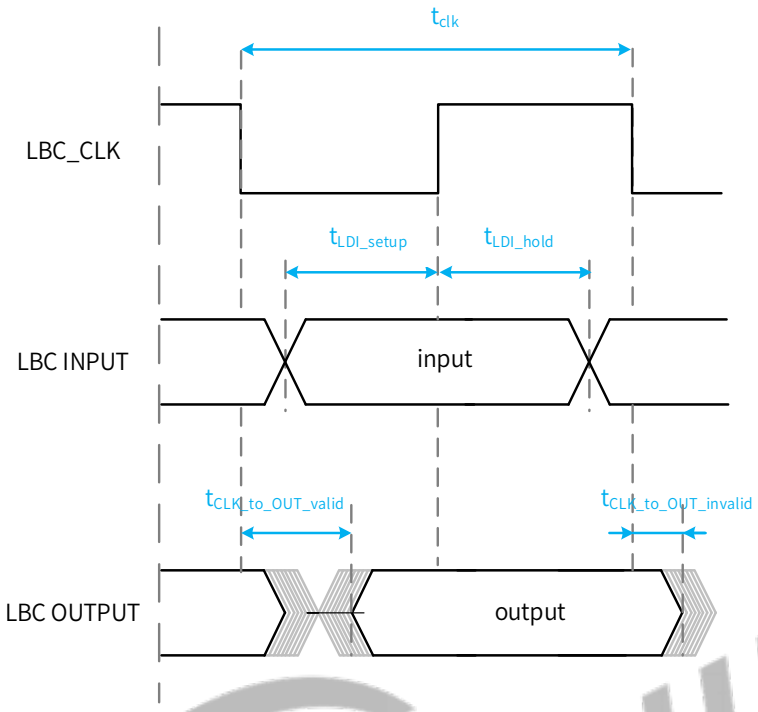
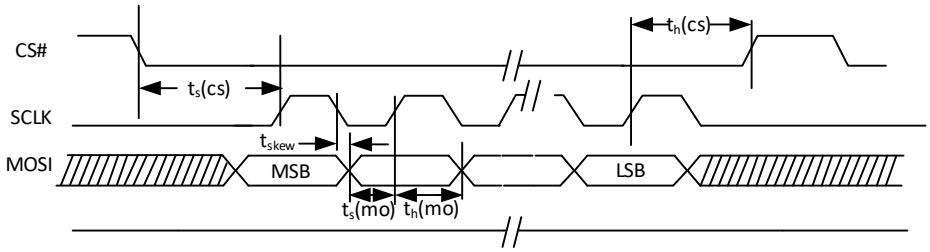



Table 5-29 Local Bus Timing Parameters

| Parameter | Min | Max | Description |
|---|---|---|---|
| $t_{clk}$ | 5ns | - | LBC_CLK period ($f_{clk}$≤200M) |
| $t_{LDI_setup}$ | 1ns | - | Input signals valid before LBC_CLK edge |
| $t_{LDI_hold}$ | 1ns | - | Input signals valid after LBC_CLK edge |
| $t_{CLK_to_OUT_valid}$ | - | 1ns | Output signals valid after LBC_CLK edge |
| $t_{CLK_to_OUT_invalid}$ | 0ns | - | Output signals invalid after LBC_CLK edge |

## 5.12.9 SPI Interface Timing

Figure 5-30 SPI Writing Timing




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Figure 5-31 SPI Reading Timing

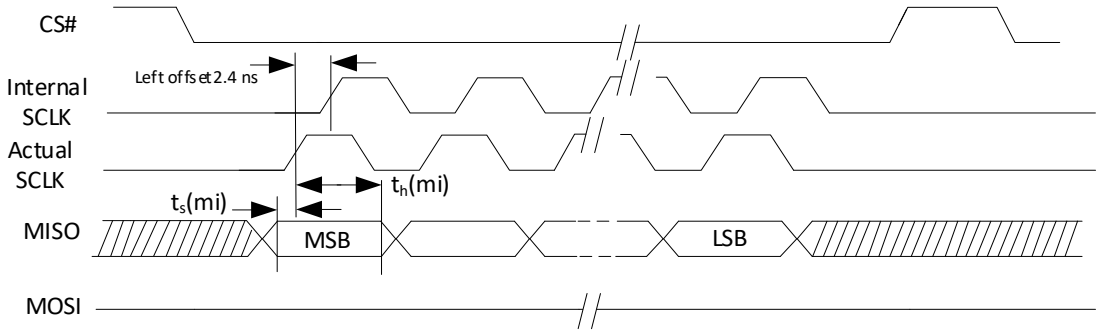



Table 5-30 SPI Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| CS# Active Setup Time | $t_s$(cs) | - | 2T(1) | - | ns |
| CS# Active Hold Time | $t_h$(cs) | - | 2T(1) | - | ns |
| CLK/Data output skew time | $t_{skew}$(2) | - | - | 1 | ns |
| Data output Delay Time | $t_v$(mo) | T(1)/2- $t_{skew}$ | - | T(1)/2- $t_{skew}$ | ns |
| Data output Hold Time | $t_h$(mo) | T(1)/2- $t_{skew}$ | - | T(1)/2- $t_{skew}$ | ns |
| Data In Setup Time | $t_s$(mi) | 0.2(master mode)<br>4.5(slave mode) | - | - | ns |
| Data In Hold Time | $t_h$(mi) | 0.2(master mode)<br>4.5(slave mode) | - | - | ns |

(1) T is the cycle of clock.

(2) $t_{skew}$ is the output skew time between SCLK and MOSI. SCLK may be either faster or slower than MOSI.

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## 5.12.10 SPI-DBI Interface Timing

Figure 5-32 DBI 3-line Serial Interface Timing

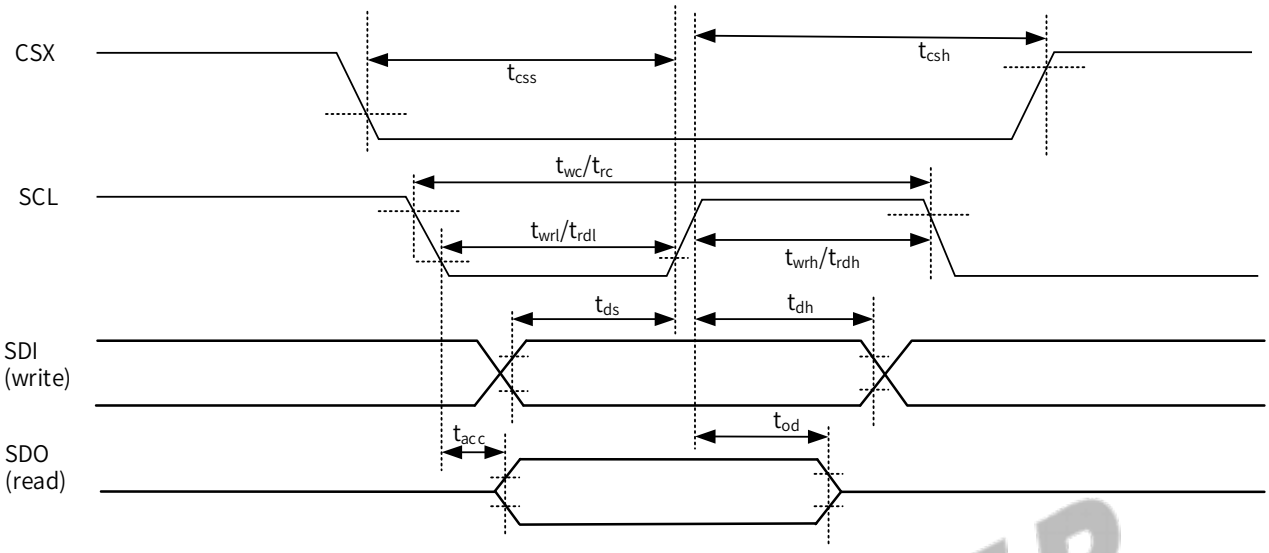



Table 5-31 DBI 3-line Serial Interface Write Timing Parameters

| Signal | Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|---|
| CSX | Chip select setup time | $t_{css}$ | 15 | | ns |
| SCL | Write cycle | $t_{wc}$ | 16 | | ns |
| | Control pulse “H” duration | $t_{wrh}$ | 7 | | ns |
| | Control pulse “L” duration | $t_{wrl}$ | 7 | | ns |
| SDI/SDO | Data setup time | $t_{ds}$ | 7(1) | | ns |
| | Data hold time | $t_{dt}$ | 7(1) | | ns |
| **Note:**<br>Range of required clock frequency: 0-60 MHz. | | | | | |

Table 5-32 DBI 3-line Serial Interface Read Timing Parameters

| Signal | Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|---|
| CSX | Chip select setup time | $t_{csh}$ | 60 | | ns |
| SCL | Read cycle | $t_{rc}$ | 150 | | ns |
| | Control pulse “H” duration | $t_{rdh}$ | 60 | | ns |
| | Control pulse “L” duration | $t_{rdl}$ | 60 | | ns |
| SDI/SDO | Read access time | $t_{racc}$ | 10(1) | 50 | ns |
| | Output disable time | $t_{od}$ | 15(1) | 50 | ns |

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| Signal | Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|---|
| Note:<br>Range of required clock frequency: 0-6.67 MHz. | | | | | |

Figure 5-33 DBI 4-line Serial Interface Timing

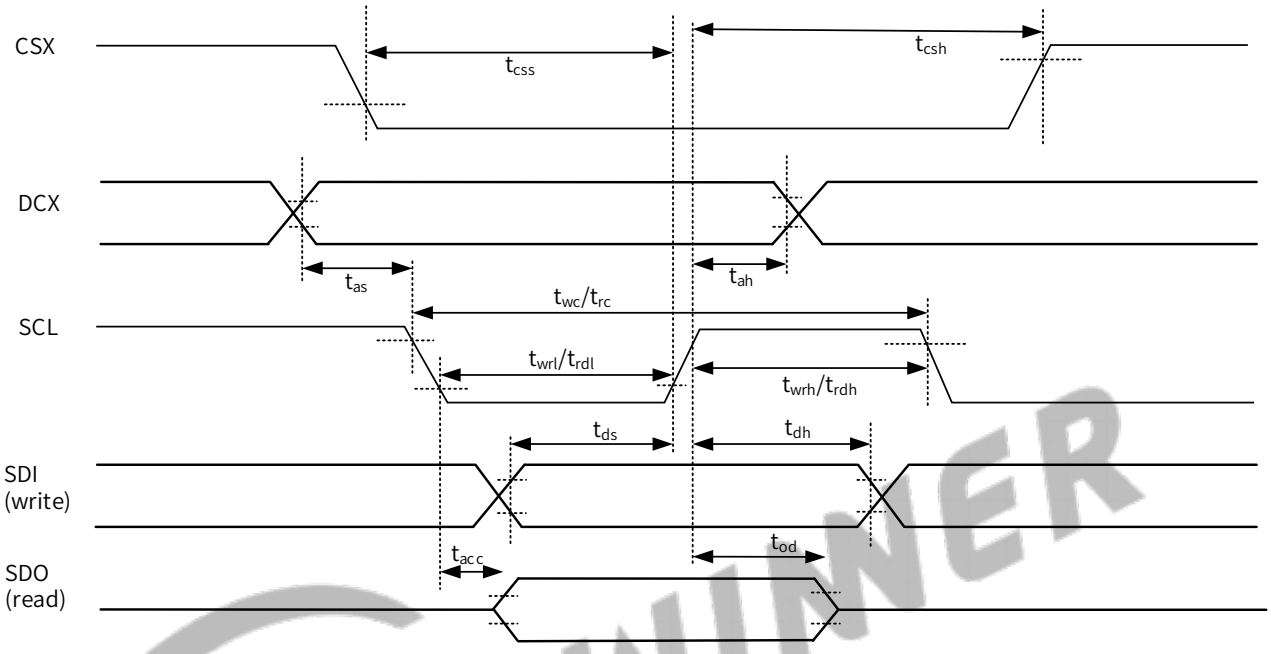



Table 5-33 DBI 4-line Serial Interface Write Timing Parameters

| Signal | Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|---|
| CSX | Chip select setup time | $t_{css}$ | 15 | | ns |
| DCX | Address setup time | $t_{as}$ | 10 | | ns |
| | Address hold time | $t_{ah}$ | 10 | | ns |
| SCL | Write cycle | $t_{wc}$ | 16 | | ns |
| | Control pulse “H” duration | $t_{wrh}$ | 7 | | ns |
| | Control pulse “L” duration | $t_{wrl}$ | 7 | | ns |
| SDI/SDO | Data setup time | $t_{ds}$ | 7(1) | | ns |
| | Data hold time | $t_{dt}$ | 7(1) | | ns |
| | Output disable time | $t_{od}$ | 15(1) | 50 | ns |
| Note:<br>Range of required clock frequency: 0-60 MHz. | | | | | |

Table 5-34 DBI 4-line Serial Interface Read Timing Parameters

| Signal | Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|---|
| CSX | Chip select setup time | $t_{csh}$ | 60 | | ns |

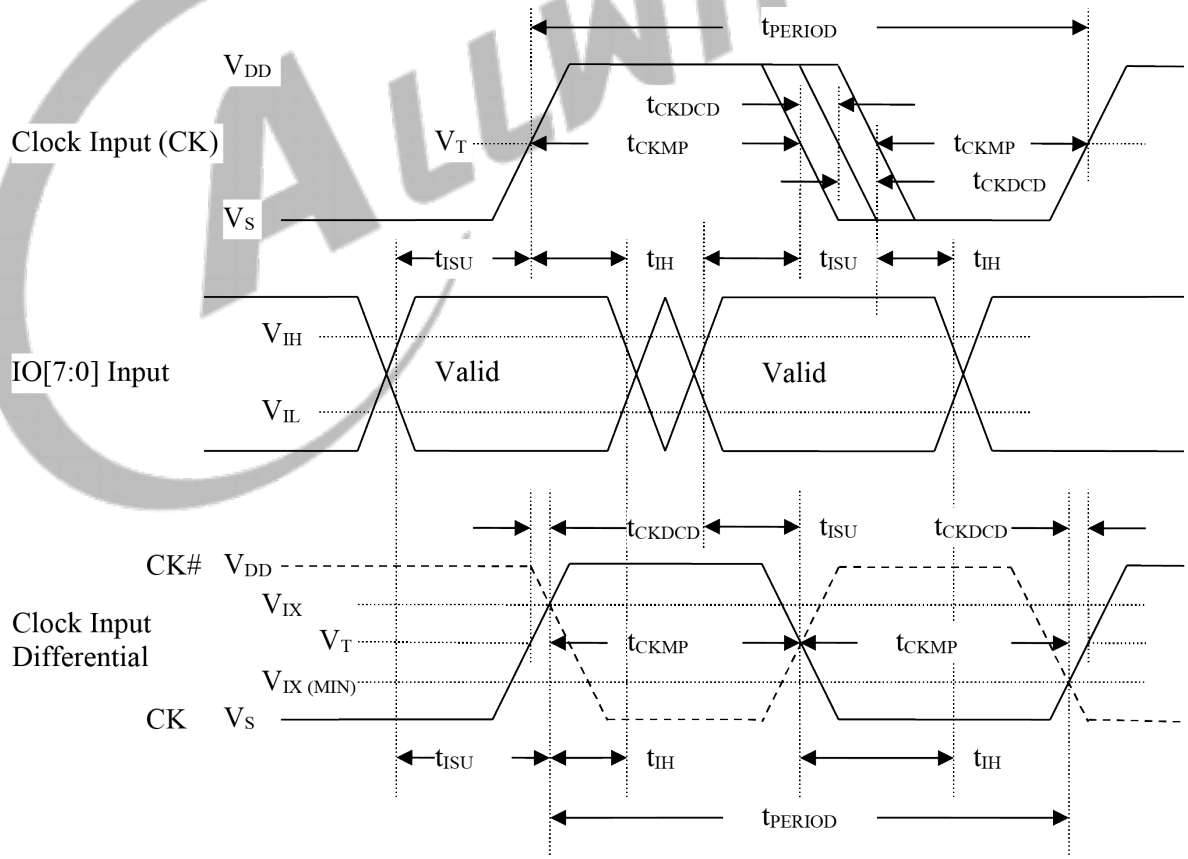
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| Signal | Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|---|
| DCX | Address setup time | $t_{as}$ | 10 | | ns |
| | Address hold time | $t_{ah}$ | 10 | | ns |
| SCL | Read cycle | $t_{rc}$ | 150 | | ns |
| | Control pulse “H” duration | $t_{rdh}$ | 60 | | ns |
| | Control pulse “L” duration | $t_{rdl}$ | 60 | | ns |
| SDI/SDO | Read access time | $t_{racc}$ | - | 50 | ns |
| | Output disable time | $t_{od}$ | 15(1) | 50 | ns |
| **Note:**<br>Range of required clock frequency: 0-6.67 MHz. | | | | | |

## 5.12.11 SPI Flash Interface Timing

### 5.12.11.1 Controller Output to Target Input Timing

Figure 5-34 xSPI Target Data Input Timing




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Table 5-35 Clock Input Threshold Levels

| Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|
| Clock Input Threshold (AC) | $V_{T\ (AC)}$ | 0.50 * $V_{DD}$ | 0.50 * $V_{DD}$ | V |
| Input differential crossing (AC) | $V_{IX\ (AC)}$ | 0.4 * $V_{DD}$ | 0.60 * $V_{DD}$ | V |

Table 5-36 xSPI Device Input Timing

| Parameter | Symbol | xSPI-333(1) | | xSPI-266(2) | | xSPI-200(3) | | Unit | Comments |
|---|---|---|---|---|---|---|---|---|---|
| | | Min | Max | Min | Max | Min | Max | | |
| **Input CK** | | | | | | | | | |
| Cycle Time Data Transfer Mode | $t_{PERIOD}$ | 6 | - | 7.5 | - | 10 | - | ns | 150 MHz (max) between the rising edges with respect to $V_T$. |
| Slew Rate | SR | 0.94 | - | 0.75 | - | 0.56 | - | V/ns | With respect to VIH/VIL. |
| Duty Cycle Distortion | $t_{CKDCD}$ | 0.0 | 0.3 | 0.0 | 0.375 | 0.0 | 0.5 | ns | Allowable deviation from an ideal 50% duty cycle with respect to $V_T$. Includes jitter and phase noise. |
| Minimum Pulse Width | $t_{CKMPW}$ | 2.7 | - | 3.375 | - | 4.5 | - | ns | With respect to $V_T$. |
| **Input Signals (Referenced to CK)** | | | | | | | | | |
| Input Setup Time (DTR) | $t_{ISUddr}$ | 0.6 | - | 0.8 | - | 1.0 | - | ns | With respect to $V_{IH}/V_{IL}$. |
| Input Hold Time (DTR) | $t_{IHddr}$ | 0.6 | - | 0.8 | - | 1.0 | - | ns | With respect to $V_{IH}/V_{IL}$. |
| Input Setup Time (STR) | $t_{ISU}$ | 1 | - | 2 | - | 2 | - | ns | With respect to $V_{IH}/V_{IL}$. |
| Input Hold Time (STR) | $t_{IH}$ | 1 | - | 2 | - | 2 | - | ns | With respect to $V_{IH}/V_{IL}$. |
| Slew Rate @ 1.8 V | SR | 0.94 | -- | 0.75 | - | 0.56 | - | V/ns | With respect to $V_{IH}/V_{IL}$ and xSPI reference load. |
| Slew Rate @ 3.0 V | SR | 1.72 | - | 1.37 | - | 1.03 | - | V/ns | With respect to $V_{IH}/V_{IL}$ and xSPI reference load. |

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| Parameter | Symbol | xSPI-333(1) | | xSPI-266(2) | | xSPI-200(3) | | Unit | Comments |
|---|---|---|---|---|---|---|---|---|---|
| | | Min | Max | Min | Max | Min | Max | | |
| Note:<br>(1) xSPI-333: Up to 166 MHz; up to 333 MT/s.<br>(2) xSPI-266: Up to 133 MHz; up to 266 MT/s.<br>(3) xSPI-200: Up to 100 MHz; up to 200 MT/s. | | | | | | | | | |

### 5.12.11.2 Target Output to Controller Input Timing

Figure 5-35 xSPI Target Data Output Timing

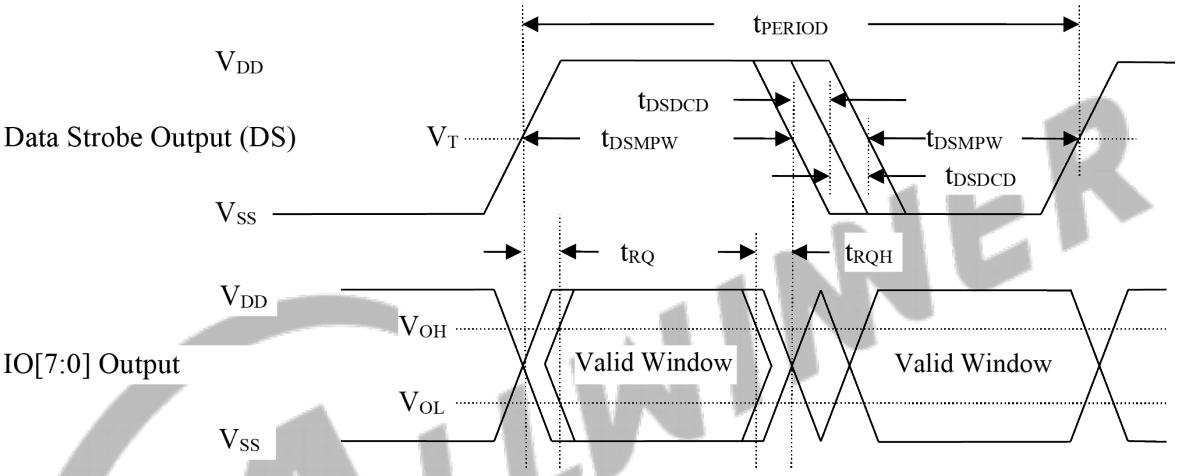



Table 5-37 xSPI Device Output Timing

| Parameter | Symbol | xSPI-200(1) | | Unit | Comments |
|---|---|---|---|---|---|
| | | Min | Max | | |
| **Data Strobe(2)** | | | | | |
| Cycle Time Data Transfer Mode | $t_{PERIOD}$ | 10 | - | ns | 100 MHz (max) between the rising edges with respect to $V_T$. |
| Duty Cycle Distortion | $t_{DSDCD}$ | 0.0 | 0.4 | ns | Allowable deviation from the input clock duty cycle distortion (tCKDCD) with respect to $V_T$. Includes jitter and phase noise. |
| Minimum Pulse Width | $t_{DSMPW}$ | 4.1 | - | ns | Minimum Pulse Width of DS is smaller than that of CK since the target is allowed to add distortion when generating DS from CK. With respect to $V_T$. |

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| Parameter | Symbol | xSPI-200(1) | | Unit | Comments |
|---|---|---|---|---|---|
| | | Min | Max | | |
| Output DATA (Referenced to DS) | | | | | |
| Output skew | $t_{RQ}$ | - | 0.8 | ns | With respect to $V_{OH}/V_{OL}$ and xSPI reference load. |
| Output hold skew | $t_{RQH}$ | - | 0.8 | ns | With respect to $V_{OH}/V_{OL}$ and xSPI reference load. |
| Note:<br>(1) xSPI-200: Up to 100 MHz; up to 200 MT/s.<br>(2) Controller CK edges are the trigger for target IO and DS edges. IO and DS edges therefore always follow their relat ed (triggering) CK edges. | | | | | |

## 5.12.12 UART Interface Timing

Figure 5-36 UART RX Timing

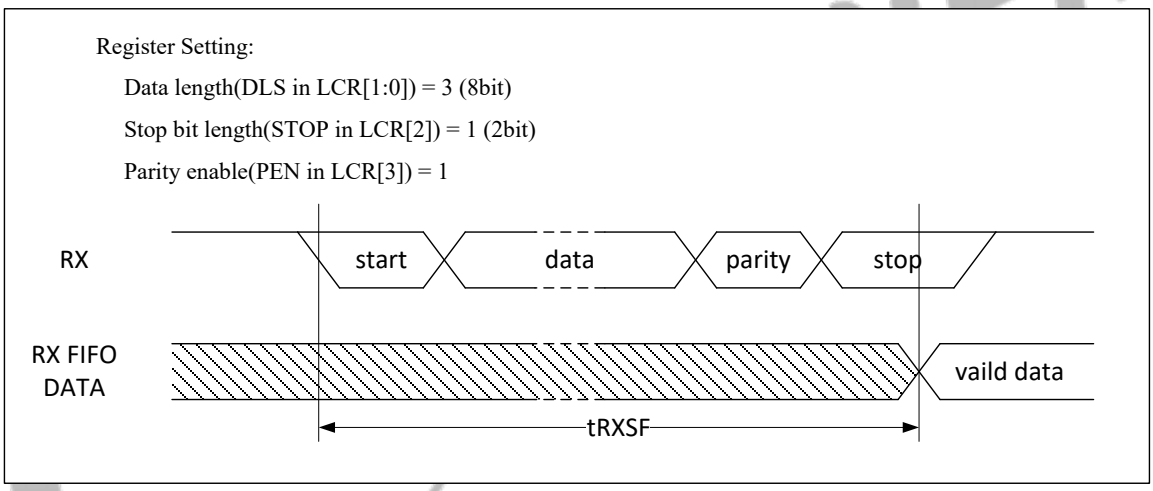
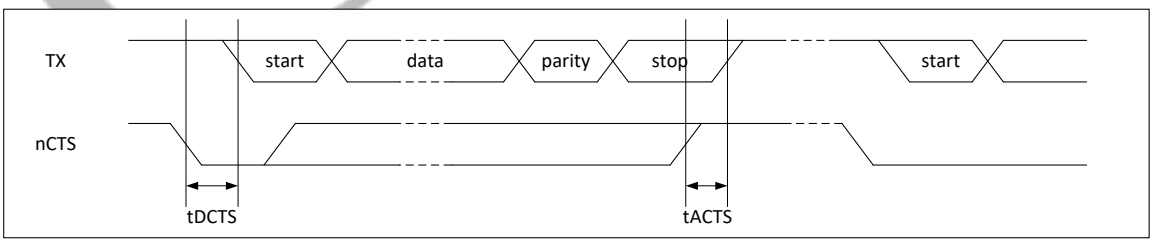



Figure 5-37 UART nCTS Timing




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Figure 5-38 UART nRTS Timing

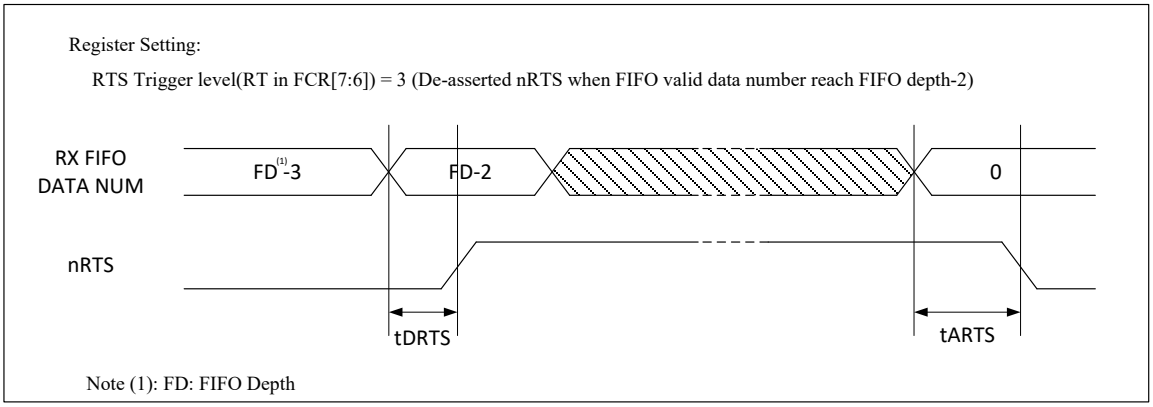



Table 5-38 UART Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| RX start to RX FIFO | tRXSF | 10.5*BRP(1) | - | 11*BRP(1) | ns |
| Delay time of de-asserted nCTS to TX strat | tDCTS | - | - | BRP(1) | ns |
| Step time of asserted nCTS to stop next transmission | tACTS | BRP(1)/4 | - | - | ns |
| Delay time of de-asserted nRTS | tDRTS | - | - | BRP(1) | ns |
| Delay time of asserted nRTS | tARTS | - | - | BRP(1) | ns |
| (1) BRP: Baud-Rate Period. | | | | | |

## 5.12.13 TWI Interface Timing

Figure 5-39 TWI Timing

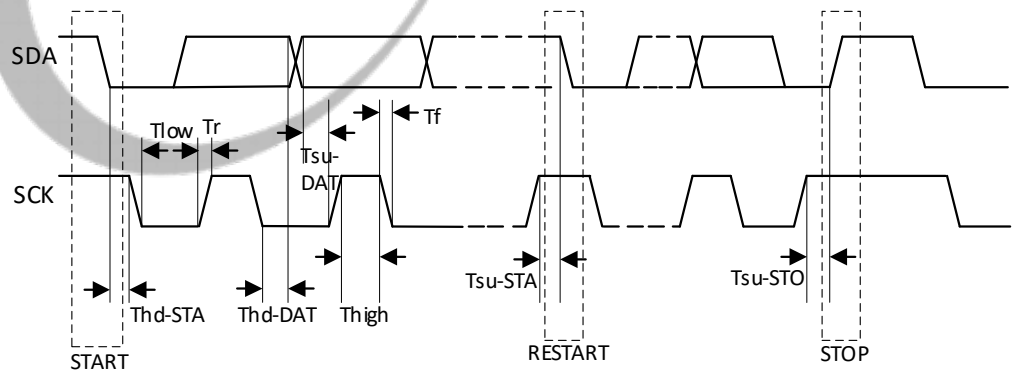



Table 5-39 TWI Timing Parameters

| Parameter | Symbol | Standard mode | | Fast mode | | Unit |
|---|---|---|---|---|---|---|
| | | Min | Max | Min | Max | |
| SCK clock frequency | Fsck | 0 | 100 | 0 | 400 | kHz |

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| Parameter | Symbol | Standard mode | | Fast mode | | Unit |
|---|---|---|---|---|---|---|
| | | Min | Max | Min | Max | |
| Setup Time In Start | Tsu-STA | 4.7 | - | 0.6 | - | us |
| Hold Time In Start | Thd-STA | 4.0 | - | 0.6 | - | us |
| Setup Time In Data | Tsu-DAT | 250 | - | 100 | - | ns |
| Hold Time In Data | Thd-DAT | 0 | - | 0 | - | us |
| Setup Time In Stop | Tsu-STO | 4.0 | - | 0.6 | - | us |
| SCK Low level Time | Tlow | 4.7 | - | 1.3 | - | us |
| SCK High level Time | Thigh | 4.0 | - | 0.6 | - | us |
| SCK/SDA Falling Time | Tf | - | 300 | 20 | 300 | ns |
| SCK/SDA Rising Time | Tr | - | 1000 | 20 | 300 | ns |

### 5.12.14 I2S/PCM Interface Timing

Figure 5-40 I2S/PCM Timing in Master Mode

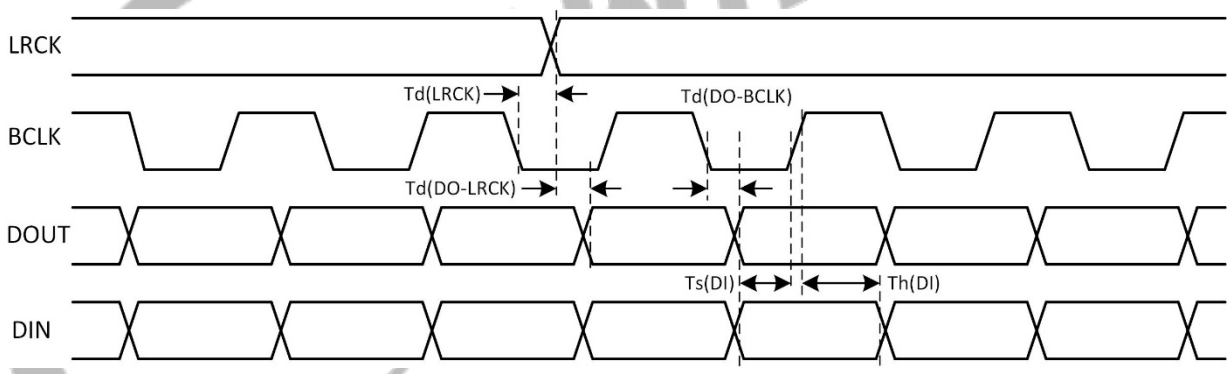



Table 5-40 I2S/PCM Timing Constants in Master Mode

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| LRCK delay | $T_d$(LRCK) | - | - | 10 | ns |
| LRCK to DOUT delay (For LJF) | $T_d$(DO-LRCK) | - | - | 10 | ns |
| BCLK to DOUT delay | $T_d$(DO-BCLK) | - | - | 10 | ns |
| DIN setup | $T_s$(DI) | 4 | - | - | ns |
| DIN hold | $T_h$(DI) | 4 | - | - | ns |
| BCLK rise time | $T_r$ | - | - | 8 | ns |
| BCLK fall time | $T_f$ | - | - | 8 | ns |

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Figure 5-41 I2S/PCM Timing in Slave Mode

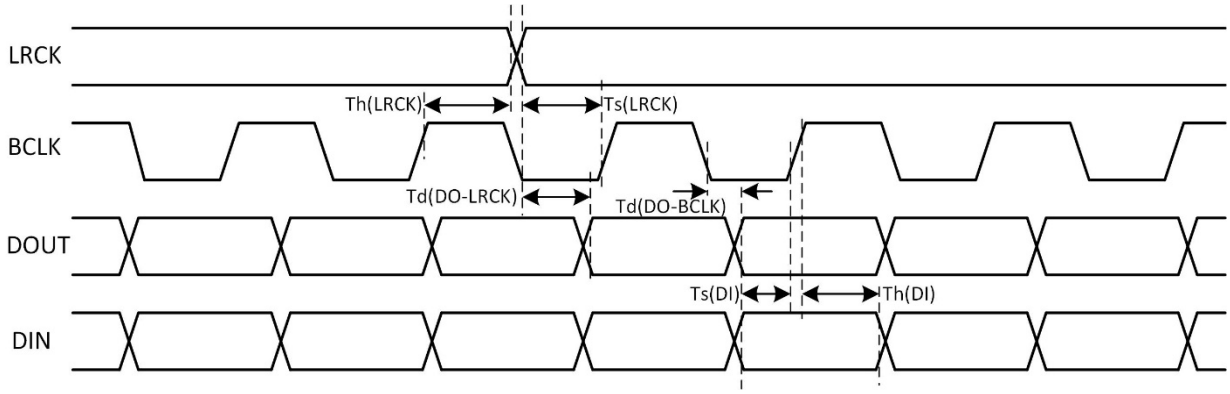



Table 5-41 Timing Constants in Slave Mode

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| LRCK setup | $T_s$(LRCK) | 4 | - | - | ns |
| LRCK hold | $T_h$(LRCK) | 4 | - | - | ns |
| LRCK to DOUT delay (For LJF) | $T_d$(DO-LRCK) | - | - | 10 | ns |
| BCLK to DOUT delay | $T_d$(DO-BCLK) | - | - | 10 | ns |
| DIN setup | $T_s$(DI) | 4 | - | - | ns |
| DIN hold | $T_h$(DI) | 4 | - | - | ns |
| BCLK rise time | $T_r$ | - | - | 4 | ns |
| BCLK fall time | $T_f$ | - | - | 4 | ns |

## 5.12.15 DMIC Interface Timing

Figure 5-42 DMIC Timing

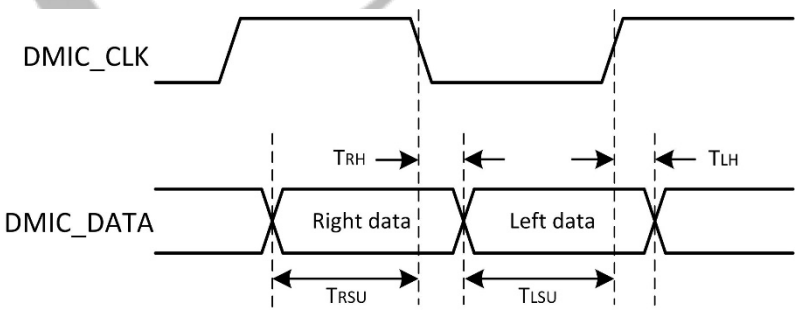



Table 5-42 DMIC Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| DMIC_DATA(Right) Setup Time to Falling Edge of DMIC_CLK | $T_{RSU}$ | 15 | - | - | ns |

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| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| DMIC_DATA(Right) Hold Time from Falling Edge of DMIC_CLK | $T_{RH}$ | 0 | - | - | ns |
| DMIC_DATA(Left) Setup Time to Rising Edge of DMIC_CLK | $T_{LSU}$ | 15 | - | - | ns |
| DMIC_DATA(Left) Hold Time From Rising edge of DMIC_CLK | $T_{LH}$ | 0 | - | - | ns |

### 5.12.16 OWA Interface Timing

Figure 5-43 OWA Timing

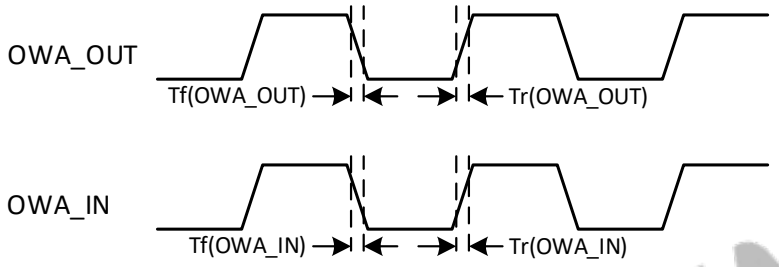



Table 5-43 OWA Timing Constants

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| OWA_OUT Rise Time | Tr(OWA_OUT) | - | - | 8 | ns |
| OWA_OUT Fall Time | Tf(OWA_OUT) | - | - | 8 | ns |
| OWA_IN Rise Time | Tr(OWA_IN) | - | - | 4 | ns |
| OWA_IN Fall Time | Tf(OWA_IN) | - | - | 4 | ns |

## 5.13 Power-Up and Power-Down Sequence

NOTE

- All “Tn” markers represent different steps during power-on and power-off process and show the minimum time interval between current step and previous step.
- The consequent steps in power-on sequence should not start before the previous step supplies have been stabilized within 90%-110% of their nominal voltage, unless stated otherwise.
- Different power rails at the same Tn step should become stable within 1ms.
- VCC25-DRAM and VDD18-LPDDR mentioned in Figure 5-44 and Figure 5-45 are external DRAM power supplies

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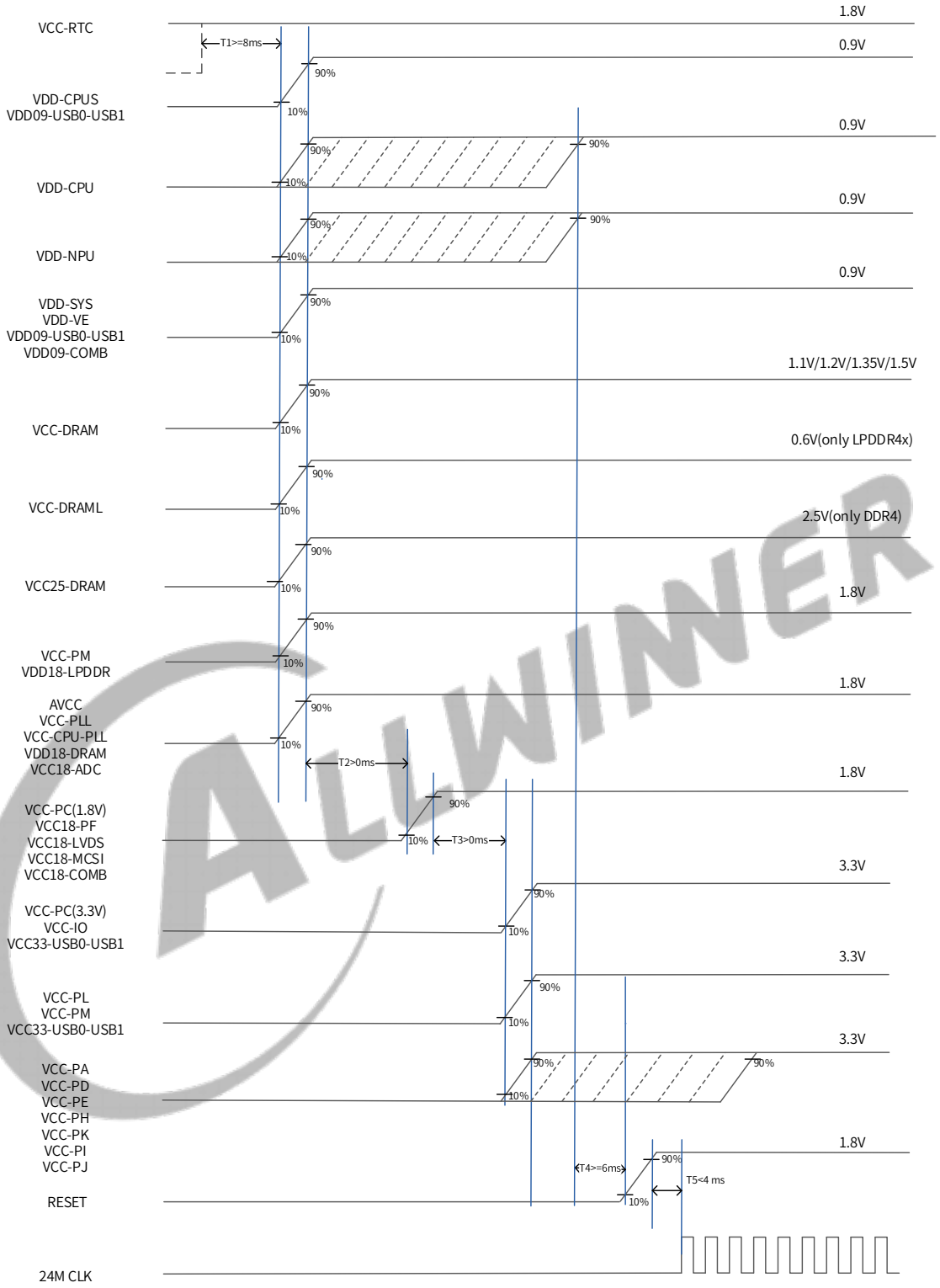
### 5.13.1 Power-Up Sequence

Figure 5-44 shows an example of the power-up sequence. The description of the power-up sequence is as follows.

- VCC18-LVDS must be ramped before VCC-PD.
- VCC18-MCSI must be ramped before 3.3 V VCC-PK.
- VCC18-ADC must be ramped before VCC-PA.
- During the entire power on sequence, the reset signal must be held on low until all other power rails (except 24 MHz CLK) are stable for at least 6 ms.
- The 24MHz clock starts oscillating and be stable within 4 ms after the reset signal is stable.

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Figure 5-44 Power-Up Sequence



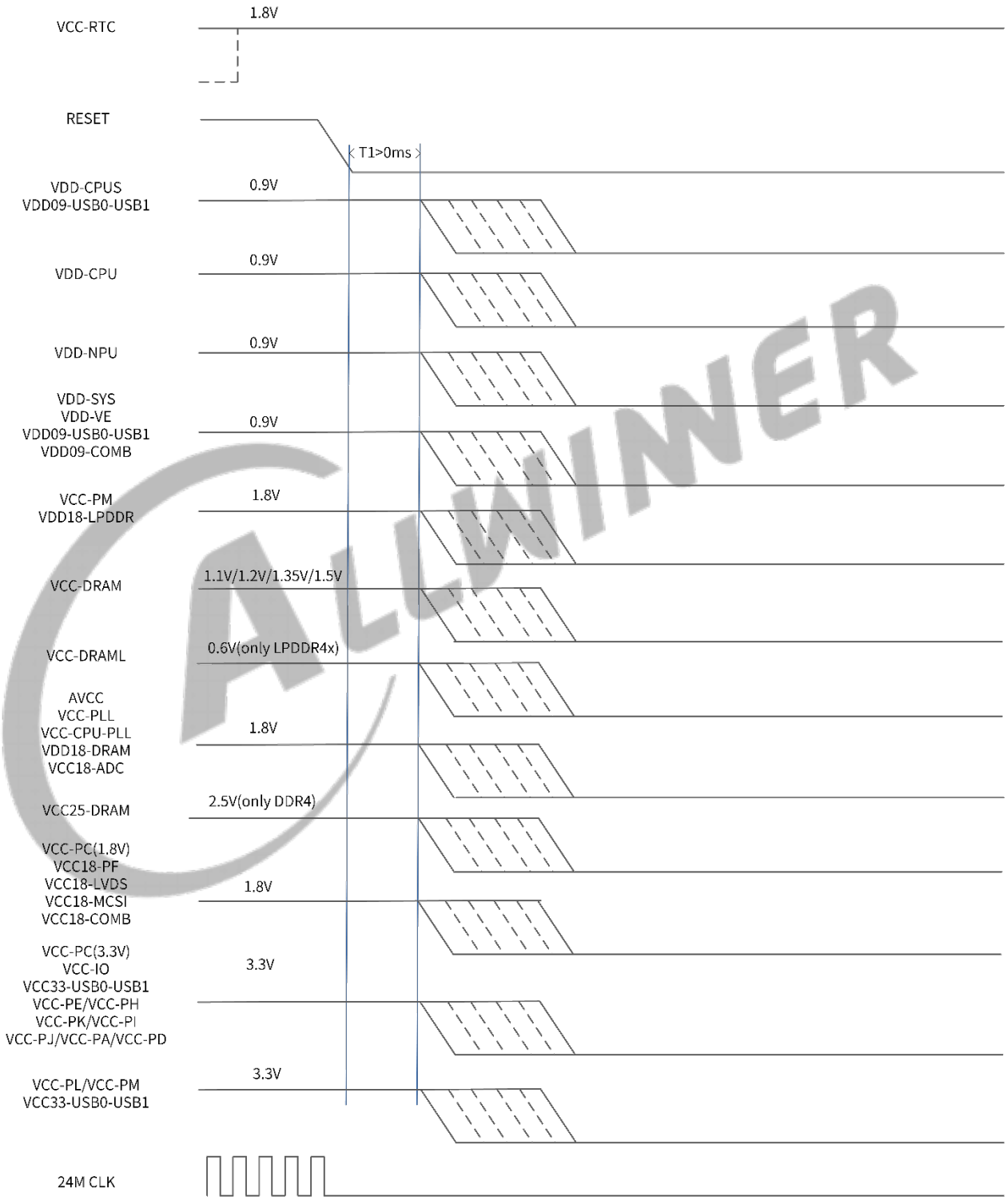

## 5.13.2 Power-Down Sequence

Figure 5-45 shows an example of the power-down sequence. The description of the power-down sequence is as follows.

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- Reset this device. After PMIC receives the power-down command, pull-down RESET.
- VCC-RTC holds high.
- After RESET ramps down, other powers start to ramp down, and the ramp rate of each power rail is generally determined by the load on that power.

Figure 5-45 Power-Down Sequence




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# 6 Temperature and Thermal Characteristics

## 6.1.1 Temperature

The following tables describe the temperature of the device

**Table 6-1 Operating and Storage Temperature**

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| $T_a$ | Ambient Operating Temperature | -40 | 85 | °C |
| $T_{STG}$ | Storage Temperature | -40 | 150 | °C |

**Table 6-2 Junction Temperature**

| Chip | Recommended Operating Temperature ($T_j$) | | Absolute Maximum Junction Temperature | Unit |
|---|---|---|---|---|
| | Min | Max | | |
| T536 | -40 | 110 | 125 | °C |

## 6.1.2 Package Thermal Characteristics

The maximum chip junction temperature ($T_{J\ max}$) must never exceed the values given in Table 6-2 Junction Temperature.

Failure to maintain a junction temperature within the range specified reduces operating lifetime, reliability, and performance, and may cause irreversible damage to the system. It is useful to calculate the exact power consumption and junction temperature to determine which the temperature will be best suited to the application. Therefore, the product should include thermal analysis and thermal design to ensure the operating junction temperature of the device is within functional limits.

The following tables show the thermal resistance characteristics of the device. These data are based on JEDEC JESD51 standard, because the actual system design and temperature could be different from JEDEC JESD51, these simulating data are a reference only and may not represent actual use-case values, please prevail in the actual application condition test.

**Table 6-3 Package Thermal Characteristics**

| Symbol | Parameter | Min | Typ(1)(2) | Max | Unit |
|---|---|---|---|---|---|
| $\theta_{JA}$ | Junction-to-Ambient Thermal Resistance | - | 23.04 | - | °C/W |
| $\theta_{JB}$ | Junction-to-Board Thermal Resistance | - | 8.64 | - | °C/W |
| $\theta_{JC}$ | Junction-to-Case Thermal Resistance | - | -(3) | - | °C/W |

(1) Reference document: JESD51-2 Integrated Circuits Thermal Test Method Environment Conditions – Natural Convection (Still Air). Available from www.jedec.org.

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(2) The testing PCB is 4 layers, 114.5 mm x 76.2 mm body, and 1.6 mm thickness. Ambient temperature is 25 °C.

(3) Due to the value of $\theta_{JC}$ approaching zero, it is unable to provide accurate test data for ED-FCCSP package.

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# 7 Pin Assignment

## 7.1.1 Pin Map

The following shows the ED-FCCSP 533 balls, 15 mm x 15 mm, 0.5 mm ball pitch, 0.3 mm ball size pin map.

**Figure 7-1 Pin Map**

FCCSP 15x15/0.3/0.5

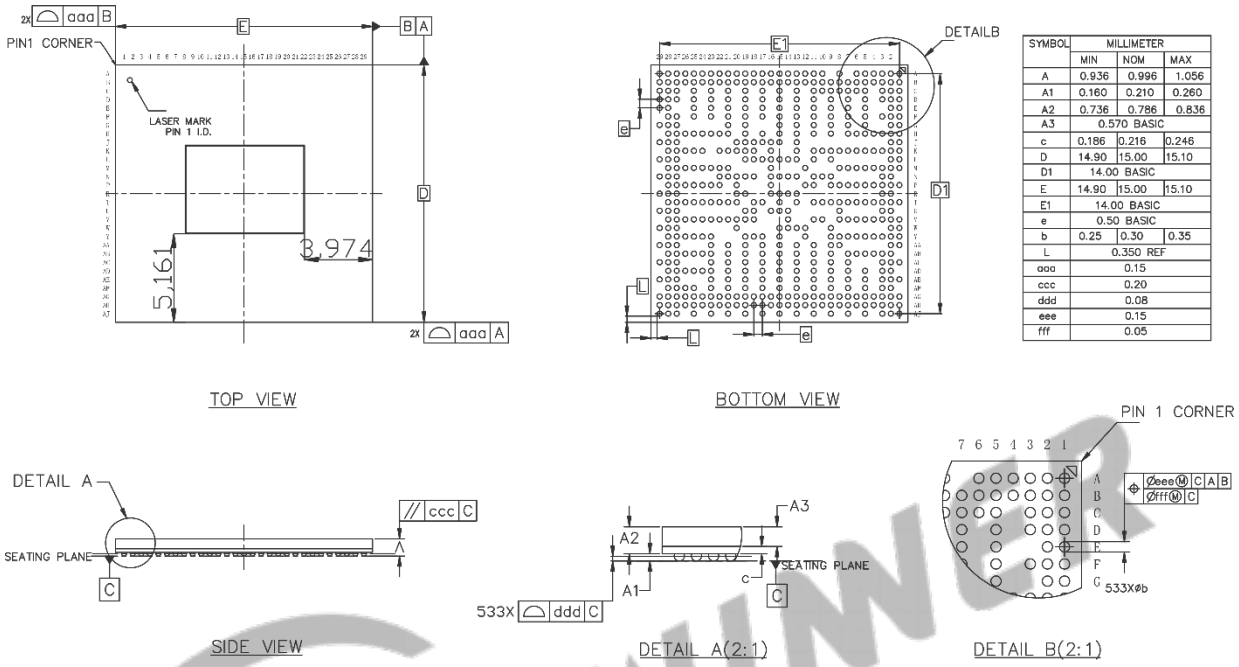
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | GND | GND | PD5 | PD3 | PD1 | PH5 | | PH15 | | PE12 | PK3 | PK5 | PK1 | PK9 | PK11 | PK7 | PK15 | PK17 | PK13 | USB0-DP | USB1-DP | PCIE-TX0-DN/USB2-U3-TXN | PCIE-RX0-DN/USB2-U3-RXN | PCIE-REF-CLKN | PCIE-REXT | PI15 | PI6 | GND | GND | A |
| B | GND | GND | PD4 | PD2 | PD0 | PH4 | PH7 | PH13 | PE0 | PE14 | PK2 | PK4 | PK0 | PK8 | PK10 | PK6 | PK14 | PK16 | PK12 | USB0-DM | USB1-DM | PCIE-TX0-DP/USB2-U3-TXP | PCIE-RX0-DP/USB2-U3-RXP | PCIE-REF-CLKP | PI16 | PI7 | PI0 | PI1 | GND | B |
| C | PD6 | PD7 | GND | | GND | PH6 | PH12 | PH14 | PE13 | PE11 | GND | | GND | | GND | | GND | | GND | | GND | | GND | | GND | | PI3 | PI2 | PI4 | C |
| D | PD8 | PD9 | GND | | PH3 | | PH11 | | PE10 | | PE9 | | PE6 | | PK21 | | PK23 | | PK18 | | PI10 | | PI13 | | PI8 | | PI5 | LINEOUTP | LINEOUTN | D |
| E | PD10 | PD11 | | | PH2 | | PH10 | | PE8 | | PE5 | | PE7 | | PK20 | | PK22 | | PK19 | | PI11 | | PI14 | | PI9 | | | PA3 | PA4 | E |
| F | PD12 | PD13 | GND | | PH1 | | PH9 | | PE2 | | PE4 | | PE17 | | GND | | GND | | GND | | PI12 | | GND | | GND | | PA2 | PA5 | | F |
| G | PD14 | PD15 | | | PH0 | | PH8 | | PE1 | | PE3 | | PE16 | | GND | | | | NC | | GND | | | | | | PA8 | PA7 | PA6 | G |
| H | PD16 | PD17 | GND | PD23 | PD22 | PD21 | GND | | GND | | GND | | PE15 | | VCC18-MCSI | | GND | | VCC18-COMB | | GND | VRA1 | AGND | PA0 | PA1 | PA9 | GPADC3-0 | TP-X2/GPADC1-1 | | H |
| J | PD18 | PD19 | | | | | | | | | | | VCC-PK | | GND | VDD09-USB0-USB1 | GND | VDD09-COMB | GND | GND | | | | | | | TP-Y1/GPADC1-2 | TP-Y2/GPADC1-3 | TP-X1/GPADC1-0 | J |
| K | PJ28 | PJ29 | PJ25 | PJ24 | PJ9 | PD20 | GND | VCC-PD | GND | VCC-PH | VCC-PE | | GND | | | VCC33-USB0-USB1 | | GND | GND | VCC-PI | AVCC | VCC-PA | | GND | GND | GPADC1-7 | GPADC1-6 | BOOT-SEL-GPADC1-4 | | K |
| L | PJ19 | GND | | | | | | | VCC18-LVDS | | GND | | VDD-SYSFB | VDD-SYS | VDD-SYS | GND | GND | GND | VDD-VE | | | | | | | | GPADC2-0 | GPADC1-8 | DDR-PARA-SEL-GPADC1-5 | L |
| M | PJ16 | PJ18 | PJ17 | PJ27 | PJ26 | PJ31 | PJ30 | GND | | | | GND | | | | GND | | | | | VCC18-ADC | AGND-ADC | VCC18-VCM-ADC | GPADC1-9 | GPADC2-1 | GPADC2-2 | GPADC2-3 | GPADC2-4 | | M |
| N | | PJ11 | PJ1 | | | | | | VDD-CPU | VDD-CPU | VDD-CPU | | GND | VDD-SYS | VDD-SYS | GND | | VDD-NPU | VDD-NPU | VDD-NPU | GND | GND | | | | | GPADC2-5 | GPADC2-6 | GPADC2-7 | N |
| P | PJ0 | PJ12 | PJ2 | PJ4 | PJ8 | PJ23 | PJ22 | VCC-PJ | VDD-CPU | | VDD-CPU | GND | | | | GND | | | | VDD-NPU | | GND | VCC-PG | PG14 | GND | GPADC2-8 | GPADC2-9 | LRADC | | P |
| R | | PJ21 | PJ20 | | | | | | | | | | VCC-CPU-PLL | CPU-PLLTEST | VDD-SYS | GND | | | VDD-NPU | VDD-NPU | | | | | | | PG3 | PG2 | PG0 | R |
| T | GND | PJ13 | PJ5 | PJ14 | PJ15 | PJ7 | PJ6 | GND | VDD-CPU | VDD-CPU | VDD-CPU | | VDD-CPUFB | | VDD-SYS | | GND | VDD-NPUFB | | GND | VCC-PL | GND | GND | PG6 | PG7 | PG12 | PG1 | GND | | T |
| U | | PJ10 | PJ3 | | | | | VCC-PC | | | | | | GND | GND | GND | | GND | GND | | | | | | | | PG13 | PG5 | PG4 | U |
| V | PC8 | PC9 | PC1 | PC0 | PC2 | PC6 | PC5 | VCC-DRAML | | VCC-DRAML | VCC-DRAML | | | VCC-DRAML | | | | SZQ | VCC18-PF | VCC-IO | VCC-PM | GND | VDD-CPUS | GND | PG11 | PG10 | PG8 | PG9 | | V |
| W | PC10 | PC11 | | | | | | | VCC-DRAM | | | VCC-DRAM | VCC-DRAM | VCC-DRAM | VCC-DRAM | | | | | VCC18-EFUSE | | | | | | | PL0 | PL2 | PL3 | W |
| Y | PC13 | PC14 | PC3 | PC4 | PC12 | GND | PC7 | GND | GND | | | VCC-DRAM | | | VDD18-DRAM | | GND | GND | GND | | | | GND | GND | PL7 | PL6 | PL5 | PL1 | | Y |
| AA | PC15 | PC16 | | | | | | GND | GND | | GND | | GND | | GND | | GND | | GND | | PLLTEST | | VCC-PLL | | | | PL10 | PL11 | PL4 | AA |
| AB | | SDQ24 | SDQ26 | GND | GND | GND | GND | | SA8 | | SA6 | | GND | | SA7 | | SDQ6 | | GND | | PB1 | | GND | VCC-RTC | PL13 | PL12 | PL9 | PL8 | | AB |
| AC | GND | SDQ27 | SDQ25 | | | | GND | | SA14 | | SACT | | SART | | SA2 | | SDQ7 | | GND | | PB0 | | PB12 | | | | PM3 | PM5 | PM4 | AC |
| AD | | SDQS3P | SDQS3N | | SDQ31 | | SBA0 | | SA15 | | SA13 | | SBG1 | | SDQ2 | | SDQ11 | | SDQ13 | | FEL | | PB5 | PB6 | GND | NC | PM0 | PM2 | | AD |
| AE | GND | SDQM3 | SDQ29 | | SDQ30 | | GND | | GND | | GND | | GND | | GND | | GND | | GND | | JTAG-SEL | | PB4 | | | | GND | NMI | PM1 | AE |
| AF | | SDQ28 | SDQ16 | | GND | | SCKE0 | | SBG0 | | SA17 | | SA9 | | SDQM0 | | SDQ9 | | SDQ12 | | PF6 | | PB3 | | PB11 | GND | RESET | X32KFOUT | | AF |
| AG | SDQ17 | SDQ18 | SDQ19 | SDQ20 | SDQ22 | SBA1 | SODT0 | SCS0 | SCKP | SA16 | SA12 | SA1 | SA10 | SDQ1 | SDQS0P | SDQ4 | SDQ8 | SDQS1P | SDQ10 | SDQ14 | PF1 | PF3 | GND | PB2 | PB8 | PB10 | GND | X32KOUT | X32KIN | AG |
| AH | GND | SDQM2 | SDQS2P | SDQ21 | SDQ23 | SA4 | SODT1 | SCS1 | SCKN | SA3 | SA5 | SA0 | SA11 | SDQ0 | SDQS0N | SDQ3 | SDQ5 | SDQS1N | SDQM1 | SDQ15 | PF0 | PF2 | PF5 | PB7 | PB14 | PB9 | DXOUT | REFCLK-OUT | GND | AH |
| AJ | GND | GND | SDQS2N | | GND | | GND | | GND | | GND | | GND | | GND | | GND | | GND | | GND | | PF4 | | PB13 | | DXIN | GND | GND | AJ |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |

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## 7.1.2 Package Dimension

The following figure shows the top, bottom, and side views of the package.

**Figure 7-2 Package Dimension**




| SYMBOL | MILLIMETER | | |
|---|---|---|---|
| | MIN | NOM | MAX |
| A | 0.936 | 0.996 | 1.056 |
| A1 | 0.160 | 0.210 | 0.260 |
| A2 | 0.736 | 0.786 | 0.836 |
| A3 | | 0.570 BASIC | |
| c | 0.186 | 0.216 | 0.246 |
| D | 14.90 | 15.00 | 15.10 |
| D1 | | 14.00 BASIC | |
| E | 14.90 | 15.00 | 15.10 |
| E1 | | 14.00 BASIC | |
| e | | 0.50 BASIC | |
| b | 0.25 | 0.30 | 0.35 |
| L | | 0.350 REF | |
| aaa | | 0.15 | |
| ccc | | 0.20 | |
| ddd | | 0.08 | |
| eee | | 0.15 | |
| fff | | 0.05 | |

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# 8 Carrier, Storage and Backing Information

## 8.1 Carrier

NOTE

- The entire chip with ED-PCCSP package can withstand a maximum plane pressure of 1500 N (approximately 966.67 PSI pressure).
- Chips with ED-FCCSP package are susceptible to stress. Please be careful not to subject the chip to excessive stress during transportation or usage.

The following table shows matrix tray carrier information.

**Table 8-1 Matrix Tray Carrier Information**

| Item | Color | Size | Note |
|---|---|---|---|
| Tray | Black | 322.6 mm x 135.9 mm x 7.62 mm | 126 Qty/Tray |
| Aluminum foil bags | Silvery white | 540 mm x 300 mm x 0.14 mm | Vacuum packing<br>Including HIC and desiccant<br>Printing: RoHS symbol |
| Pearl cotton cushion (Vacuum bag) | White | 12 mm x 680 mm x 185 mm | |
| Pearl cotton cushion (The Gap between vacuum bag and inner box) | White | Left-Right:<br>12 mm x 180 mm x 85 mm<br>Front-Back:<br>12 mm x 350 mm x 70 mm | |
| Inner Box | White | 396 mm x 196 mm x 96 mm | Printing: RoHS symbol<br>10 Tray/Inner box |
| Carton | White | 420 mm x 410 mm x 320 mm | 6 Inner box/Carton |

The following table shows packing quantity.

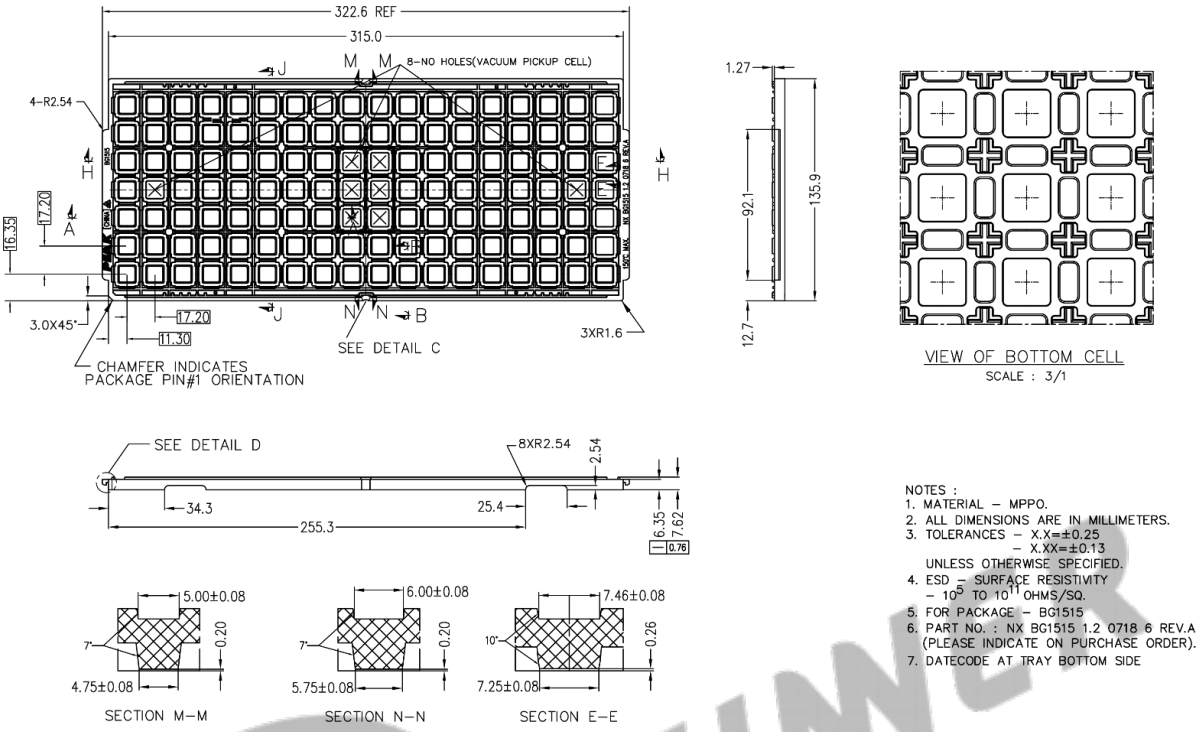
**Table 8-2 Packing Quantity Information**

| Sample | Size | Qty/Tray | Tray/Inner Box | Full Inner Box Qty | Inner Box/Carton | Full Carton Qty |
|---|---|---|---|---|---|---|
| T536 | 15 mm x 15 mm | 126 | 10 | 1260 | 6 | 7560 |

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The following figure shows tray dimension drawing.

**Figure 8-1 Tray Dimension Drawing**




# 8.2 Storage

Reliability is affected if any condition specified in section 8.2.2 and section 8.2.3 has been exceeded.

## 8.2.1 Moisture Sensitivity Level (MSL)

A package's MSL indicates its ability to withstand exposure after it is removed from its shipment bag, a low MSL device sample can be exposed on the factory floor longer than a high MSL device sample. Table 8-3 defines all MSL.

**Table 8-3 MSL Summary**

| MSL | Out-of-bag floor life | Comments |
|---|---|---|
| 1 | Unlimited | ⩽30°C / 85%RH |
| 2 | 1 year | ⩽30°C / 60%RH |
| 2a | 4 weeks | ⩽30°C / 60%RH |
| 3 | 168 hours | ⩽30°C / 60%RH |
| 4 | 72 hours | ⩽30°C / 60%RH |
| 5 | 48 hours | ⩽30°C / 60%RH |
| 5a | 24 hours | ⩽30°C / 60%RH |
| 6 | Time on Label (TOL) | ⩽30°C / 60%RH |

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NOTE

The T536 device samples are classified as MSL3.

### 8.2.2 Bagged Storage Conditions

The following table defines the shelf life.

**Table 8-4 Bagged Storage Conditions**

| | |
|---|---|
| Packing Mode | Vacuum packing |
| Storage Temperature | 20 26°C |
| Storage Humidity | 40% 60%RH |
| Shelf Life | 12 months |

### 8.2.3 Out-of-bag Duration

It is defined by the device MSL rating. The out-of-bag duration is as follows.

**Table 8-5 Out-of-bag Duration**

| | |
|---|---|
| Storage Temperature | 20 26°C |
| Storage Humidity | 40% 60%RH |
| Moisture Sensitive Level (MSL) | 3 |
| Floor Life | 168 hours |

For no mention of storage rules in this document, refer to the latest *IPC/JEDEC J-STD-020C*.

## 8.3 Baking

It is not necessary to bake chips if the conditions specified in section 8.2.2 and section 8.2.3 have not been exceeded. It is necessary to bake chips if any condition specified in section 8.2.2 and section 8.2.3 has been exceeded.

It is necessary to bake chips if the storage humidity condition has been exceeded, we recommend that the device sample removed from its shipment bag for more than 2 days shall be baked to guarantee production.

Baking conditions: 125°C, 8 hours, nitrogen protection. Note that the baking should not exceed 1 times due to a risk of deformation.

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# 9 Reflow Profile

All Allwinner chips provided for clients are lead-free RoHS-compliant products.

The reflow profile recommended in this document is a lead-free reflow profile that is suitable for pure lead-free technology of lead-free solder paste. If customers need to use lead solder paste, contact Allwinner FAE.

The following figure shows the appropriate reflow profile.

**Figure 9-1 Lead-free Reflow Profile**

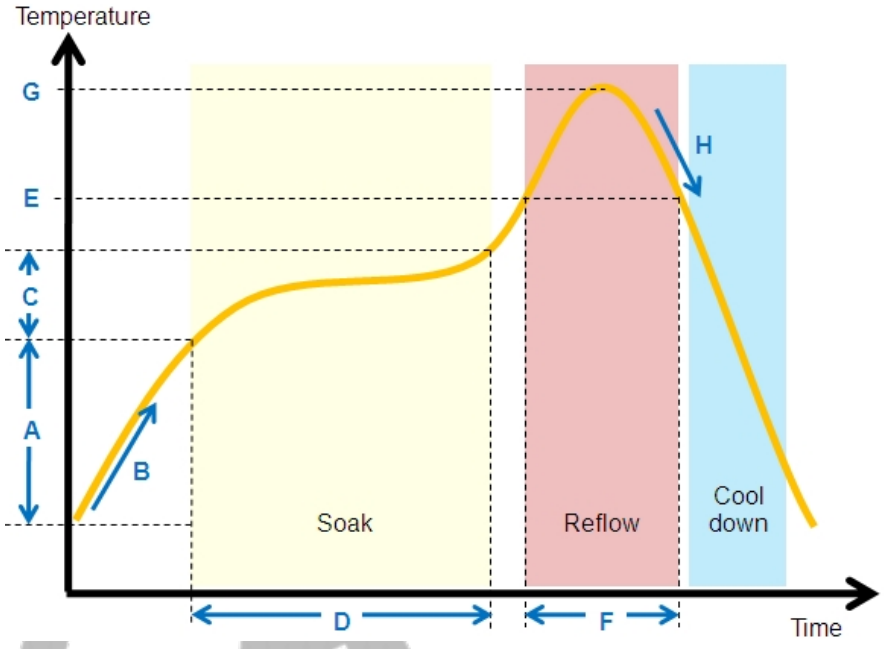



**Table 9-1 Lead-free Reflow Profile Conditions**

| | QTI typical SMT reflow profile conditions (for reference only) | |
|---|---|---|
| | Step | Reflow condition |
| Environment | N2 purge reflow usage (yes/no) | Yes, N2 purge used |
| | If yes, O2 ppm level | O2 < 1500 ppm |
| A | Preheat ramp up temperature range | 25 °C -> 150 °C |
| B | Preheat ramp up rate | 1.5–2.5 °C/s |
| C | Soak temperature range | 150 °C -> 190 °C |
| D | Soak time | 80–110 s |
| E | Liquidus temperature | 217°C |
| F | Time above liquidus | 60–90 s |
| G | Peak temperature | 240–250 °C |

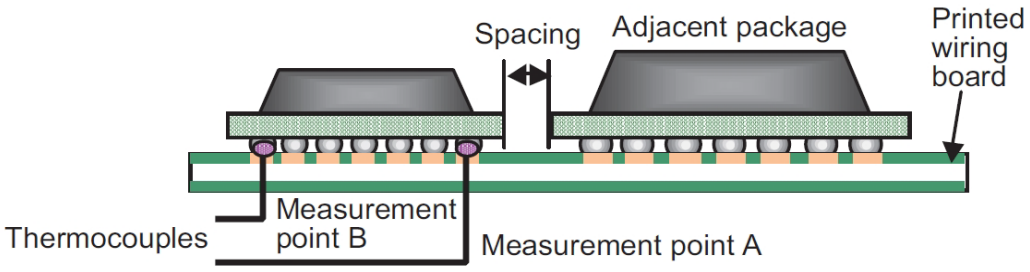
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| | QTI typical SMT reflow profile conditions (for reference only) | |
|---|---|---|
| | Step | Reflow condition |
| H | Cool down temperature rate | ⩽4°C/s |

The method of measuring the reflow soldering process is as follows.

Fix the thermocouple probe of the temperature measuring line at the connection point between the pin (solderable end) of the packaged device and the pad by using high-temperature solder wire or high-temperature tape, fix the packaged device at the pad by using high-temperature tape or other methods, and cover over the thermocouple probe.

Figure 9-2 Measuring the Reflow Soldering Process




NOTE

To measure the temperature of the QFP-packaged chip, place the temperature probe directly at the pin.

If possible, the more accurate measuring way is to drill the packaged device, or drill the PCB, and fix the thermocouple probe through the drilled hole at the pad.

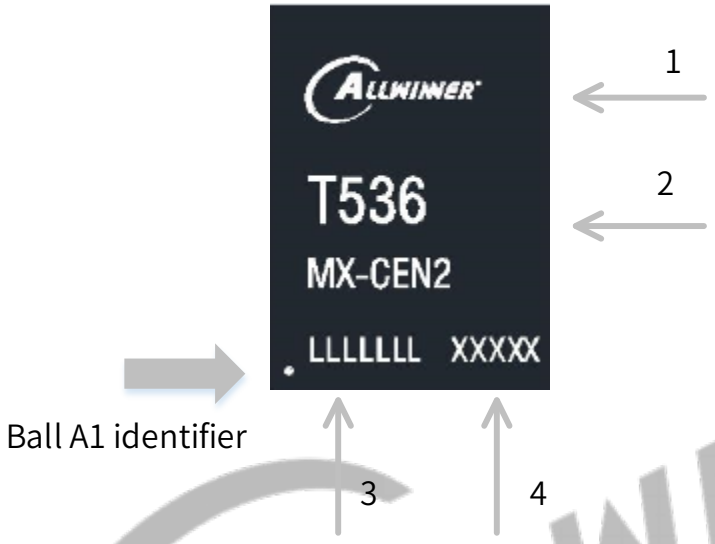
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# 10 Part Marking

## 10.1 T536MX-CEN2

The following figure shows the T536MX-CEN2 marking.

**Figure 10-1 T536MX-CEN2 Marking**




The following table describes the T536MX-CEN2 marking definitions.

**Table 10-1 T536MX-CEN2 Marking Definitions**

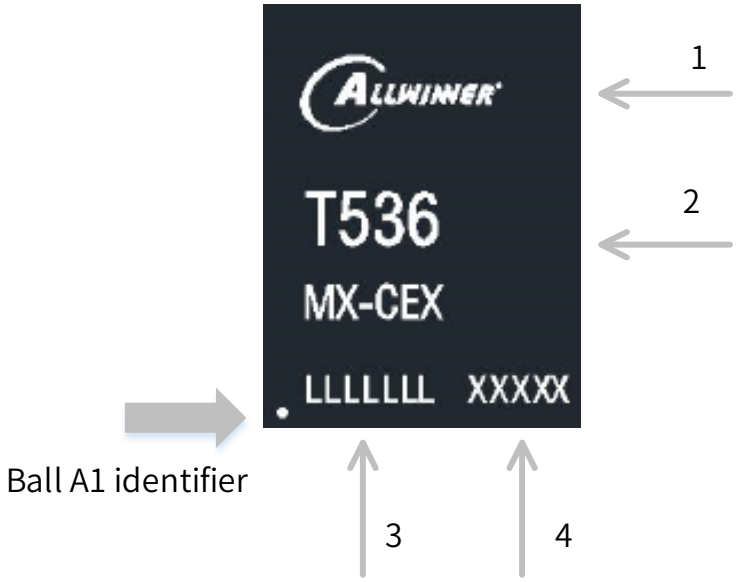
| No. | Marking | Description | Fixed/Dynamic |
|---|---|---|---|
| 1 | ALLWINNER | Allwinner logo or name | Fixed |
| 2 | T536MX-CEN2 | Product name | Fixed |
| 3 | LLLLLLL | Lot number | Dynamic |
| 4 | XXXXX | Date code | Dynamic |

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## 10.2 T536MX-CEX

The following figure shows the T536MX-CEX marking.

**Figure 10-2 T536MX-CEX Marking**




The following table describes the T536MX-CEX marking definitions.

**Table 10-2 T536MX-CEX Marking Definitions**

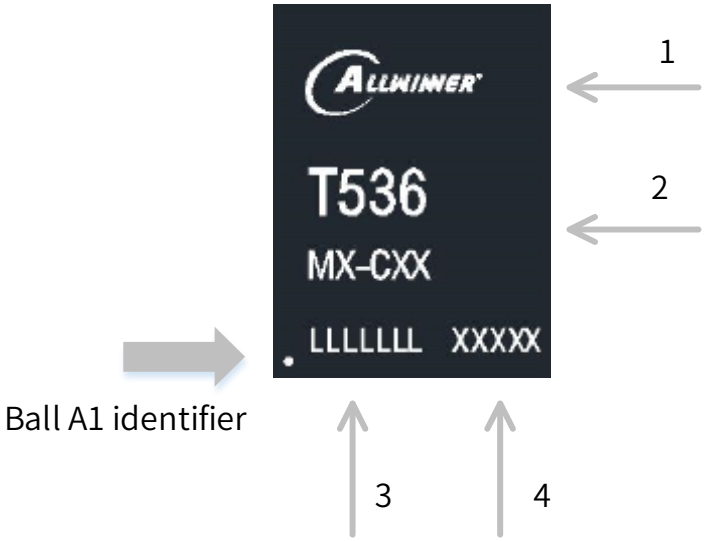
| No. | Marking | Description | Fixed/Dynamic |
|---|---|---|---|
| 1 | ALLWINNER | Allwinner logo or name | Fixed |
| 2 | T536MX-CEX | Product name | Fixed |
| 3 | LLLLLLL | Lot number | Dynamic |
| 4 | XXXXX | Date code | Dynamic |

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## 10.3 T536MX-CXX

The following figure shows the T536MX-CXX marking.

**Figure 10-3 T536MX-CXX Marking**




The following table describes the T536MX-CXX marking definitions.

**Table 10-3 T536MX-CXX Marking Definitions**

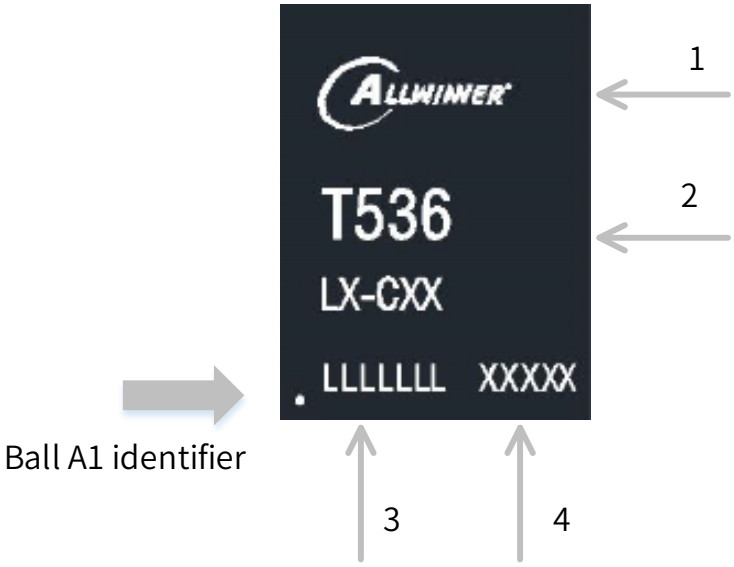
| No. | Marking | Description | Fixed/Dynamic |
|---|---|---|---|
| 1 | ALLWINNER | Allwinner logo or name | Fixed |
| 2 | T536MX-CXX | Product name | Fixed |
| 3 | LLLLLLL | Lot number | Dynamic |
| 4 | XXXXX | Date code | Dynamic |

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## 10.4 T536LX-CXX

The following figure shows the T536LX-CXX marking.

**Figure 10-4 T536LX-CXX Marking**




The following table describes the T536LX-CXX marking definitions.

**Table 10-4 T536LX-CXX Marking Definitions**

| No. | Marking | Description | Fixed/Dynamic |
|---|---|---|---|
| 1 | ALLWINNER | Allwinner logo or name | Fixed |
| 2 | T536LX-CXX | Product name | Fixed |
| 3 | LLLLLLL | Lot number | Dynamic |
| 4 | XXXXX | Date code | Dynamic |

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